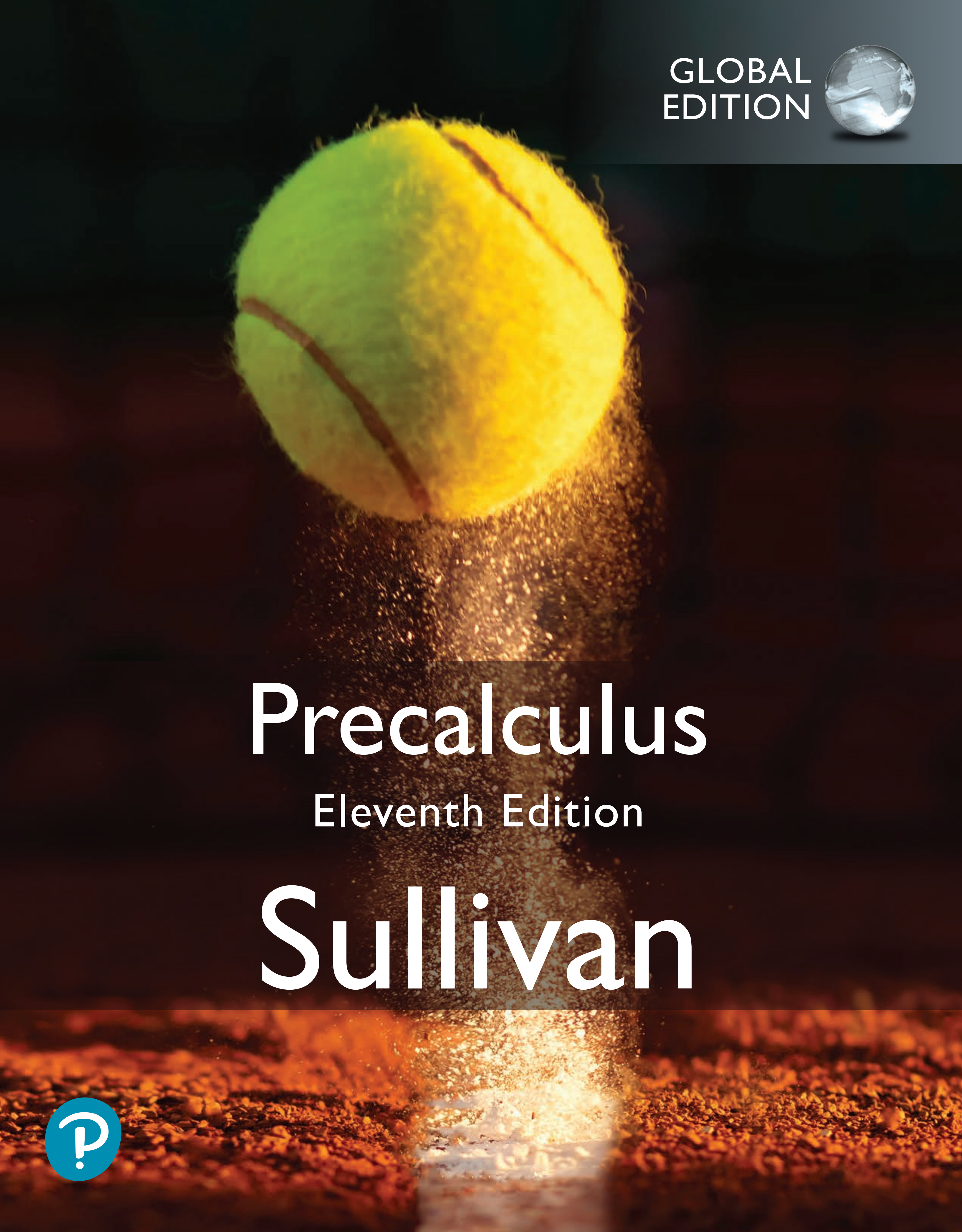

GLOBAL EDITION

Precalculus

Eleventh Edition

Sullivan

To the Student

As you begin, you may feel anxious about the number of theorems, definitions, procedures, and equations. You may wonder if you can learn it all in time. Don't worry–your concerns are normal. This textbook was written with you in mind. If you attend class, work hard, and read and study this text, you will build the knowledge and skills you need to be successful. Here's how you can use the text to your benefit.

Read Carefully

When you get busy, it's easy to skip reading and go right to the problems. Don't … the text has a large number of examples and clear explanations to help you break down the mathematics into easy-to-understand steps. Reading will provide you with a clearer understanding, beyond simple memorization. Read before class (not after) so you can ask questions about anything you didn't understand. You'll be amazed at how much more you'll get out of class if you do this.

Use the Features

I use many different methods in the classroom to communicate. Those methods, when incorporated into the text, are called "features." The features serve many purposes, from providing timely review of material you learned before (just when you need it) to providing organized review sessions to help you prepare for quizzes and tests. Take advantage of the features and you will master the material.

To make this easier, we've provided a brief guide to getting the most from this text. Refer to "Prepare for Class," "Practice," and "Review" at the front of the text. Spend fifteen minutes reviewing the guide and familiarizing yourself with the features by flipping to the page numbers provided. Then, as you read, use them. This is the best way to make the most of your text.

Please do not hesitate to contact me through Pearson, with any questions, comments, or suggestions for improving this text. I look forward to hearing from you, and good luck with all of your studies.

Best Wishes!

Michael Sullivan

Prepare for Class "Read the Book"

Feature	Description	Benefit	Page
Every Chapter Opener begins with . . .			
Chapter-Opening Topic & Project	Each chapter begins with a discussion of a topic of current interest and ends with a related project.	The project lets you apply what you learned to solve a problem related to the topic.	294
Internet-based Project	The projects allow for the integration of spreadsheet technology that you will need to be a productive member of the workforce.	The projects give you an opportunity to collaborate and use mathematics to deal with issues of current interest.	396
Every Section begins with . . .			
LEARNING OBJECTIVES 2	Each section begins with a list of objectives. Objectives also appear in the text where the objective is covered.	These focus your study by emphasizing what's most important and where to find it.	319
Sections contain . . .			
PREPARING FOR THIS SECTION	Most sections begin with a list of key concepts to review with page numbers.	Ever forget what you've learned? This feature highlights previously learned material to be used in this section. Review it, and you'll always be prepared to move forward.	315
Now Work the 'Are You Prepared?' Problems	Problems that assess whether you have the prerequisite knowledge for the upcoming section.	Not sure you need the Preparing for This Section review? Work the 'Are You Prepared?' problems. If you get one wrong, you'll know exactly what you need to review and where to review it!	315, 326
Now Work PROBLEMS	These follow most examples and direct you to a related exercise.	We learn best by doing. You'll solidify your understanding of examples if you try a similar problem right away, to be sure you understand what you've just read.	322, 327
WARNING	Warnings are provided in the text.	These point out common mistakes and help you to avoid them.	349
Exploration and Seeing the Concept	These graphing utility activities foreshadow a concept or solidify a concept just presented.	You will obtain a deeper and more intuitive understanding of theorems and definitions.	310, 335
In Words	These provide alternative descriptions of select definitions and theorems.	Does math ever look foreign to you? This feature translates math into plain English.	332
Calculus	These appear next to information essential for the study of calculus.	Pay attention–if you spend extra time now, you'll do better later!	90, 299, 322
SHOWCASE EXAMPLES	These examples provide "how-to" instruction by offering a guided, step-by-step approach to solving a problem.	With each step presented on the left and the mathematics displayed on the right, you can immediately see how each step is used.	261
Model It! Examples and Problems	These examples and problems require you to build a mathematical model from either a verbal description or data. The homework Model It! problems are marked by purple headings.	It is rare for a problem to come in the form "*Solve the following equation*." Rather, the equation must be developed based on an explanation of the problem. These problems require you to develop models to find a solution to the problem.	339, 368
NEW! Need to Review?	These margin notes provide a just-in-time reminder of a concept needed now, but covered in an earlier section of the book. Each note is back-referenced to the chapter, section and page where the concept was originally discussed.	Sometimes as you read, you encounter a word or concept you know you've seen before, but don't remember exactly what it means. This feature will point you to where you first learned the word or concept. A quick review now will help you see the connection to what you are learning for the first time and make remembering easier the next time.	308

Practice “Work the Problems”

Feature	Description	Benefit	Page
‘Are You Prepared?’ Problems	These assess your retention of the prerequisite material you’ll need. Answers are given at the end of the section exercises. This feature is related to the Preparing for This Section feature.	Do you always remember what you’ve learned? Working these problems is the best way to find out. If you get one wrong, you’ll know exactly what you need to review and where to review it!	332, 340
Concepts and Vocabulary	These short-answer questions, mainly Fill-in-the-Blank, Multiple-Choice and True/False items, assess your understanding of key definitions and concepts in the current section.	It is difficult to learn math without knowing the language of mathematics. These problems test your understanding of the formulas and vocabulary.	326
Skill Building	Correlated with section examples, these problems provide straightforward practice.	It’s important to dig in and develop your skills. These problems provide you with ample opportunity to do so.	326–328
Applications and Extensions	These problems allow you to apply your skills to real-world problems. They also allow you to extend concepts learned in the section.	You will see that the material learned within the section has many uses in everyday life.	329–331
NEW! Challenge Problems	These problems have been added in most sections and appear at the end of the Application and Extensions exercises. They are intended to be thought-provoking, requiring some ingenuity to solve.	Are you a student who likes being challenged? Then the Challenge Problems are for you! Your professor might also choose to assign a challenge problem as a group project. The ability to work with a team is a highly regarded skill in the working world.	331
Explaining Concepts: Discussion and Writing	“Discussion and Writing” problems are colored red. They support class discussion, verbalization of mathematical ideas, and writing and research projects.	To verbalize an idea, or to describe it clearly in writing, shows real understanding. These problems nurture that understanding. Many are challenging, but you’ll get out what you put in.	331
Retain Your Knowledge	These problems allow you to practice content learned earlier in the course.	Remembering how to solve all the different kinds of problems that you encounter throughout the course is difficult. This practice helps you remember.	331
Now Work PROBLEMS	Many examples refer you to a related homework problem. These related problems are marked by a pencil and orange numbers.	If you get stuck while working problems, look for the closest Now Work problem, and refer to the related example to see if it helps.	324, 325
Review Exercises	Every chapter concludes with a comprehensive list of exercises to pratice. Use the list of objectives to determine the objective and examples that correspond to the problems.	Work these problems to ensure that you understand all the skills and concepts of the chapter. Think of it as a comprehensive review of the chapter.	391–394

Review "Study for Quizzes and Tests"

Feature	Description	Benefit	Page
The Chapter Review at the end of each chapter contains . . .			
Things to Know	A detailed list of important theorems, formulas, and definitions from the chapter.	Review these and you'll know the most important material in the chapter!	389–390
You Should Be Able to . . .	Contains a complete list of objectives by section, examples that illustrate the objective, and practice exercises that test your understanding of the objective.	Do the recommended exercises and you'll have mastered the key material. If you get something wrong, go back and work through the objective listed and try again.	390–391
Review Exercises	These provide comprehensive review and practice of key skills, matched to the Learning Objectives for each section.	Practice makes perfect. These problems combine exercises from all sections, giving you a comprehensive review in one place.	391–394
Chapter Test	About 15–20 problems that can be taken as a Chapter Test. Be sure to take the Chapter Test under test conditions—no notes!	Be prepared. Take the sample practice test under test conditions. This will get you ready for your instructor's test. If you get a problem wrong, you can watch the Chapter Test Prep Video.	394
Cumulative Review	These problem sets appear at the end of each chapter, beginning with Chapter 2. They combine problems from previous chapters, providing an ongoing cumulative review. When you use them in conjunction with the Retain Your Knowledge problems, you will be ready for the final exam.	These problem sets are really important. Completing them will ensure that you are not forgetting anything as you go. This will go a long way toward keeping you primed for the final exam.	395
Chapter Projects	The Chapter Projects apply to what you've learned in the chapter. Additional projects are available on the Instructor's Resource Center (IRC).	The Chapter Projects give you an opportunity to use what you've learned in the chapter to the opening topic. If your instructor allows, these make excellent opportunities to work in a group, which is often the best way to learn math.	396
Internet-Based Projects	In selected chapters, a Web-based project is given.	These projects give you an opportunity to collaborate and use mathematics to deal with issues of current interest by using the Internet to research and collect data.	396

To the Memory of
My Mother and Father

Precalculus

Eleventh Edition

Global Edition

Michael Sullivan
Chicago State University

Pearson

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Appendix A Review A1

Appendix B Graphing Utilities B1

Three Distinct Series

Students have different goals, learning styles, and levels of preparation. Instructors have different teaching philosophies, styles, and techniques. Rather than write one series to fit all, the Sullivans have written three distinct series. All share the same goal—to develop a high level of mathematical understanding and an appreciation for the way mathematics can describe the world around us. The manner of reaching that goal, however, differs from series to series.

Flagship Series, Eleventh Edition

The Flagship Series is the most traditional in approach yet modern in its treatment of precalculus mathematics. In each text, needed review material is included, and is referenced when it is used. Graphing utility coverage is optional and can be included or excluded at the discretion of the instructor: *College Algebra, Algebra & Trigonometry, Trigonometry: A Unit Circle Approach, Precalculus.*

Enhanced with Graphing Utilities Series, Seventh Edition

This series provides a thorough integration of graphing utilities into topics, allowing students to explore mathematical concepts and encounter ideas usually studied in later courses. Many examples show solutions using algebra side-by-side with graphing techniques. Using technology, the approach to solving certain problems differs from the Flagship Series, while the emphasis on understanding concepts and building strong skills is maintained: *College Algebra, Algebra & Trigonometry, Precalculus*.

Concepts through Functions Series, Fourth Edition

This series differs from the others, utilizing a functions approach that serves as the organizing principle tying concepts together. Functions are introduced early in various formats. The approach supports the Rule of Four, which states that functions can be represented symbolically, numerically, graphically, and verbally. Each chapter introduces a new type of function and then develops all concepts pertaining to that particular function. The solutions of equations and inequalities, instead of being developed as stand-alone topics, are developed in the context of the underlying functions. Graphing utility coverage is optional and can be included or excluded at the discretion of the instructor: *College Algebra; Precalculus, with a Unit Circle Approach to Trigonometry; Precalculus, with a Right Triangle Approach to Trigonometry*.

The Flagship Series

College Algebra, Eleventh Edition

This text provides a contemporary approach to college algebra, with three chapters of review material preceding the chapters on functions. Graphing calculator usage is provided, but is optional. After completing this book, a student will be adequately prepared for trigonometry, finite mathematics, and business calculus.

Algebra & Trigonometry, Eleventh Edition

This text contains all the material in *College Algebra*, but also develops the trigonometric functions using a right triangle approach and shows how it relates to the unit circle approach. Graphing techniques are emphasized, including a thorough discussion of polar coordinates, parametric equations, and conics using polar coordinates. Vectors in the plane, sequences, induction, and the binomial theorem are also presented. Graphing calculator usage is provided, but is optional. After completing this book, a student will be adequately prepared for finite mathematics, business calculus, and engineering calculus.

Precalculus, Eleventh Edition

This text contains one review chapter before covering the traditional precalculus topics of polynomial, rational, exponential, and logarithmic functions and their graphs. The trigonometric functions are introduced using a unit circle approach and showing how it relates to the right triangle approach. Graphing techniques are emphasized, including a thorough discussion of polar coordinates, parametric equations, and conics using polar coordinates. Vectors in the plane and in space, including the dot and cross products, sequences, induction, and the binomial theorem are also presented. Graphing calculator usage is provided, but is optional. The final chapter provides an introduction to calculus, with a discussion of the limit, the derivative, and the integral of a function. After completing this book, a student will be adequately prepared for finite mathematics, business calculus, and engineering calculus.

Trigonometry: a Unit Circle Approach, Eleventh Edition

This text, designed for stand-alone courses in trigonometry, develops the trigonometric functions using a unit circle approach and shows how it relates to the right triangle approach. Vectors in the plane and in space, including the dot and cross products, are presented. Graphing techniques are emphasized, including a thorough discussion of polar coordinates, parametric equations, and conics using polar coordinates. Graphing calculator usage is provided, but is optional. After completing this book, a student will be adequately prepared for finite mathematics, business calculus, and engineering calculus.

Preface to the Instructor

As a professor of mathematics at an urban public university for 35 years, I understand the varied needs of precalculus students. Students range from being underprepared with little mathematical background and a fear of mathematics, to being highly prepared and motivated. For some, this is their final course in mathematics. For others, it is preparation for future mathematics courses. I have written this text with both groups in mind.

A tremendous benefit of authoring a successful series is the broad-based feedback I receive from instructors and students who have used previous editions. I am sincerely grateful for their support. Virtually every change to this edition is the result of their thoughtful comments and suggestions. I hope that I have been able to take their ideas and, building upon a successful foundation of the tenth edition, make this series an even better learning and teaching tool for students and instructors.

Features in the Eleventh Edition

A descriptive list of the many special features of *Precalculus* can be found in the front of this text. This list places the features in their proper context, as building blocks of an overall learning system that has been carefully crafted over the years to help students get the most out of the time they put into studying. Please take the time to review it and to discuss it with your students at the beginning of your course. My experience has been that when students use these features, they are more successful in the course.

- **Updated! Retain Your Knowledge Problems** These problems, which were new to the previous edition, are based on the article *"To Retain New Learning, Do the Math,"* published in the *Edurati Review*. In this article, Kevin Washburn suggests that "the more students are required to recall new content or skills, the better their memory will be." The Retain Your Knowledge problems were so well received that they have been expanded in this edition. Moreover, while the focus remains to help students maintain their skills, in most sections, problems were chosen that preview skills required to succeed in subsequent sections or in calculus. These are easily identified by the calculus icon ( ). All answers to Retain Your Knowledge problems are given in the back of the text and all are assignable in MyLab Math.
- **Guided Lecture Notes** Ideal for online, emporium/redesign courses, inverted classrooms, or traditional lecture classrooms. These lecture notes help students take thorough, organized, and understandable notes as they watch the Author in Action videos. They ask students to complete definitions, procedures, and examples based on the content of the videos and text. In addition, experience suggests that students learn by doing and understanding the why/how of the concept or property. Therefore, many sections will have an exploration activity to motivate student learning. These explorations introduce the topic and/or connect it to either a real-world application or a previous section. For example, when the vertical-line test is discussed in Section 2.2, after the theorem statement, the notes ask the students to explain why the vertical-line test works by using the definition of a function. This challenge helps students process the information at a higher level of understanding.
- **Illustrations** Many of the figures have captions to help connect the illustrations to the explanations in the body of the text.
- **Graphing Utility Screen Captures** In several instances we have added Desmos screen captures along with the TI-84 Plus C screen captures. These updated screen captures provide alternate ways of visualizing concepts and making connections between equations, data and graphs in full color.
- **Chapter Projects**, which apply the concepts of each chapter to a real-world situation, have been enhanced to give students an up-to-the-minute experience. Many of these projects are new, requiring the student to research information online in order to solve problems.
- **Exercise Sets** The exercises in the text have been reviewed and analyzed, some have been removed, and new ones have been added. All time-sensitive problems have been updated to the most recent information available. The problem sets remain classified according to purpose.

  The ***'Are You Prepared?'*** problems have been improved to better serve their purpose as a just-in-time review of concepts that the student will need to apply in the upcoming section.

  The ***Concepts and Vocabulary*** problems have been expanded to cover each objective of the section. These multiple-choice, fill-in-the-blank, and True/False exercises have been written to also serve as reading quizzes.

  Skill Building problems develop the student's computational skills with a large selection of exercises that are directly related to the objectives of the section. ***Mixed Practice*** problems offer a comprehensive assessment of skills that relate to more than one objective. Often these require skills learned earlier in the course.

  Applications and Extensions problems have been updated. Further, many new application-type exercises have been added, especially ones involving information and data drawn from sources the student will recognize, to improve relevance and timeliness.

  At the end of Applications and Extensions, we have a collection of one or more ***Challenge Problems***. These problems, as the title suggests, are intended to be thought-provoking, requiring some ingenuity to solve. They can be used for group work or to challenge students.

The ***Explaining Concepts: Discussion and Writing*** exercises provide opportunity for classroom discussion and group projects.

Updated! *Retain Your Knowledge* has been improved and expanded. The problems are based on material learned earlier in the course, especially calculus-related material. They serve to keep information that has already been learned "fresh" in the mind of the student.

NEW Need to Review? These margin notes provide a just-in-time reminder of a concept needed now, but covered in an earlier section of the book. Each note includes a reference to the chapter, section and page where the concept was originally discussed.

Content Changes to the 11th edition

- ***Challenge Problems*** have been added in most sections at the end of the Application and Extensions exercises. Challenge Problems are intended to be thought-provoking problems that require some ingenuity to solve. They can be used to challenge students or for group work.
- **Need to Review?** These margin notes provide a just-in-time review for a concept needed now, but covered in an earlier section of the book. Each note is back-referenced to the chapter, section and page where the concept was originally discussed.
- Additional **Retain Your Knowledge** exercises, whose purpose is to keep learned material fresh in a student's mind, have been added to each section. Many of these new problems preview skills required for calculus or for concepts needed in subsequent sections.
- **Desmos screen captures** have been added throughout the text. This is done to recognize that graphing technology expands beyond graphing calculators.
- Examples and exercises throughout the text have been augmented to reflect a broader selection of STEM applications.
- Concepts and Vocabulary exercises have been expanded to cover each objective of a section.
- Skill building exercises have been expanded to assess a wider range of difficulty.
- Applied problems and those based on real data have been updated where appropriate.

Appendix A

- Section A.10 Objective 3 now includes rationalizing the numerator
  - NEW Example 6 Rationalizing Numerators
  - Problems 69-76 provide practice.
- Section A.10 Exercises now include more practice in simplifying radicals

Chapter 1

- NEW Section 1.2 Example 9 Testing an Equation for Symmetry
- Section 1.3 has been reorganized to treat the slope-intercept form of the equation of a line before finding an equation of a line using two points.

Chapter 2

- NEW Section 2.1 Objective 1 Describe a Relation
- NEW Section 2.2 Example 4 Expending Energy
- NEW Section 2.4 Example 4 Analyzing a Piecewise-defined Function
- NEW Example 1 Describing a Relation demonstrates using the Rule of Four to express a relation numerically, as a mapping, and graphically given a verbal description.

Chapter 3

- Section 3.3 introduces the concept of concavity for a quadratic function
- NEW Section 3.3 Example 3 Graphing a Quadratic Function Using Its Vertex, Axis, and Intercepts
- Section 3.3 Example 8 Analyzing the Motion of a Projectile (formerly in Section 3.4)
- NEW Section 3.4 Example 4 Fitting a Quadratic Function to Data

Chapter 4

- Section 4.1 has been revised and split into two sections:
  - 4.1 Polynomial Functions
  - 4.2 Graphing Polynomial Functions; Models
- NEW Section 4.2 Example 2 Graphing a Polynomial Function (a 4th degree polynomial function)

Chapter 5

- Section 5.2 now finds and verifies inverse functions analytically and graphically.

Chapter 6

- NEW Section 6.1 Example 6 Field Width of a Digital Lens Reflex Camera Lens
- Section 6.4 and 6.5 were reorganized for increased clarity.

Chapter 7

- Sections 7.1 and 7.2 were reorganized for increased clarity.

Chapter 9

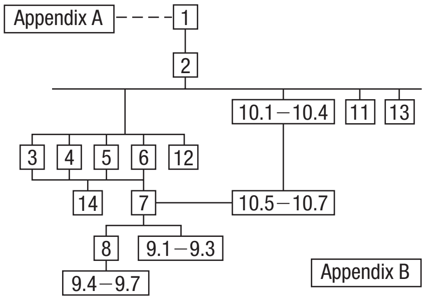
- Section 9.3 The complex plane; DeMoivre's Theorem, was rewritten to support the exponential form of a complex number.
  - Euler's Formula is introduced to express a complex number in exponential form.
  - The exponential form of a complex number is used to compute products and quotients.
  - DeMoivre's Theorem is expressed using the exponential form of a complex number.
  - The exponential form is used to find complex roots.

Chapter 11

- NEW Section 11.5 Example 1 Identifying Proper and Improper Rational Expressions

Using the Eleventh Edition Effectively with Your Syllabus

To meet the varied needs of diverse syllabi, this text contains more content than is likely to be covered in a *Precalculus* course. As the chart illustrates, this text has been organized with flexibility of use in mind. Within a given chapter, certain sections are optional (see the details that follow the figure below) and can be omitted without loss of continuity.



Appendix A Review

This chapter consists of review material. It may be used as the first part of the course or later as a just-in-time review when the content is required. Specific references to this chapter occur throughout the text to assist in the review process.

Chapter 1 Graphs

This chapter lays the foundation for functions.

Chapter 2 Functions and Their Graphs

Perhaps the most important chapter. Section 2.6 is optional.

Chapter 3 Linear and Quadratic Functions

Topic selection depends on your syllabus. Sections 3.2 and 3.4 may be omitted without loss of continuity.

Chapter 4 Polynomial and Rational Functions

Topic selection depends on your syllabus.

Chapter 5 Exponential and Logarithmic Functions

Sections 5.1–5.6 follow in sequence. Sections 5.7, 5.8, and 5.9 are optional.

Chapter 6 Trigonometric Functions

Section 6.6 may be omitted in a brief course.

Chapter 7 Analytic Trigonometry

Section 7.7 may be omitted in a brief course.

Chapter 8 Applications of Trigonometric Functions

Sections 8.4 and 8.5 may be omitted in a brief course.

Chapter 9 Polar Coordinates; Vectors

Sections 9.1–9.3 and Sections 9.4–9.7 are independent and may be covered separately.

Chapter 10 Analytic Geometry

Sections 10.1–10.4 follow in sequence. Sections 10.5, 10.6, and 10.7 are independent of each other, but each requires Sections 10.1–10.4.

Chapter 11 Systems of Equations and Inequalities

Sections 11.2–11.7 may be covered in any order, but each requires Section 11.1. Section 11.8 requires Section 11.7.

Chapter 12 Sequences; Induction; The Binomial Theorem

There are three independent parts: Sections 12.1–12.3; Section 12.4; and Section 12.5.

Chapter 13 Counting and Probability

The sections follow in sequence.

Chapter 14 A Preview of Calculus: The Limit, Derivative, and Integral of a Function

If time permits, coverage of this chapter will give your students a beneficial head start in calculus.

Acknowledgments

Textbooks are written by authors, but evolve from an idea to final form through the efforts of many people. It was Don Dellen who first suggested this text and series to me. Don is remembered for his extensive contributions to publishing and mathematics.

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Pearson MyLab

Get the Most Out of MyLab Math

Math courses are continuously evolving to help today's students succeed. It's more challenging than ever to support students with a wide range of backgrounds, learner styles, and math anxieties. The flexibility to build a course that fits instructors' individual course formats—with a variety of content options and multimedia resources all in one place—has made MyLab Math the market-leading solution for teaching and learning mathematics since its inception.

Preparedness

One of the biggest challenges in College Algebra, Trigonometry, and Precalculus is making sure students are adequately prepared with prerequisite knowledge. For a student, having the essential algebra skills upfront in this course can dramatically increase success.

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Assignments

10/18/19 11:59pm	Chapter 4 Skills Check
10/18/19 11:59pm	Chapter 4 Skills Review Homework
04/01/20 11:59pm	Chapter 5 Skills Check
04/01/20 11:59pm	Chapter 5 Skills Review Homework
09/14/20 11:59pm	Chapter 6 Skills Check
09/14/20 11:59pm	Chapter 6 Skills Review Homework

Pearson MyLab

Resources for Success

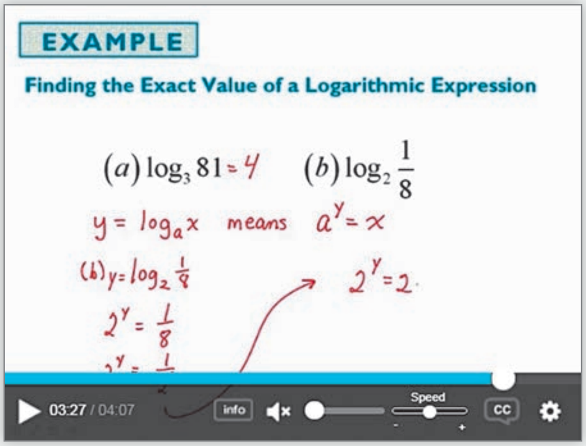
MyLab Math Online Course for Precalculus, 11th Edition by Michael Sullivan (access code required)

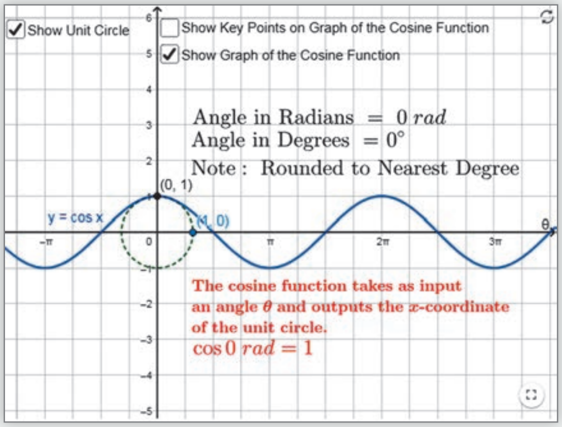
MyLab™ Math is tightly integrated with each author's style, offering a range of author-created multimedia resources, so your students have a consistent experience.

Video Program and Resources

Author in Action Videos are actual classroom lectures by Michael Sullivan III with fully worked-out examples.

- **Video assessment** questions are available to assign in MyLab Math for key videos.
- **Updated!** The corresponding **Guided Lecture Notes** assist students in taking thorough, organized, and understandable notes while watching Author in Action videos.



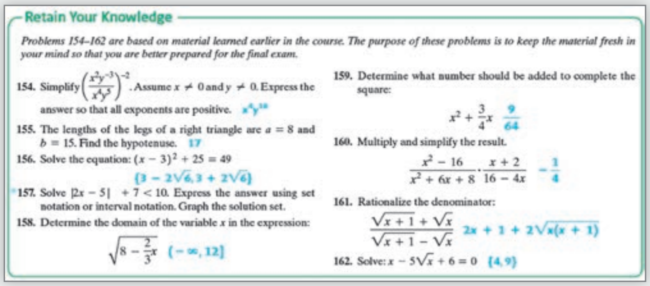


Guided Visualizations

New! Guided Visualizations, created in GeoGebra by Michael Sullivan III, bring mathematical concepts to life, helping students visualize the concept through directed exploration and purposeful manipulation. Assignable in MyLab Math with assessment questions to check students' conceptual understanding.

Retain Your Knowledge Exercises

Updated! Retain Your Knowledge Exercises, assignable in MyLab Math, improve students' recall of concepts learned earlier in the course. New for the 11th Edition, additional exercises will be included that will have an emphasis on content that students will build upon in the immediate upcoming section.



Retain Your Knowledge

Problems 154–162 are based on material learned earlier in the course. The purpose of these problems is to keep the material fresh in your mind so that you are better prepared for the final exam.

154. Simplify $\left(\frac{x^2y^{-3}}{x^4y^5}\right)^{-2}$. Assume $x \neq 0$ and $y \neq 0$. Express the answer so that all exponents are positive. x^4y^{16}
155. The lengths of the legs of a right triangle are $a = 8$ and $b = 15$. Find the hypotenuse. 17
156. Solve the equation: $(x - 3)^2 + 25 = 49$ $\{3 - 2\sqrt{6}, 3 + 2\sqrt{6}\}$
157. Solve $|2x - 5| + 7 < 10$. Express the answer using set notation or interval notation. Graph the solution set.
158. Determine the domain of the variable x in the expression: $\sqrt{8 - \frac{2}{3}x}$ $(-\infty, 12]$
159. Determine what number should be added to complete the square: $x^2 + \frac{3}{4}x$ $\frac{9}{64}$
160. Multiply and simplify the result. $\frac{x^2 - 16}{x^2 + 6x + 8} \cdot \frac{x + 2}{16 - 4x}$ $-\frac{1}{4}$
161. Rationalize the denominator: $\frac{\sqrt{x+1} + \sqrt{x}}{\sqrt{x+1} - \sqrt{x}}$ $2x + 1 + 2\sqrt{x(x+1)}$
162. Solve: $x - 5\sqrt{x} + 6 = 0$ $\{4, 9\}$

https://mlm.pearson.com/global/

Resources for Success

Pearson MyLab

Instructor Resources

Online resources can be downloaded from the Instructor Resource Center.

Instructor's Solutions Manual

Includes fully worked solutions to all exercises in the text.

Learning Catalytics Question Library

Questions written by Michael Sullivan III are available within MyLab Math to deliver through Learning Catalytics to engage students in your course.

Powerpoint® Lecture Slides

Fully editable slides correlate to the textbook.

Mini Lecture Notes

Includes additional examples and helpful teaching tips, by section.

Testgen®

TestGen (www.pearsoned.com/testgen) enables instructors to build, edit, print, and administer tests using a computerized bank of questions developed to cover all the objectives of the text.

Online Chapter Projects

Additional projects that give students an opportunity to apply what they learned in the chapter.

Student Resources

Additional resources to enhance student success.

Lecture Video

Author in Action videos are actual classroom lectures with fully worked out examples presented by Michael Sullivan, III. All video is assignable within MyLab Math.

Chapter Test Prep Videos

Students can watch instructors work through step-by-step solutions to all chapter test exercises from the text. These are available in MyLab Math and on YouTube.

Guided Lecture Notes

These lecture notes assist students in taking thorough, organized, and understandable notes while watching Author in Action videos. Students actively participate in learning the how/why of important concepts through explorations and activities. The Guided Lecture Notes are available as PDF's and customizable Word files in MyLab Math. They can also be packaged with the text and the MyLab Math access code.

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1

How to Value a House

Two things to consider in valuing a home: (1) How does it compare to similar nearby homes that have sold recently? (2) What value do you place on the advertised features and amenities?

The Zestimate® home value is a good starting point in figuring out the value of a home. It shows you how the home compares relative to others in the area, but you then need to add in all the other qualities that only someone who has seen the house knows.

Looking at "comps"

Knowing whether an asking price is fair will be important when you're ready to make an offer on a house. It will be even more important when your mortgage lender hires an appraiser to determine whether the house is worth the loan you're after.

Check on Zillow to see recent sales of similar, or comparable, homes in the area. Print them out and keep these "comps." You'll be referring to them quite a bit.

Note that "recent sales" usually means within the past six months. A sales price from a year ago probably bears little or no relation to what is going on in your area right now. In fact, some lenders will not accept comps older than three months.

Market activity also determines how easy or difficult it is to find accurate comps. In a "hot" or busy market, you're likely to have lots of comps to choose from. In a less active market finding reasonable comps becomes harder. And if the home you're looking at has special design features, finding a comparable property is harder still. It's also necessary to know what's going on in a given sub-segment. Maybe large, high-end homes are selling like hotcakes, but owners of smaller houses are staying put, or vice versa.

Source: *http://luthersanchez.com/2016/03/09/how-to-value-a-house/*

—See the Internet-based Chapter Project—

A Look Back

Appendix A reviews skills from intermediate algebra.

A Look Ahead

Here we connect algebra and geometry using the rectangular coordinate system. In the 1600s, algebra had developed to the point that René Descartes (1596–1650) and Pierre de Fermat (1601–1665) were able to use rectangular coordinates to translate geometry problems into algebra problems, and vice versa. This enabled both geometers and algebraists to gain new insights into their subjects, which had been thought to be separate but now were seen as connected.

Outline

1.1 The Distance and Midpoint Formulas

PREPARING FOR THIS SECTION *Before getting started, review the following:*

- Algebra Essentials (Section A.1, pp. A1–A10)
- Geometry Essentials (Section A.2, pp. A14–A19)

Now Work the **'Are You Prepared?'** problems on page 42.

OBJECTIVES
1 Use the Distance Formula (p. 39)
2 Use the Midpoint Formula (p. 41)

Rectangular Coordinates

We locate a point on the real number line by assigning it a single real number, called the *coordinate of the point*. For work in a two-dimensional plane, we locate points by using two numbers.

Begin with two real number lines located in the same plane: one horizontal and the other vertical. The horizontal line is called the ***x*-axis**, the vertical line the ***y*-axis**, and the point of intersection the **origin *O***. See Figure 1. Assign coordinates to every point on these number lines using a convenient scale. In mathematics, we usually use the same scale on each axis, but in applications, different scales appropriate to the application may be used.

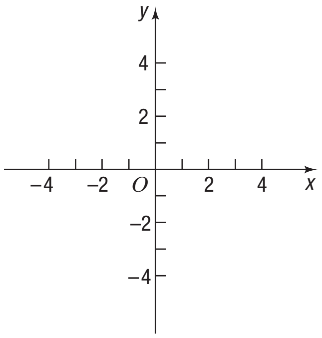


Figure 1 *xy*-Plane

The origin O has a value of 0 on both the x-axis and the y-axis. Points on the x-axis to the right of O are associated with positive real numbers, and those to the left of O are associated with negative real numbers. Points on the y-axis above O are associated with positive real numbers, and those below O are associated with negative real numbers. In Figure 1, the x-axis and y-axis are labeled as x and y, respectively, and an arrow at the end of each axis is used to denote the positive direction.

The coordinate system described here is called a **rectangular** or **Cartesian*** **coordinate system**. The x-axis and y-axis lie in a *plane* called the ***xy*-plane**, and the x-axis and y-axis are referred to as the **coordinate axes**.

Any point P in the xy-plane can be located by using an **ordered pair** (x, y) of real numbers. Let x denote the signed distance of P from the y-axis (*signed* means that if P is to the right of the y-axis, then $x > 0$, and if P is to the left of the y-axis, then $x < 0$); and let y denote the signed distance of P from the x-axis. The ordered pair (x, y), also called the **coordinates** of P, gives us enough information to locate the point P in the plane.

For example, to locate the point whose coordinates are $(-3, 1)$, go 3 units along the x-axis to the left of O and then go straight up 1 unit. We **plot** this point by placing a dot at this location. See Figure 2, in which the points with coordinates $(-3, 1)$, $(-2, -3)$, $(3, -2)$, and $(3, 2)$ are plotted.

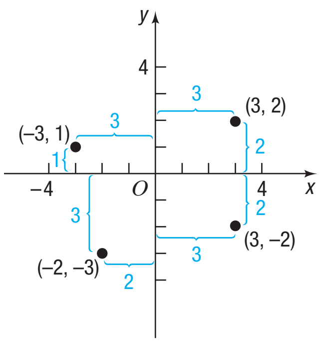


Figure 2

The origin has coordinates $(0, 0)$. Any point on the x-axis has coordinates of the form $(x, 0)$, and any point on the y-axis has coordinates of the form $(0, y)$.

If (x, y) are the coordinates of a point P, then x is called the ***x*-coordinate**, or **abscissa**, of P, and y is the ***y*-coordinate**, or **ordinate**, of P. We identify the point P by its coordinates (x, y) by writing $P = (x, y)$. Usually, we will simply say "the point (x, y)" rather than "the point whose coordinates are (x, y)."

The coordinate axes partition the xy-plane into four sections called **quadrants**, as shown in Figure 3. In quadrant I, both the x-coordinate and the y-coordinate of all points are positive; in quadrant II, x is negative and y is positive; in quadrant III, both x and y are negative; and in quadrant IV, x is positive and y is negative. Points on the coordinate axes belong to no quadrant.

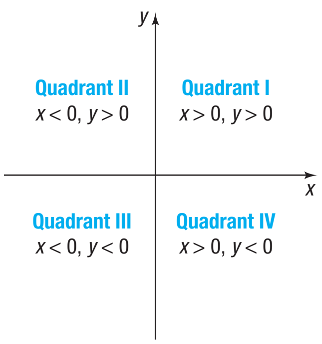


Figure 3

Now Work PROBLEM 15

*Named after René Descartes (1596–1650), a French mathematician, philosopher, and theologian.

COMMENT On a graphing calculator, you can set the scale on each axis. Once this has been done, you obtain the **viewing rectangle**. See Figure 4 for a typical viewing rectangle. You should now read Section B.1, *The Viewing Rectangle*.

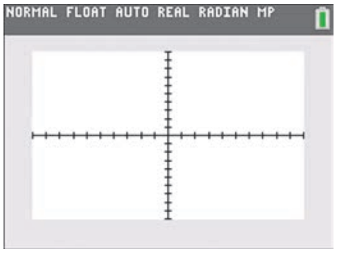


Figure 4 TI-84 Plus C Standard Viewing Rectangle ■

1 Use the Distance Formula

If the same units of measurement (such as inches, centimeters, and so on) are used for both the x-axis and y-axis, then all distances in the xy-plane can be measured using this unit of measurement.

EXAMPLE 1 Finding the Distance between Two Points

Find the distance d between the points $(1, 3)$ and $(5, 6)$.

Solution First plot the points $(1, 3)$ and $(5, 6)$ and connect them with a line segment. See Figure 5(a). To find the length d, begin by drawing a horizontal line segment from $(1,3)$ to $(5,3)$ and a vertical line segment from $(5,3)$ to $(5,6)$, forming a right triangle, as shown in Figure 5(b). One leg of the triangle is of length 4 (since $|5 - 1| = 4$), and the other is of length 3 (since $|6 - 3| = 3$). By the Pythagorean Theorem, the square of the distance d that we seek is

Need to Review?
The Pythagorean Theorem and its converse are discussed in Section A.2, pp. A14–A15.

$$d^2 = 4^2 + 3^2 = 16 + 9 = 25$$
$$d = \sqrt{25} = 5$$

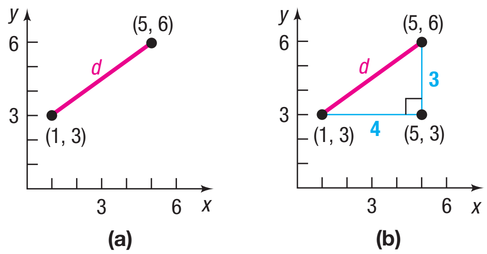


Figure 5

The **distance formula** provides a straightforward method for computing the distance between two points.

THEOREM Distance Formula

The distance between two points $P_1 = (x_1, y_1)$ and $P_2 = (x_2, y_2)$, denoted by $d(P_1, P_2)$, is

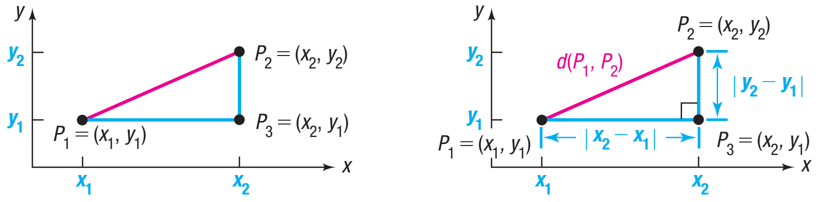
$$d(P_1, P_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \qquad \textbf{(1)}$$

In Words
To compute the distance between two points, find the difference of the x-coordinates, square it, and add this to the square of the difference of the y-coordinates. The square root of this sum is the distance.

Proof of the Distance Formula Let (x_1, y_1) denote the coordinates of point P_1 and let (x_2, y_2) denote the coordinates of point P_2.

- Assume that the line joining P_1 and P_2 is neither horizontal nor vertical. Refer to Figure 6(a) on the next page. The coordinates of P_3 are (x_2, y_1). The horizontal distance from P_1 to P_3 equals the absolute value of the difference of the x-coordinates, $|x_2 - x_1|$. The vertical distance from P_3 to P_2 equals the absolute value of the difference of the y-coordinates, $|y_2 - y_1|$. See Figure 6(b). The distance $d(P_1, P_2)$ is the length of the hypotenuse of the right triangle, so, by the Pythagorean Theorem, it follows that

$$\begin{aligned}[d(P_1, P_2)]^2 &= |x_2 - x_1|^2 + |y_2 - y_1|^2 \\ &= (x_2 - x_1)^2 + (y_2 - y_1)^2 \\ d(P_1, P_2) &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}\end{aligned}$$

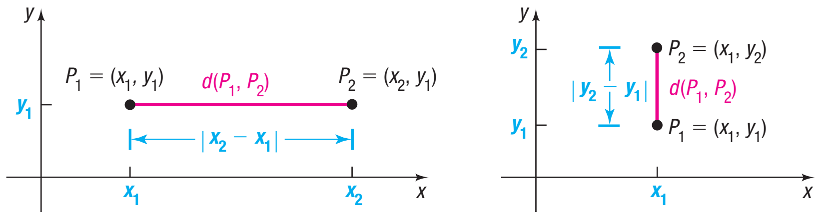


(a) (b)

Figure 6

- If the line joining P_1 and P_2 is horizontal, then the y-coordinate of P_1 equals the y-coordinate of P_2; that is, $y_1 = y_2$. Refer to Figure 7(a). In this case, the distance formula (1) still works, because for $y_1 = y_2$, it reduces to

$$d(P_1, P_2) = \sqrt{(x_2 - x_1)^2 + 0^2} = \sqrt{(x_2 - x_1)^2} = |x_2 - x_1|$$



(a) (b)

Figure 7

- A similar argument holds if the line joining P_1 and P_2 is vertical. See Figure 7(b). ■

EXAMPLE 2 Using the Distance Formula

Find the distance d between the points $(-4, 5)$ and $(3, 2)$.

Solution Using the distance formula, equation (1), reveals that the distance d is

$$\begin{aligned}d = \sqrt{[3 - (-4)]^2 + (2 - 5)^2} &= \sqrt{7^2 + (-3)^2} \\ &= \sqrt{49 + 9} = \sqrt{58} \approx 7.62\end{aligned}$$

Now Work PROBLEMS 19 AND 23

The distance between two points $P_1 = (x_1, y_1)$ and $P_2 = (x_2, y_2)$ is never a negative number. Also, the distance between two points is 0 only when the points are identical—that is, when $x_1 = x_2$ and $y_1 = y_2$. And, because $(x_2 - x_1)^2 = (x_1 - x_2)^2$ and $(y_2 - y_1)^2 = (y_1 - y_2)^2$, it makes no difference whether the distance is computed from P_1 to P_2 or from P_2 to P_1; that is, $d(P_1, P_2) = d(P_2, P_1)$.

The introduction to this chapter mentioned that rectangular coordinates enable us to translate geometry problems into algebra problems, and vice versa. The next example shows how algebra (the distance formula) can be used to solve geometry problems.

EXAMPLE 3 Using Algebra to Solve a Geometry Problem

Consider the three points $A = (-2, 1)$, $B = (2, 3)$, and $C = (3, 1)$.

(a) Plot each point and form the triangle ABC.
(b) Find the length of each side of the triangle.
(c) Show that the triangle is a right triangle.
(d) Find the area of the triangle.

Solution

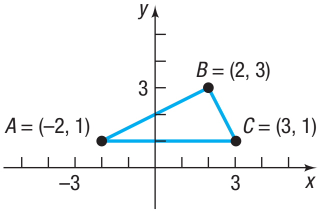


Figure 8

(a) Figure 8 shows the points A, B, C and the triangle ABC.

(b) To find the length of each side of the triangle, use the distance formula, equation (1).

$$d(A, B) = \sqrt{[2 - (-2)]^2 + (3 - 1)^2} = \sqrt{16 + 4} = \sqrt{20} = 2\sqrt{5}$$
$$d(B, C) = \sqrt{(3 - 2)^2 + (1 - 3)^2} = \sqrt{1 + 4} = \sqrt{5}$$
$$d(A, C) = \sqrt{[3 - (-2)]^2 + (1 - 1)^2} = \sqrt{25 + 0} = 5$$

(c) If the sum of the squares of the lengths of two of the sides equals the square of the length of the third side, then the triangle is a right triangle. Looking at Figure 8, it seems reasonable to conjecture that the angle at vertex B might be a right angle. We shall check to see whether

$$[d(A, B)]^2 + [d(B, C)]^2 = [d(A, C)]^2$$

Using the results in part (b) yields

$$[d(A, B)]^2 + [d(B, C)]^2 = (2\sqrt{5})^2 + (\sqrt{5})^2$$
$$= 20 + 5 = 25 = [d(A, C)]^2$$

It follows from the converse of the Pythagorean Theorem that triangle ABC is a right triangle.

(d) Because the right angle is at vertex B, the sides AB and BC form the base and height of the triangle. Its area is

$$\text{Area} = \frac{1}{2} \cdot \text{Base} \cdot \text{Height} = \frac{1}{2} \cdot 2\sqrt{5} \cdot \sqrt{5} = 5 \text{ square units}$$

Now Work PROBLEM 33

2 Use the Midpoint Formula

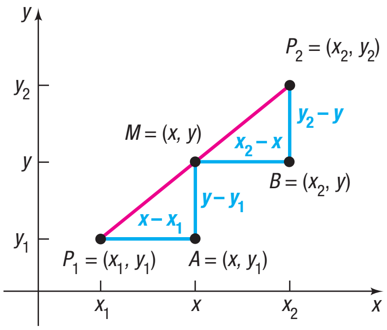


Figure 9

We now derive a formula for the coordinates of the **midpoint of a line segment**. Let $P_1 = (x_1, y_1)$ and $P_2 = (x_2, y_2)$ be the endpoints of a line segment, and let $M = (x, y)$ be the point on the line segment that is the same distance from P_1 as it is from P_2. See Figure 9. The triangles P_1AM and MBP_2 are congruent. [Do you see why? $d(P_1, M) = d(M, P_2)$ is given; also, $\angle AP_1M = \angle BMP_2$* and $\angle P_1MA = \angle MP_2B$. So, we have angle–side–angle.] Because triangles P_1AM and MBP_2 are congruent, corresponding sides are equal in length. That is,

$$x - x_1 = x_2 - x \quad \text{and} \quad y - y_1 = y_2 - y$$
$$2x = x_1 + x_2 \qquad\qquad 2y = y_1 + y_2$$
$$x = \frac{x_1 + x_2}{2} \qquad\qquad y = \frac{y_1 + y_2}{2}$$

*A postulate from geometry states that the transversal $\overline{P_1P_2}$ forms congruent corresponding angles with the parallel line segments $\overline{P_1A}$ and $\overline{MB}$.

In Words
To find the midpoint of a line segment, average the x-coordinates of the endpoints, and average the y-coordinates of the endpoints.

THEOREM Midpoint Formula

The midpoint $M = (x, y)$ of the line segment from $P_1 = (x_1, y_1)$ to $P_2 = (x_2, y_2)$ is

$$M = (x, y) = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right) \tag{2}$$

EXAMPLE 4 **Finding the Midpoint of a Line Segment**

Find the midpoint of the line segment from $P_1 = (-5, 5)$ to $P_2 = (3, 1)$. Plot the points P_1 and P_2 and their midpoint.

Solution Use the midpoint formula (2) with $x_1 = -5$, $y_1 = 5$, $x_2 = 3$, and $y_2 = 1$. The coordinates (x, y) of the midpoint M are

$$x = \frac{x_1 + x_2}{2} = \frac{-5 + 3}{2} = -1 \quad \text{and} \quad y = \frac{y_1 + y_2}{2} = \frac{5 + 1}{2} = 3$$

That is, $M = (-1, 3)$. See Figure 10.

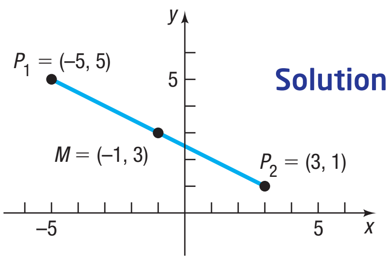


Figure 10

Now Work PROBLEM 39

1.1 Assess Your Understanding

'Are You Prepared?' *Answers are given at the end of these exercises. If you get a wrong answer, read the pages listed in red.*

1. On the real number line, the origin is assigned the number ________. (p. A4)
2. If -3 and 5 are the coordinates of two points on the real number line, the distance between these points is ________. (pp. A5–A6)
3. If 3 and 4 are the legs of a right triangle, the hypotenuse is ________. (p. A14)
4. Use the converse of the Pythagorean Theorem to show that a triangle whose sides are of lengths 11, 60, and 61 is a right triangle. (pp. A14–A15)
5. The area A of a triangle whose base is b and whose altitude is h is $A =$ ________. (p. A15)
6. ***True or False*** Two triangles are congruent if two angles and the included side of one equals two angles and the included side of the other. (pp. A16–A17)

Concepts and Vocabulary

7. If (x, y) are the coordinates of a point P in the xy-plane, then x is called the ________________ of P, and y is the ________________ of P.
8. The coordinate axes partition the xy-plane into four sections called ____________.
9. If three distinct points P, Q, and R all lie on a line, and if $d(P, Q) = d(Q, R)$, then Q is called the ____________ of the line segment from P to R.
10. ***True or False*** The distance between two points is sometimes a negative number.
11. ***True or False*** The point $(-1, 4)$ lies in quadrant IV of the Cartesian plane.
12. ***True or False*** The midpoint of a line segment is found by averaging the x-coordinates and averaging the y-coordinates of the endpoints.
13. ***Multiple Choice*** Which of the following statements is true for a point (x, y) that lies in quadrant III?
    (a) Both x and y are positive.
    (b) Both x and y are negative.
    (c) x is positive, and y is negative.
    (d) x is negative, and y is positive.
14. ***Multiple Choice*** Choose the expression that equals the distance between two points (x_1, y_1) and (x_2, y_2).
    (a) $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
    (b) $\sqrt{(x_2 + x_1)^2 - (y_2 + y_1)^2}$
    (c) $\sqrt{(x_2 - x_1)^2 - (y_2 - y_1)^2}$
    (d) $\sqrt{(x_2 + x_1)^2 + (y_2 + y_1)^2}$

Skill Building

In Problems 15 and 16, plot each point in the xy-plane. State which quadrant or on what coordinate axis each point lies.

15. **(a)** $A = (-3, 2)$ **(b)** $B = (6, 0)$ **(c)** $C = (-2, -2)$ **(d)** $D = (6, 5)$ **(e)** $E = (0, -3)$ **(f)** $F = (6, -3)$

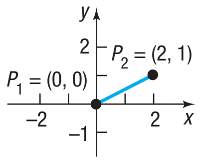
16. **(a)** $A = (1, 4)$ **(b)** $B = (-3, -4)$ **(c)** $C = (-3, 4)$ **(d)** $D = (4, 1)$ **(e)** $E = (0, 1)$ **(f)** $F = (-3, 0)$

17. Plot the points $(0, 3)$, $(1, 3)$, $(-2, 3)$, $(5, 3)$, and $(-4, 3)$. Describe the set of all points of the form $(x, 3)$, where x is a real number.

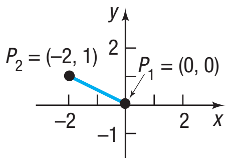
18. Plot the points $(2, 0)$, $(2, -3)$, $(2, 4)$, $(2, 1)$, and $(2, -1)$. Describe the set of all points of the form $(2, y)$, where y is a real number.

In Problems 19–32, find the distance d between the points P_1 and P_2.

19.



20.



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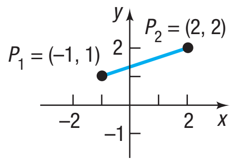
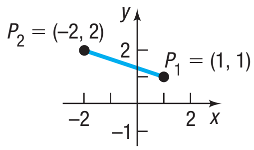


22.



23. $P_1 = (3, -4)$; $P_2 = (5, 4)$

24. $P_1 = (-1, 0)$; $P_2 = (2, 4)$

25. $P_1 = (2, -3)$; $P_2 = (4, 2)$

26. $P_1 = (-7, 3)$; $P_2 = (4, 0)$

27. $P_1 = (-4, -3)$; $P_2 = (6, 2)$

28. $P_1 = (5, -2)$; $P_2 = (6, 1)$

29. $P_1 = (1.2, 2.3)$; $P_2 = (-0.3, 1.1)$

30. $P_1 = (-0.2, 0.3)$; $P_2 = (2.3, 1.1)$

31. $P_1 = (a, a)$; $P_2 = (0, 0)$

32. $P_1 = (a, b)$; $P_2 = (0, 0)$

In Problems 33–38, plot each point and form the triangle ABC. Show that the triangle is a right triangle. Find its area.

33. $A = (-2, 5)$; $B = (1, 3)$; $C = (-1, 0)$

34. $A = (-2, 5)$; $B = (12, 3)$; $C = (10, -11)$

35. $A = (-6, 3)$; $B = (3, -5)$; $C = (-1, 5)$

36. $A = (-5, 3)$; $B = (6, 0)$; $C = (5, 5)$

37. $A = (4, -3)$; $B = (4, 1)$; $C = (2, 1)$

38. $A = (4, -3)$; $B = (0, -3)$; $C = (4, 2)$

In Problems 39–46, find the midpoint of the line segment joining the points P_1 and P_2.

39. $P_1 = (3, -4)$; $P_2 = (5, 4)$

40. $P_1 = (-2, 0)$; $P_2 = (2, 4)$

41. $P_1 = (2, -3)$; $P_2 = (4, 2)$

42. $P_1 = (-1, 4)$; $P_2 = (8, 0)$

43. $P_1 = (-4, -3)$; $P_2 = (2, 2)$

44. $P_1 = (7, -5)$; $P_2 = (9, 1)$

45. $P_1 = (a, a)$; $P_2 = (0, 0)$

46. $P_1 = (a, b)$; $P_2 = (0, 0)$

Applications and Extensions

47. If the point $(3, 8)$ is shifted 2 units to the right and 4 units down, what are its new coordinates?

48. If the point $(-1, 6)$ is shifted 2 units to the left and 4 units up, what are its new coordinates?

49. Find all points having an x-coordinate of 4 whose distance from the point $(-4, 2)$ is 10.

(a) By using the Pythagorean Theorem.
(b) By using the distance formula.

50. Find all points having a y-coordinate of -6 whose distance from the point $(1, 2)$ is 17.

(a) By using the Pythagorean Theorem.
(b) By using the distance formula.

51. Find all points on the x-axis that are 12 units from the point $(5, -6)$.

52. Find all points on the y-axis that are 6 units from the point $(4, -3)$.

53. Suppose that $A = (2, 5)$ are the coordinates of a point in the xy-plane.

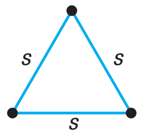
(a) Find the coordinates of the point if A is shifted 2 units to the right and 3 units down.
(b) Find the coordinates of the point if A is shifted 1 unit to the left and 6 units up.

54. Plot the points $A = (-1, 8)$ and $M = (2, 3)$ in the xy-plane. If M is the midpoint of a line segment AB, find the coordinates of B.

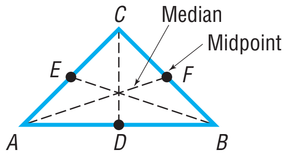
55. The midpoint of the line segment from P_1 to P_2 is $(-6, 5)$. If $P_1 = (-6, 3)$. what is P_2?

56. The midpoint of the line segment from P_1 to P_2 is $(5, -4)$. If $P_2 = (7, -2)$, what is P_1?

57. **Geometry** An **equilateral triangle** has three sides of equal length. If two vertices of an equilateral triangle are $(0, 4)$ and $(0, 0)$ find the third vertex. How many of these triangles are possible?



58. **Geometry** The **medians** of a triangle are the line segments from each vertex to the midpoint of the opposite side (see the figure). Find the lengths of the medians of the triangle with vertices at $A = (0, 0)$, $B = (6, 0)$, and $C = (4, 4)$.



In Problems 59–62, find the length of each side of the triangle determined by the three points P_1, P_2, and P_3. State whether the triangle is an isosceles triangle, a right triangle, neither of these, or both. (An ***isosceles triangle*** *is one in which at least two of the sides are of equal length.)*

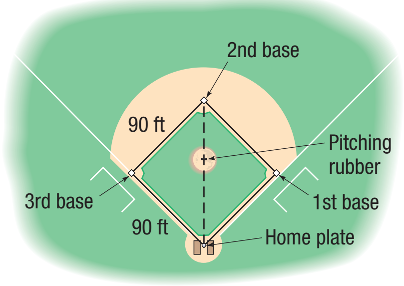
59. $P_1 = (-1, 4)$; $P_2 = (6, 2)$; $P_3 = (4, -5)$

60. $P_1 = (4, 2)$; $P_2 = (10, 4)$; $P_3 = (6, -4)$

61. $P_1 = (7, 2)$; $P_2 = (-4, 0)$; $P_3 = (4, 6)$

62. $P_1 = (-8, -3)$, $P_2 = (0, 15)$, $P_3 = (5, 2)$

63. **Baseball** A major league baseball "diamond" is actually a square 90 feet on a side (see the figure). What is the distance directly from home plate to second base (the diagonal of the square)?



64. **Little League Baseball** The layout of a Little League playing field is a square 60 feet on a side. How far is it directly from home plate to second base (the diagonal of the square)?

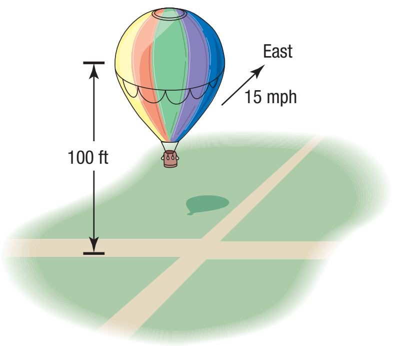
***Source**: 2018 Little League Baseball Official Regulations, Playing Rules, and Operating Policies*

65. **Baseball** Refer to Problem 63. Overlay a rectangular coordinate system on a major league baseball diamond so that the origin is at home plate, the positive x-axis lies in the direction from home plate to first base, and the positive y-axis lies in the direction from home plate to third base.
    (a) What are the coordinates of first base, second base, and third base? Use feet as the unit of measurement.
    (b) If the right fielder is located at (310, 15) how far is it from the right fielder to second base?
    (c) If the center fielder is located at (300, 300), how far is it from the center fielder to third base?

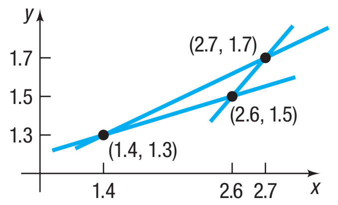
66. **Little League Baseball** Refer to Problem 64. Overlay a rectangular coordinate system on a Little League baseball diamond so that the origin is at home plate, the positive x-axis lies in the direction from home plate to first base, and the positive y-axis lies in the direction from home plate to third base.
    (a) What are the coordinates of first base, second base, and third base? Use feet as the unit of measurement.
    (b) If the right fielder is located at (180, 20), how far is it from the right fielder to second base?
    (c) If the center fielder is located at (220, 220), how far is it from the center fielder to third base?

67. **Distance between Moving Objects** A Ford Focus and a Freightliner Cascadia truck leave an intersection at the same time. The Focus heads east at an average speed of 60 miles per hour, while the Cascadia heads south at an average speed of 45 miles per hour. Find an expression for their distance apart d (in miles) at the end of t hours.

68. **Distance of a Moving Object from a Fixed Point** A hot-air balloon, headed due east at an average speed of 15 miles per hour and at a constant altitude of 100 feet, passes over an intersection (see the figure). Find an expression for the distance d (measured in feet) from the balloon to the intersection t seconds later.



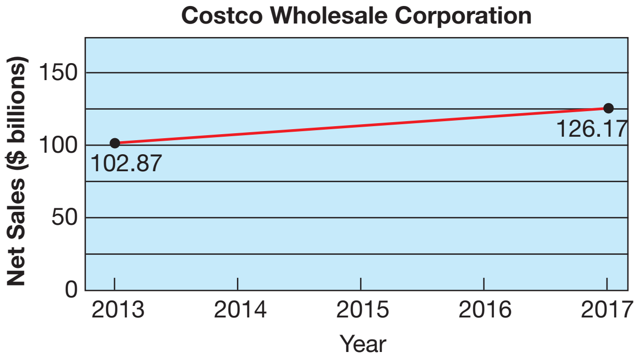
69. **Drafting Error** When a draftsman draws three lines that are to intersect at one point, the lines may not intersect as intended and subsequently will form an **error triangle**. If this error triangle is long and thin, one estimate for the location of the desired point is the midpoint of the shortest side. The figure shows one such error triangle.



(a) Find an estimate for the desired intersection point.
(b) Find the distance from (1.4, 1.3) to the midpoint found in part (a).

70. Net Sales The figure illustrates the net sales growth of Costco Wholesale Corporation from 2013 through 2017. Use the midpoint formula to estimate the net sales of Costco Wholesale Corporation in 2015. How does your result compare to the reported value of \$113.67 billion?

Source: *Costco Wholesale Corporation 2017 Annual Report*



71. Poverty Threshold A poverty threshold represents the minimum annual household income for a family not to be considered poor. In 1995, the poverty threshold for a family of four with two children under the age of 18 years was \$15,598. In 2005, the poverty threshold for a family of four with two children under the age of 18 years was \$19,508. Assuming poverty thresholds increase in a straight-line fashion, use the midpoint formula to estimate the poverty threshold of a family of four with two children under the age of 18 in 2000.

72. ***Challenge Problem*** **Geometry** Verify that the points $(0,0)$, $(a,0)$, and $\left(\frac{a}{2}, \frac{\sqrt{3}a}{2}\right)$ are the vertices of an equilateral triangle. Then show that the midpoints of the three sides are the vertices of a second equilateral triangle.

73. ***Challenge Problem*** **Geometry** A point P is equidistant from $(-5, 1)$ and $(4, -4)$. Find the coordinates of P if its y-coordinate is twice its x-coordinate.

74. ***Challenge Problem*** **Geometry** Find the midpoint of each diagonal of a square with side of length s. Draw the conclusion that the diagonals of a square intersect at their midpoints.

[**Hint**: Use $(0, 0)$, $(0, s)$, $(s, 0)$, and (s, s) as the vertices of the square.]

75. ***Challenge Problem*** **Geometry** For any parallelogram, prove that the sum of the squares of the lengths of the sides equals the sum of the squares of the lengths of the diagonals.

[**Hint**: Use $(0, 0)$, $(a, 0)$, $(a + b, c)$, and (b, c) as the vertices of the parallelogram. Assume a, b, and c are positive.]

Explaining Concepts: Discussion and Writing

76. Write a paragraph that describes a Cartesian plane. Then write a second paragraph that describes how to plot points in the Cartesian plane. Your paragraphs should include the terms "coordinate axes," "ordered pair," "coordinates," "plot," "x-coordinate," and "y-coordinate."

'Are You Prepared?' Answers

1. 0 **2.** 8 **3.** 5 **4.** $11^2 + 60^2 = 121 + 3600 = 3721 = 61^2$ **5.** $\frac{1}{2}bh$ **6.** True

1.2 Graphs of Equations in Two Variables; Intercepts; Symmetry

PREPARING FOR THIS SECTION *Before getting started, review the following:*

- Solving Linear Equations (Section A.6, pp. A44–A45)
- Solve a Quadratic Equation by Factoring (Section A.6, pp. A47–A48)

Now Work the **'Are You Prepared?'** problems on page 53.

OBJECTIVES
1. Graph Equations by Plotting Points (p. 46)
2. Find Intercepts from a Graph (p. 48)
3. Find Intercepts from an Equation (p. 48)
4. Test an Equation for Symmetry with Respect to the x-Axis, the y-Axis, and the Origin (p. 49)
5. Know How to Graph Key Equations (p. 51)

1 Graph Equations by Plotting Points

An **equation in two variables**, say x and y, is a statement in which two expressions involving x and y are equal. The expressions are called the **sides** of the equation. Since an equation is a statement, it may be true or false, depending on the value of the variables. Any values of x and y that result in a true statement are said to **satisfy** the equation.

For example, the following are all equations in two variables x and y:

$$x^2 + y^2 = 5 \qquad 2x - y = 6 \qquad y = 2x + 5 \qquad x^2 = y$$

The first of these, $x^2 + y^2 = 5$, is satisfied for $x = 1, y = 2$, since $1^2 + 2^2 = 5$. Other choices of x and y, such as $x = -1, y = -2$, also satisfy this equation. It is not satisfied for $x = 2$ and $y = 3$, since $2^2 + 3^2 = 4 + 9 = 13 \neq 5$.

The **graph of an equation in two variables** x and y consists of the set of points in the xy-plane whose coordinates (x, y) satisfy the equation.

Graphs play an important role in helping us to visualize the relationships that exist between two variables or quantities. Table 1 shows the average price of gasoline in the United States for the years 1991–2017 (adjusted for inflation). If we plot these data using year as the x-coordinate and price as the y-coordinate, and then connect the points (year, price), we obtain Figure 11.

Table 1 Average Price of Gasoline

Year	Price	Year	Price	Year	Price
1991	1.98	2000	2.11	2009	2.68
1992	1.90	2001	1.97	2010	3.12
1993	1.81	2002	1.83	2011	3.84
1994	1.78	2003	2.07	2012	3.87
1995	1.78	2004	2.40	2013	3.68
1996	1.87	2005	2.85	2014	3.48
1997	1.83	2006	3.13	2015	2.51
1998	1.55	2007	3.31	2016	2.19
1999	1.67	2008	3.70	2017	2.38

***Source**: U.S. Energy Information Administration (https://www.eia.gov/dnav/pet/pet_pri_gnd_dcus_nus_a.htm)*

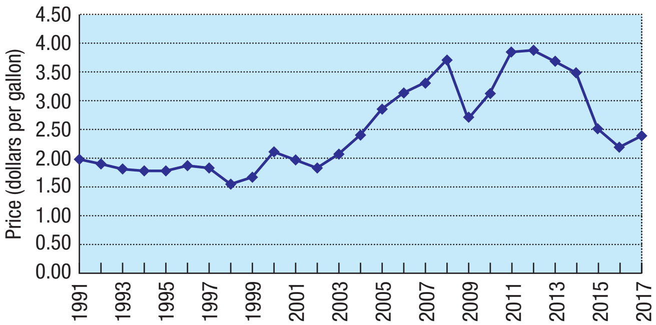


Figure 11

EXAMPLE 1 Determining Whether a Point Is on the Graph of an Equation

Determine whether the points are on the graph of the equation $2x - y = 6$.

(a) $(2, 3)$ (b) $(2, -2)$

Solution

(a) For the point $(2, 3)$, check to see whether $x = 2, y = 3$ satisfies the equation $2x - y = 6$.

$$2x - y = 2 \cdot 2 - 3 = 4 - 3 = 1 \neq 6$$

The equation is not satisfied, so the point $(2, 3)$ is not on the graph of $2x - y = 6$.

(b) For the point $(2, -2)$,

$$2x - y = 2 \cdot 2 - (-2) = 4 + 2 = 6$$

The equation is satisfied, so the point $(2, -2)$ is on the graph of $2x - y = 6$.

Now Work PROBLEM 13

EXAMPLE 2 Graphing an Equation by Plotting Points

Graph the equation: $y = 2x + 5$

Solution The graph consists of all points (x, y) that satisfy the equation. To locate some of these points (and get an idea of the pattern of the graph), assign some numbers to x, and find corresponding values for y.

If	Then	Point on Graph
$x = 0$	$y = 2 \cdot 0 + 5 = 5$	$(0, 5)$
$x = 1$	$y = 2 \cdot 1 + 5 = 7$	$(1, 7)$
$x = -5$	$y = 2 \cdot (-5) + 5 = -5$	$(-5, -5)$
$x = 10$	$y = 2 \cdot 10 + 5 = 25$	$(10, 25)$

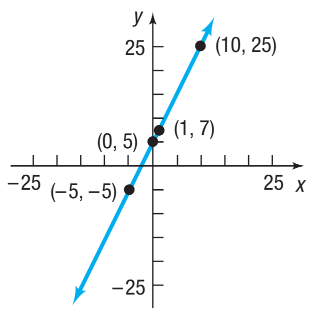


Figure 12 $y = 2x + 5$

By plotting these points and then connecting them, we obtain the graph (a *line*) of the equation $y = 2x + 5$, as shown in Figure 12.

EXAMPLE 3 Graphing an Equation by Plotting Points

Graph the equation: $y = x^2$

Solution Table 2 provides several points on the graph of $y = x^2$. Plotting these points and connecting them with a smooth curve gives the graph (a *parabola*) shown in Figure 13.

Table 2

x	$y = x^2$	(x, y)
-4	16	$(-4, 16)$
-3	9	$(-3, 9)$
-2	4	$(-2, 4)$
-1	1	$(-1, 1)$
0	0	$(0, 0)$
1	1	$(1, 1)$
2	4	$(2, 4)$
3	9	$(3, 9)$
4	16	$(4, 16)$

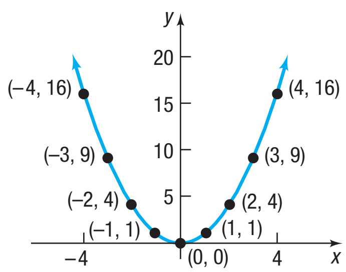


Figure 13 $y = x^2$

The graphs of the equations shown in Figures 12 and 13 do not show all points. For example, in Figure 12, the point $(20, 45)$ is a part of the graph of $y = 2x + 5$, but it is not shown. Since the graph of $y = 2x + 5$ can be extended out indefinitely, we use arrows to indicate that the pattern shown continues. It is important, when showing a graph, to present enough of the graph so that any viewer of the illustration will "see" the rest of it as an obvious continuation of what is actually there. This is referred to as a **complete graph**.

One way to obtain the complete graph of an equation is to plot enough points on the graph for a pattern to become evident. Then these points are connected with a smooth curve following the suggested pattern. But how many points are sufficient? Sometimes knowledge about the equation tells us. For example, we will learn in the next section that if an equation is of the form $y = mx + b$, then its graph is a line. In this case, only two points are needed to obtain the complete graph.

COMMENT Another way to obtain the graph of an equation is to use a graphing utility. Read Section B.2, *Using a Graphing Utility to Graph Equations*. ■

One purpose of this text is to investigate the properties of equations in order to decide whether a graph is complete. Sometimes we shall graph equations by plotting points. Shortly, we shall investigate various techniques that will enable us to graph an equation without plotting so many points.

Two techniques that sometimes reduce the number of points required to graph an equation involve finding *intercepts* and checking for *symmetry*.

2 Find Intercepts from a Graph

The points, if any, at which a graph crosses or touches the coordinate axes are called the **intercepts** of the graph. See Figure 14. The x-coordinate of a point at which the graph crosses or touches the x-axis is an ***x*-intercept**, and the y-coordinate of a point at which the graph crosses or touches the y-axis is a ***y*-intercept**.

In Words
Intercepts are points (ordered pairs). An x-intercept or a y-intercept is a number. For example, the point (3, 0) is an intercept; the number 3 is an x-intercept.

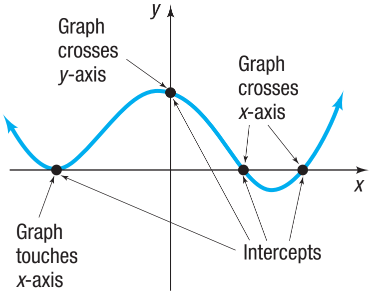


Figure 14

EXAMPLE 4 **Finding Intercepts from a Graph**

Find the intercepts of the graph in Figure 15. What are its x-intercepts? What are its y-intercepts?

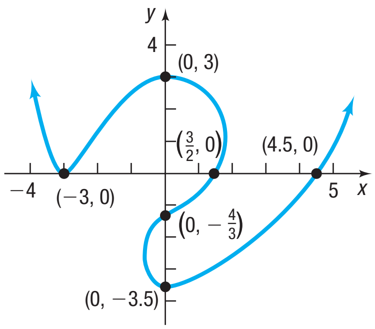


Figure 15

Solution The intercepts of the graph are the points

$$(-3, 0), \quad (0, 3), \quad \left(\frac{3}{2}, 0\right), \quad \left(0, -\frac{4}{3}\right), \quad (0, -3.5), \quad (4.5, 0)$$

The x-intercepts are -3, $\frac{3}{2}$, and 4.5; the y-intercepts are -3.5, $-\frac{4}{3}$, and 3.

In Example 4, notice that intercepts are listed as ordered pairs, and the x-intercepts and the y-intercepts are listed as numbers. We use this distinction throughout the text.

Now Work PROBLEM 41(a)

3 Find Intercepts from an Equation

The intercepts of a graph can be found from its equation by using the fact that points on the x-axis have y-coordinates equal to 0, and points on the y-axis have x-coordinates equal to 0.

COMMENT For many equations, finding intercepts may not be so easy. In such cases, a graphing utility can be used. Read the first part of Section B.3, *Using a Graphing Utility to Locate Intercepts and Check for Symmetry*, to find out how to locate intercepts using a graphing utility. ■

Procedure for Finding Intercepts

- To find the x-intercept(s), if any, of the graph of an equation, let $y = 0$ in the equation and solve for x, where x is a real number.
- To find the y-intercept(s), if any, of the graph of an equation, let $x = 0$ in the equation and solve for y, where y is a real number.

EXAMPLE 5 **Finding Intercepts from an Equation**

Find the x-intercept(s) and the y-intercept(s) of the graph of $y = x^2 - 4$. Then graph $y = x^2 - 4$ by plotting points.

Solution To find the x-intercept(s), let $y = 0$ and obtain the equation

$$\begin{aligned} x^2 - 4 &= 0 && y = x^2 - 4 \text{ with } y = 0 \\ (x + 2)(x - 2) &= 0 && \text{Factor.} \\ x + 2 = 0 \quad \text{or} \quad x - 2 &= 0 && \text{Use the Zero-Product Property.} \\ x = -2 \quad \text{or} \quad x &= 2 && \text{Solve.} \end{aligned}$$

The equation has two solutions, -2 and 2. The x-intercepts are -2 and 2.

To find the y-intercept(s), let $x = 0$ in the equation.

$$\begin{aligned} y &= x^2 - 4 \\ &= 0^2 - 4 = -4 \end{aligned}$$

The y-intercept is -4.

Since $x^2 \geq 0$ for all x, we deduce from the equation $y = x^2 - 4$ that $y \geq -4$ for all x. This information, the intercepts, and the points from Table 3 enable us to graph $y = x^2 - 4$. See Figure 16.

Table 3

x	$y = x^2 - 4$	(x, y)
-3	5	$(-3, 5)$
-1	-3	$(-1, -3)$
1	-3	$(1, -3)$
3	5	$(3, 5)$

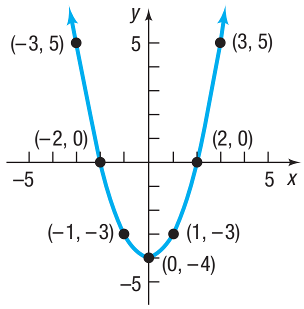


Figure 16 $y = x^2 - 4$

Now Work PROBLEM 23

4 Test an Equation for Symmetry with Respect to the x-Axis, the y-Axis, and the Origin

Another helpful tool for graphing equations by hand involves *symmetry*, particularly symmetry with respect to the x-axis, the y-axis, and the origin.

Symmetry often occurs in nature. Consider the picture of the butterfly. Do you see the symmetry?

DEFINITION Symmetry with Respect to the x-Axis

A graph is **symmetric with respect to the x-axis** if, for every point (x, y) on the graph, the point $(x, -y)$ is also on the graph.

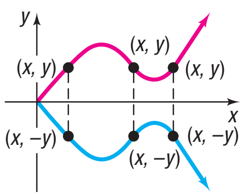


Figure 17 Symmetry with respect to the x-axis

Figure 17 illustrates the definition. Note that when a graph is symmetric with respect to the x-axis, the part of the graph above the x-axis is a reflection (or mirror image) of the part below it, and vice versa.

EXAMPLE 6 **Points Symmetric with Respect to the x-Axis**

If a graph is symmetric with respect to the x-axis, and the point $(3, 2)$ is on the graph, then the point $(3, -2)$ is also on the graph.

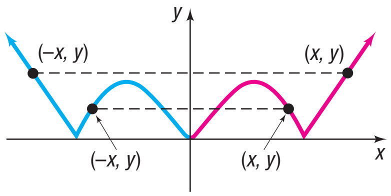


Figure 18 Symmetry with respect to the y-axis

DEFINITION Symmetry with Respect to the y-Axis

A graph is **symmetric with respect to the y-axis** if, for every point (x, y) on the graph, the point $(-x, y)$ is also on the graph.

Figure 18 illustrates the definition. When a graph is symmetric with respect to the y-axis, the part of the graph to the right of the y-axis is a reflection of the part to the left of it, and vice versa.

EXAMPLE 7 Points Symmetric with Respect to the y-Axis

If a graph is symmetric with respect to the y-axis and the point $(5, 8)$ is on the graph, then the point $(-5, 8)$ is also on the graph.

DEFINITION Symmetry with Respect to the Origin

A graph is **symmetric with respect to the origin** if, for every point (x, y) on the graph, the point $(-x, -y)$ is also on the graph.

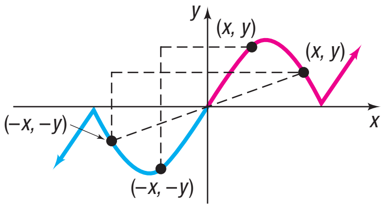


Figure 19 Symmetry with respect to the origin

Figure 19 illustrates the definition. Symmetry with respect to the origin may be viewed in three ways:

- As a reflection about the y-axis, followed by a reflection about the x-axis
- As a projection along a line through the origin so that the distances from the origin are equal
- As half of a complete revolution about the origin

EXAMPLE 8 Points Symmetric with Respect to the Origin

If a graph is symmetric with respect to the origin, and the point $(4, -2)$ is on the graph, then the point $(-4, 2)$ is also on the graph.

Now Work PROBLEMS 31 AND 41(b)

When the graph of an equation is symmetric with respect to a coordinate axis or the origin, the number of points that you need to plot in order to see the pattern is reduced. For example, if the graph of an equation is symmetric with respect to the y-axis, then once points to the right of the y-axis are plotted, an equal number of points on the graph can be obtained by reflecting them about the y-axis. Because of this, before we graph an equation, we should first determine whether it has any symmetry. The following tests are used for this purpose.

Tests for Symmetry

To test the graph of an equation for symmetry with respect to the

- *x-Axis* Replace y by $-y$ in the equation and simplify. If an equivalent equation results, the graph of the equation is symmetric with respect to the x-axis.
- *y-Axis* Replace x by $-x$ in the equation and simplify. If an equivalent equation results, the graph of the equation is symmetric with respect to the y-axis.
- *Origin* Replace x by $-x$ and y by $-y$ in the equation and simplify. If an equivalent equation results, the graph of the equation is symmetric with respect to the origin.

EXAMPLE 9 Testing an Equation for Symmetry

Test $4x^2 + 9y^2 = 36$ for symmetry.

Solution

x-Axis: To test for symmetry with respect to the x-axis, replace y by $-y$. Since $4x^2 + 9(-y)^2 = 36$ is equivalent to $4x^2 + 9y^2 = 36$, the graph of the equation is symmetric with respect to the x-axis.

y-Axis: To test for symmetry with respect to the y-axis, replace x by $-x$. Since $4(-x)^2 + 9y^2 = 36$ is equivalent to $4x^2 + 9y^2 = 36$, the graph of the equation is symmetric with respect to the y-axis.

Origin: To test for symmetry with respect to the origin, replace x by $-x$ and y by $-y$. Since $4(-x)^2 + 9(-y)^2 = 36$ is equivalent to $4x^2 + 9y^2 = 36$, the graph of the equation is symmetric with respect to the origin.

EXAMPLE 10 Testing an Equation for Symmetry

Test $y = \dfrac{4x^2}{x^2 + 1}$ for symmetry.

Solution

x-Axis: To test for symmetry with respect to the x-axis, replace y by $-y$. Since $-y = \dfrac{4x^2}{x^2 + 1}$ is not equivalent to $y = \dfrac{4x^2}{x^2 + 1}$, the graph of the equation is not symmetric with respect to the x-axis.

y-Axis: To test for symmetry with respect to the y-axis, replace x by $-x$. Since $y = \dfrac{4(-x)^2}{(-x)^2 + 1} = \dfrac{4x^2}{x^2 + 1}$ is equivalent to $y = \dfrac{4x^2}{x^2 + 1}$, the graph of the equation is symmetric with respect to the y-axis.

Origin: To test for symmetry with respect to the origin, replace x by $-x$ and y by $-y$.

$$-y = \frac{4(-x)^2}{(-x)^2 + 1} \quad \text{Replace } x \text{ by } -x \text{ and } y \text{ by } -y.$$

$$-y = \frac{4x^2}{x^2 + 1} \quad \text{Simplify.}$$

$$y = -\frac{4x^2}{x^2 + 1} \quad \text{Multiply both sides by } -1.$$

Since the result is not equivalent to the original equation, the graph of the equation $y = \dfrac{4x^2}{x^2 + 1}$ is not symmetric with respect to the origin.

Seeing the Concept

Figure 20 shows the graph of $4x^2 + 9y^2 = 36$ using Desmos. Do you see the symmetry with respect to the x-axis, the y-axis, and the origin?

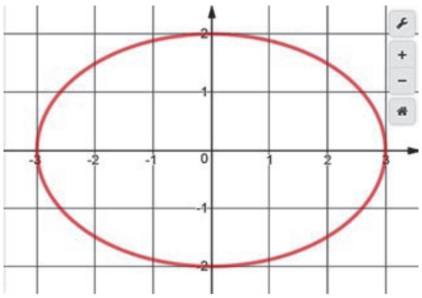


Figure 20 $4x^2 + 9y^2 = 36$

Seeing the Concept

Figure 21 shows the graph of $y = \dfrac{4x^2}{x^2 + 1}$ using a TI-84 Plus C graphing calculator. Do you see the symmetry with respect to the y-axis?

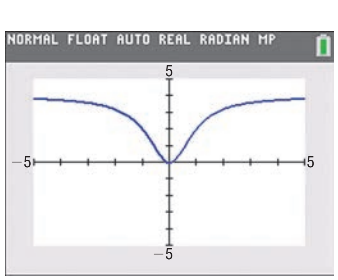


Figure 21 $y = \dfrac{4x^2}{x^2 + 1}$

Now Work PROBLEM 61

5 Know How to Graph Key Equations

The next three examples use intercepts, symmetry, and point plotting to obtain the graphs of key equations. It is important to know the graphs of these key equations because we use them later. The first of these is $y = x^3$.

EXAMPLE 11 Graphing the Equation $y = x^3$ by Finding Intercepts, Checking for Symmetry, and Plotting Points

Graph the equation $y = x^3$ by plotting points. Find any intercepts and check for symmetry first.

Solution First, find the intercepts. When $x = 0$, then $y = 0$; and when $y = 0$, then $x = 0$. The origin $(0, 0)$ is the only intercept. Now test for symmetry.

x-Axis: Replace y by $-y$. Since $-y = x^3$ is not equivalent to $y = x^3$, the graph is not symmetric with respect to the x-axis.

y-Axis: Replace x by $-x$. Since $y = (-x)^3 = -x^3$ is not equivalent to $y = x^3$, the graph is not symmetric with respect to the y-axis.

Origin: Replace x by $-x$ and y by $-y$. Since $-y = (-x)^3 = -x^3$ is equivalent to $y = x^3$ (multiply both sides by -1), the graph is symmetric with respect to the origin.

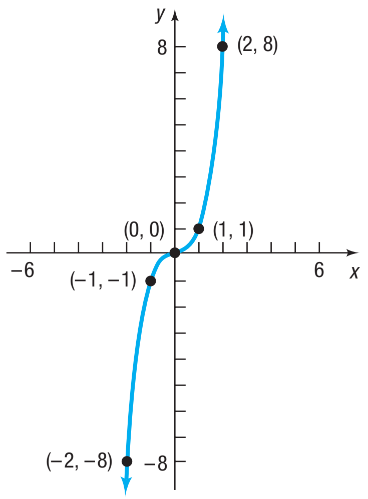


Figure 22 $y = x^3$

To graph $y = x^3$, use the equation to obtain several points on the graph. Because of the symmetry, we need to locate only points on the graph for which $x \geq 0$. See Table 4. Since $(1, 1)$ is on the graph, and the graph is symmetric with respect to the origin, the point $(-1, -1)$ is also on the graph. Plot the points from Table 4 and use the symmetry. Figure 22 shows the graph.

Table 4

x	$y = x^3$	(x, y)
0	0	(0, 0)
1	1	(1, 1)
2	8	(2, 8)
3	27	(3, 27)

EXAMPLE 12 Graphing the Equation $x = y^2$

(a) Graph the equation $x = y^2$. Find any intercepts and check for symmetry first.
(b) Graph $x = y^2$, $y \geq 0$.

Solution
(a) The lone intercept is $(0, 0)$. The graph is symmetric with respect to the x-axis. (Do you see why? Replace y by $-y$.) Figure 23(a) shows the graph.
(b) If we restrict y so that $y \geq 0$, the equation $x = y^2$, $y \geq 0$, may be written equivalently as $y = \sqrt{x}$. The portion of the graph of $x = y^2$ in quadrant I is therefore the graph of $y = \sqrt{x}$. See Figure 23(b).

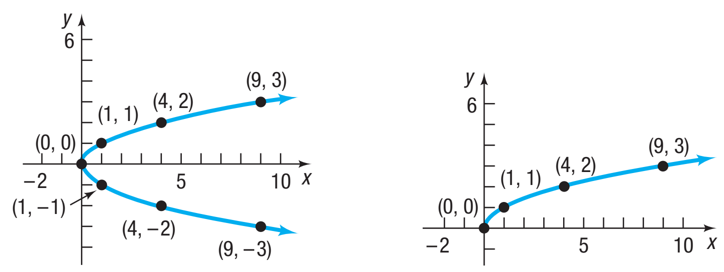


Figure 23 **(a)** $x = y^2$ **(b)** $y = \sqrt{x}$

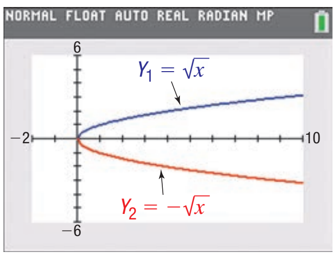


Figure 24

COMMENT To see the graph of the equation $x = y^2$ on a graphing calculator, you will need to graph two equations: $Y_1 = \sqrt{x}$ and $Y_2 = -\sqrt{x}$. We discuss why in Chapter 2. See Figure 24. ■

EXAMPLE 13 Graphing the Equation $y = \frac{1}{x}$

Graph the equation $y = \frac{1}{x}$. First, find any intercepts and check for symmetry.

Solution

Check for intercepts first. If we let $x = 0$, we obtain 0 in the denominator, which makes y undefined. We conclude that there is no y-intercept. If we let $y = 0$, we get the equation $\frac{1}{x} = 0$, which has no solution. We conclude that there is no x-intercept. The graph of $y = \frac{1}{x}$ does not cross or touch the coordinate axes.

Next check for symmetry:

x-Axis: Replacing y by $-y$ yields $-y = \frac{1}{x}$, which is not equivalent to $y = \frac{1}{x}$.

y-Axis: Replacing x by $-x$ yields $y = \frac{1}{-x} = -\frac{1}{x}$, which is not equivalent to $y = \frac{1}{x}$.

Origin: Replacing x by $-x$ and y by $-y$ yields $-y = -\frac{1}{x}$, which is equivalent to $y = \frac{1}{x}$. The graph is symmetric with respect to the origin.

Table 5

x	$y = \frac{1}{x}$	(x, y)
$\frac{1}{10}$	10	$\left(\frac{1}{10}, 10\right)$
$\frac{1}{3}$	3	$\left(\frac{1}{3}, 3\right)$
$\frac{1}{2}$	2	$\left(\frac{1}{2}, 2\right)$
1	1	$(1, 1)$
2	$\frac{1}{2}$	$\left(2, \frac{1}{2}\right)$
3	$\frac{1}{3}$	$\left(3, \frac{1}{3}\right)$
10	$\frac{1}{10}$	$\left(10, \frac{1}{10}\right)$

Now set up Table 5, listing several points on the graph. Because of the symmetry with respect to the origin, we use only positive values of x. From Table 5 we infer that if x is a large and positive number, then $y = \frac{1}{x}$ is a positive number close to 0. We also infer that if x is a positive number close to 0, then $y = \frac{1}{x}$ is a large and positive number. Armed with this information, we can graph the equation.

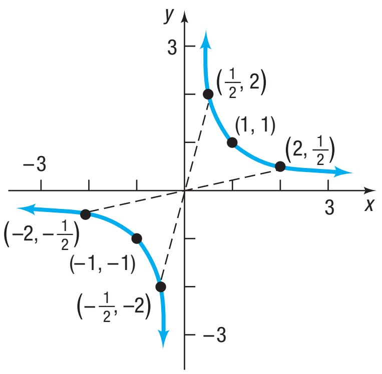


Figure 25 $y = \frac{1}{x}$

Figure 25 illustrates some of these points and the graph of $y = \frac{1}{x}$. Observe how the absence of intercepts and the existence of symmetry with respect to the origin were utilized.

COMMENT Refer to Example 2 in Section B.3, for the graph of $y = \frac{1}{x}$ using a graphing calculator. ■

1.2 Assess Your Understanding

'Are You Prepared?' *Answers are given at the end of these exercises. If you get a wrong answer, read the pages listed in red.*

1. Solve the equation $2(x + 3) - 1 = -7$. (pp. A44–A45)

2. Solve the equation $x^2 - 9 = 0$. (pp. A47–A48)

Concepts and Vocabulary

3. The points, if any, at which a graph crosses or touches the coordinate axes are called ____________.

4. The x-intercepts of the graph of an equation are those x-values for which __________.

5. If for every point (x, y) on the graph of an equation the point $(-x, y)$ is also on the graph, then the graph is symmetric with respect to the __________.

6. If the graph of an equation is symmetric with respect to the y-axis and -4 is an x-intercept of this graph, then ___ is also an x-intercept.

7. If the graph of an equation is symmetric with respect to the origin and $(3, -4)$ is a point on the graph, then __________ is also a point on the graph.

8. ***True or False*** To find the y-intercepts of the graph of an equation, let $x = 0$ and solve for y.

9. ***True or False*** The y-coordinate of a point at which the graph crosses or touches the x-axis is an x-intercept.

10. ***True or False*** If a graph is symmetric with respect to the x-axis, then it cannot be symmetric with respect to the y-axis.

11. ***Multiple Choice*** Given that the intercepts of a graph are $(-4, 0)$ and $(0, 5)$, choose the statement that is true.

(a) The y-intercept is -4, and the x-intercept is 5.
(b) The y-intercepts are -4 and 5.
(c) The x-intercepts are -4 and 5.
(d) The x-intercept is -4, and the y-intercept is 5.

12. ***Multiple Choice*** To test whether the graph of an equation is symmetric with respect to the origin, replace __________ in the equation and simplify. If an equivalent equation results, then the graph is symmetric with respect to the origin.

(a) x by $-x$ **(b)** y by $-y$
(c) x by $-x$ and y by $-y$ **(d)** x by $-y$ and y by $-x$

Skill Building

In Problems 13–18, determine which of the given points are on the graph of the equation.

13. Equation: $y = x^4 - \sqrt{x}$
Points: $(0, 0)$; $(1, 1)$; $(2, 4)$

14. Equation: $y = x^3 - 2\sqrt{x}$
Points: $(0, 0)$; $(1, 1)$; $(1, -1)$

15. Equation: $y^3 = x + 1$
Points: $(1, 2)$; $(0, 1)$; $(-1, 0)$

16. Equation: $y^2 = x^2 + 9$
Points: $(0, 3)$; $(3, 0)$; $(-3, 0)$

17. Equation: $x^2 + 4y^2 = 4$
Points: $(0, 1)$; $(2, 0)$; $\left(2, \frac{1}{2}\right)$

18. Equation: $x^2 + y^2 = 4$
Points: $(0, 2)$; $(-2, 2)$; $(\sqrt{2}, \sqrt{2})$

In Problems 19–30, find the intercepts and graph each equation by plotting points. Be sure to label the intercepts.

19. $y = x - 6$ **20.** $y = x + 2$ **21.** $y = 3x - 9$ **22.** $y = 2x + 8$
23. $y = x^2 - 1$ **24.** $y = x^2 - 9$ **25.** $y = -x^2 + 1$ **26.** $y = -x^2 + 4$
27. $5x + 2y = 10$ **28.** $2x + 3y = 6$ **29.** $4x^2 + y = 4$ **30.** $9x^2 + 4y = 36$

In Problems 31–40, plot each point. Then plot the point that is symmetric to it with respect to (a) the x-axis; (b) the y-axis; (c) the origin.

31. $(3, 4)$ **32.** $(5, 3)$ **33.** $(4, -2)$ **34.** $(-2, 1)$ **35.** $(-1, -1)$
36. $(5, -2)$ **37.** $(4, 0)$ **38.** $(-3, -4)$ **39.** $(-3, 0)$ **40.** $(0, -3)$

In Problems 41–52, the graph of an equation is given. (a) Find the intercepts. (b) Indicate whether the graph is symmetric with respect to the x-axis, the y-axis, the origin or none of these.

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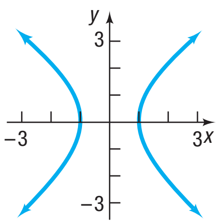

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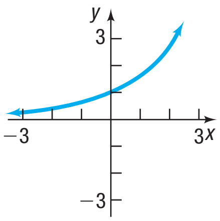

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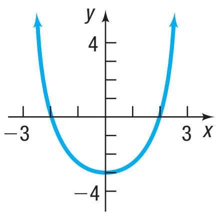

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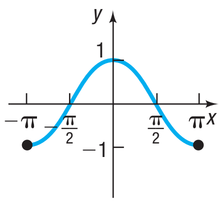

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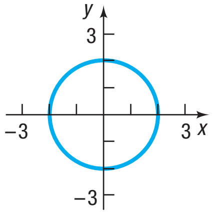

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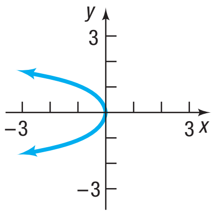

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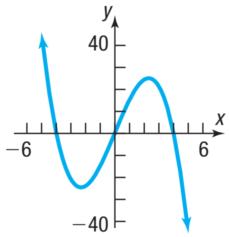

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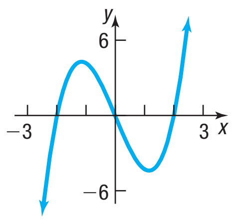

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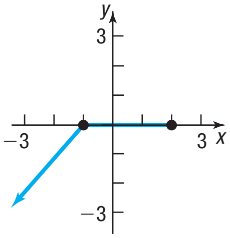

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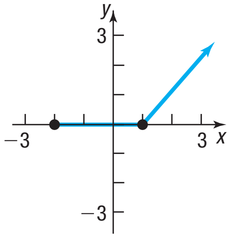

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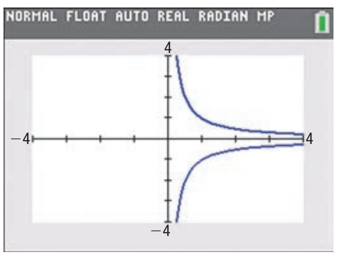

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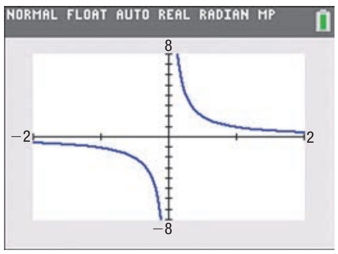
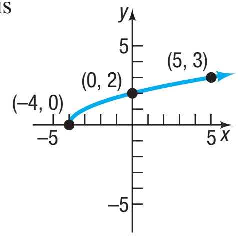
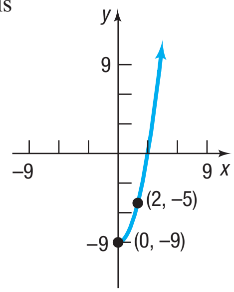


In Problems 53–56, draw a complete graph so that it has the type of symmetry indicated.

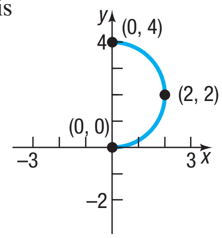
53. x-axis



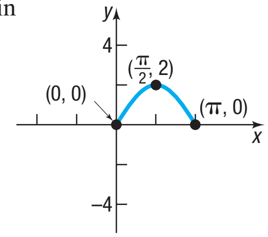
54. y-axis



55. y-axis



56. Origin



In Problems 57–72, list the intercepts and test for symmetry.

57. $y^2 = x + 9$
58. $y^2 = x + 16$
59. $y = \sqrt[5]{x}$
60. $y = \sqrt[3]{x}$
61. $x^2 + y - 9 = 0$
62. $x^2 - y - 4 = 0$
63. $4x^2 + y^2 = 4$
64. $25x^2 + 4y^2 = 100$
65. $y = x^4 - 1$
66. $y = x^3 - 64$
67. $y = x^2 + 4$
68. $y = x^2 - 2x - 8$
69. $y = \dfrac{x^2 - 4}{2x}$
70. $y = \dfrac{4x}{x^2 + 16}$
71. $y = \dfrac{x^4 + 1}{2x^5}$
72. $y = \dfrac{-x^3}{x^2 - 9}$

In Problems 73–76, graph each equation.

73. $x = y^2$
74. $y = x^3$
75. $y = \dfrac{1}{x}$
76. $y = \sqrt{x}$

77. If $(a, -5)$ is a point on the graph of $y = x^2 + 6x$, what is a?

78. If $(a, 4)$ is a point on the graph of $y = x^2 + 3x$, what is a?

Applications and Extensions

79. Given that the point $(2, 6)$ is on the graph of an equation that is symmetric with respect to the origin, what other point is on the graph?

80. If the graph of an equation is symmetric with respect to the y-axis and 6 is an x-intercept of this graph, name another x-intercept.

81. If the graph of an equation is symmetric with respect to the y-axis and 1 is an x-intercept of this graph, name another x-intercept.

82. If the graph of an equation is symmetric with respect tothe x-axis and 2 is a y-intercept, name another y-intercept.

83. Microphones In studios and on stages, cardioid microphones are often preferred for the richness they add to voices and for their ability to reduce the level of sound from the sides and rear of the microphone. Suppose one such cardioids pattern is given by the equation $(x^2 + y^2 - 9x)^2 = 81x^2 + 81y^2$.

(a) Find the intercepts of the graph of the equation.
(b) Test for symmetry with respect to the x-axis, y-axis, and origin.

Source: *www.notaviva.com*

84. Solar Energy The solar electric generating systems at Kramer Junction, California, use parabolic troughs to heat a heat-transfer fluid to a high temperature. This fluid is used to generate steam that drives a power conversion system to produce electricity. For troughs 7.5 feet wide, an equation for the cross section is $16y^2 = 120x - 225$.

(a) Find the intercepts of the graph of the equation.
(b) Test for symmetry with respect to the x-axis, the y-axis, and the origin.

Source: *U.S. Department of Energy*

85. ***Challenge Problem*** **Lemniscate** For a nonzero constant a, find the intercepts of the graph of $(x^2 + y^2)^2 = a^2(x^2 - y^2)$. Then test for symmetry with respect to the x-axis, the y-axis, and the origin.

86. ***Challenge Problem*** **Limaçon** For nonzero constants a and b, find the intercepts of the graph of

$$(x^2 + y^2 - ax)^2 = b^2(x^2 + y^2)$$

Then test for symmetry with respect to the x-axis, the y-axis, and the origin.

Explaining Concepts: Discussion and Writing

87. **(a)** Graph $y = \sqrt{x^2}$, $y = x$, $y = |x|$, and $y = (\sqrt{x})^2$, noting which graphs are the same.
(b) Explain why the graphs of $y = \sqrt{x^2}$ and $y = |x|$ are the same.
(c) Explain why the graphs of $y = x$ and $y = (\sqrt{x})^2$ are not the same.
(d) Explain why the graphs of $y = \sqrt{x^2}$ and $y = x$ are not the same.

88. Explain what is meant by a complete graph.

89. Draw a graph of an equation that contains two x-intercepts; at one the graph crosses the x-axis, and at the other the graph touches the x-axis.

90. Make up an equation with the intercepts (2, 0), (4, 0), and (0, 1). Compare your equation with a friend's equation. Comment on any similarities.

91. Draw a graph that contains the points

$$(-2, -1), (0, 1), (1, 3), \text{ and } (3, 5)$$

Compare your graph with those of other students. Are most of the graphs almost straight lines? How many are "curved"? Discuss the various ways in which these points might be connected.

92. An equation is being tested for symmetry with respect to the x-axis, the y-axis, and the origin. Explain why, if two of these symmetries are present, the remaining one must also be present.

93. Draw a graph that contains the points $(-2, 5)$, $(-1, 3)$, and (0, 2) and is symmetric with respect to the y-axis. Compare your graph with those of other students; comment on any similarities. Can a graph contain these points and be symmetric with respect to the x-axis? the origin? Why or why not?

'Are You Prepared?' Answers

1. $\{-6\}$ **2.** $\{-3, 3\}$

1.3 Lines

OBJECTIVES
1 Calculate and Interpret the Slope of a Line (p. 56)
2 Graph Lines Given a Point and the Slope (p. 59)
3 Find the Equation of a Vertical Line (p. 60)
4 Use the Point-Slope Form of a Line; Identify Horizontal Lines (p. 60)
5 Use the Slope-Intercept Form of a Line (p. 61)
6 Find an Equation of a Line Given Two Points (p. 63)
7 Graph Lines Written in General Form Using Intercepts (p. 63)
8 Find Equations of Parallel Lines (p. 64)
9 Find Equations of Perpendicular Lines (p. 65)

In this section we study a certain type of equation that contains two variables, called a *linear equation,* and its graph, a *line*.

1 Calculate and Interpret the Slope of a Line

Consider the staircase illustrated in Figure 26. Each step contains exactly the same horizontal **run** and the same vertical **rise**. The ratio of the rise to the run, called the *slope*, is a numerical measure of the steepness of the staircase. For example, if the run is increased and the rise remains the same, the staircase becomes less steep. If the run is kept the same but the rise is increased, the staircase becomes more steep. This important characteristic of a line is best defined using rectangular coordinates.

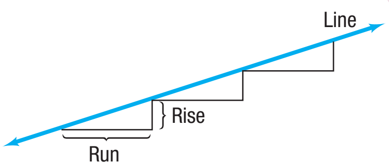


Figure 26

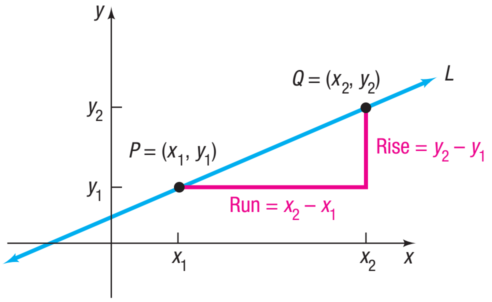
DEFINITION Slope

Let $P = (x_1, y_1)$ and $Q = (x_2, y_2)$ be two distinct points. If $x_1 \neq x_2$, the **slope *m*** of the nonvertical line L containing P and Q is defined by the formula

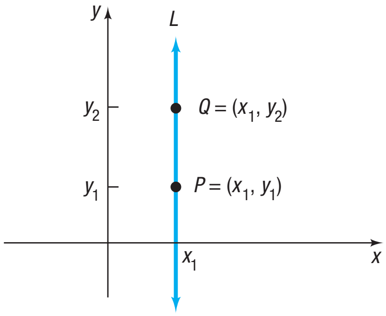
$$m = \frac{y_2 - y_1}{x_2 - x_1} \qquad x_1 \neq x_2 \qquad \textbf{(1)}$$

If $x_1 = x_2$, then L is a **vertical line** and the slope m of L is **undefined** (since this results in division by 0).

Figure 27(a) illustrates the slope of a nonvertical line; Figure 27(b) illustrates a vertical line.



(a) Slope of L is $m = \dfrac{y_2 - y_1}{x_2 - x_1}$, $x_1 \neq x_2$



(b) Slope is undefined; L is vertical

Figure 27

In Words
The symbol Δ is the Greek uppercase letter delta. In mathematics, Δ is read "change in," so $\frac{\Delta y}{\Delta x}$ is read "change in y divided by change in x."

As Figure 27(a) illustrates, the slope m of a nonvertical line may be viewed as

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\text{Rise}}{\text{Run}} \quad \text{or as} \quad m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\text{Change in } y}{\text{Change in } x} = \frac{\Delta y}{\Delta x}$$

That is, the slope m of a nonvertical line measures the amount y changes when x changes from x_1 to x_2. The expression $\frac{\Delta y}{\Delta x}$ is called the **average rate of change** of y with respect to x.

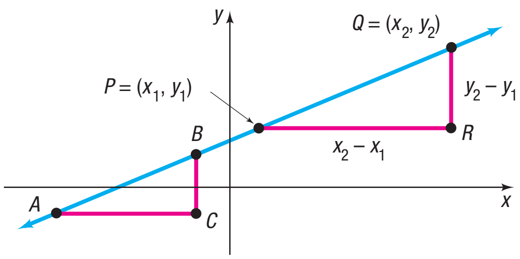
Two comments about computing the slope of a nonvertical line may prove helpful:

- Any two distinct points on the line can be used to compute the slope of the line. (See Figure 28 for justification.) Since any two distinct points can be used to compute the slope of a line, the average rate of change of a line is always the same number.

Figure 28
Triangles ABC and PQR are similar (equal angles), so ratios of corresponding sides are proportional. Then the slope using P and Q is

$$\frac{y_2 - y_1}{x_2 - x_1} = \frac{d(B, C)}{d(A, C)},$$

which is the slope using A and B.



- The slope of a line may be computed from $P = (x_1, y_1)$ to $Q = (x_2, y_2)$ or from Q to P because

$$\frac{y_2 - y_1}{x_2 - x_1} = \frac{y_1 - y_2}{x_1 - x_2}$$

EXAMPLE 1 Finding and Interpreting the Slope of a Line Given Two Points

The slope m of the line containing the points $(1, 2)$ and $(5, -3)$ may be computed as

$$m = \frac{-3 - 2}{5 - 1} = \frac{-5}{4} = -\frac{5}{4} \quad \text{or as} \quad m = \frac{2 - (-3)}{1 - 5} = \frac{5}{-4} = -\frac{5}{4}$$

For every 4-unit change in x, y will change by -5 units. That is, if x increases by 4 units, then y will decrease by 5 units. The average rate of change of y with respect to x is $-\frac{5}{4}$.

Now Work PROBLEMS 13 AND 19

EXAMPLE 2 Finding the Slopes of Various Lines Containing the Same Point (2, 3)

Compute the slopes of the lines L_1, L_2, L_3, and L_4 containing the following pairs of points. Graph all four lines on the same set of coordinate axes.

$$\begin{aligned} &L_1: \quad P = (2, 3) \qquad Q_1 = (-1, -2) \\ &L_2: \quad P = (2, 3) \qquad Q_2 = (3, -1) \\ &L_3: \quad P = (2, 3) \qquad Q_3 = (5, 3) \\ &L_4: \quad P = (2, 3) \qquad Q_4 = (2, 5) \end{aligned}$$

Solution Let m_1, m_2, m_3, and m_4 denote the slopes of the lines L_1, L_2, L_3, and L_4, respectively. Then

$$m_1 = \frac{-2 - 3}{-1 - 2} = \frac{-5}{-3} = \frac{5}{3} \qquad \text{A rise of 5 divided by a run of 3}$$

$$m_2 = \frac{-1 - 3}{3 - 2} = \frac{-4}{1} = -4$$

$$m_3 = \frac{3 - 3}{5 - 2} = \frac{0}{3} = 0$$

m_4 is undefined because $x_1 = x_2 = 2$

The graphs of these lines are in Figure 29.

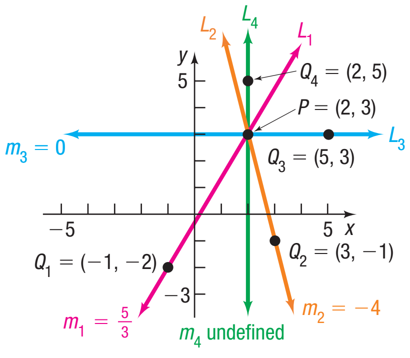


Figure 29

Figure 29 illustrates the following facts:

- When the slope of a line is positive, the line slants upward from left to right (L_1).
- When the slope of a line is negative, the line slants downward from left to right (L_2).
- When the slope is 0, the line is horizontal (L_3).
- When the slope is undefined, the line is vertical (L_4).

Seeing the Concept

On the same screen, graph the following equations:

$$Y_1 = 0 \qquad \text{Slope of line is 0.}$$

$$Y_2 = \frac{1}{4}x \qquad \text{Slope of line is } \frac{1}{4}.$$

$$Y_3 = \frac{1}{2}x \qquad \text{Slope of line is } \frac{1}{2}.$$

$$Y_4 = x \qquad \text{Slope of line is 1.}$$

$$Y_5 = 2x \qquad \text{Slope of line is 2.}$$

$$Y_6 = 6x \qquad \text{Slope of line is 6.}$$

See Figure 30.

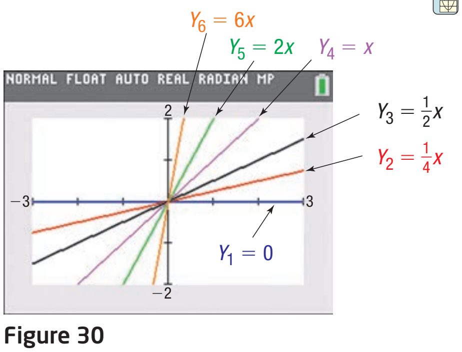


Figure 30

Seeing the Concept

On the same screen, graph the following equations:

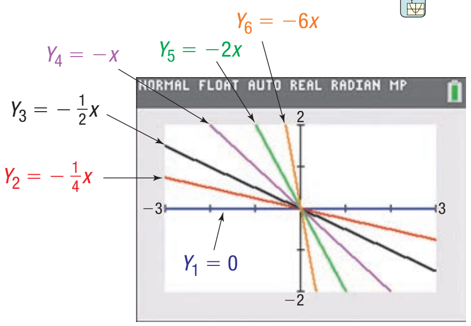


Figure 31

$$Y_1 = 0 \quad \text{Slope of line is } 0.$$

$$Y_2 = -\frac{1}{4}x \quad \text{Slope of line is } -\frac{1}{4}.$$

$$Y_3 = -\frac{1}{2}x \quad \text{Slope of line is } -\frac{1}{2}.$$

$$Y_4 = -x \quad \text{Slope of line is } -1.$$

$$Y_5 = -2x \quad \text{Slope of line is } -2.$$

$$Y_6 = -6x \quad \text{Slope of line is } -6.$$

See Figure 31.

Figures 30 and 31 illustrate that the closer the line is to the vertical position, the greater the magnitude of the slope.

2 Graph Lines Given a Point and the Slope

EXAMPLE 3 Graphing a Line Given a Point and a Slope

Draw a graph of the line that contains the point (3, 2) and has a slope of:

(a) $\frac{3}{4}$ (b) $-\frac{4}{5}$

Solution

(a) Slope $= \frac{\text{Rise}}{\text{Run}}$. The slope $\frac{3}{4}$ means that for every horizontal change (run) of 4 units to the right, there is a vertical change (rise) of 3 units. Start at the point (3, 2) and move 4 units to the right and 3 units up, arriving at the point (7, 5). Drawing the line through the points (7, 5) and (3, 2) gives the graph. See Figure 32.

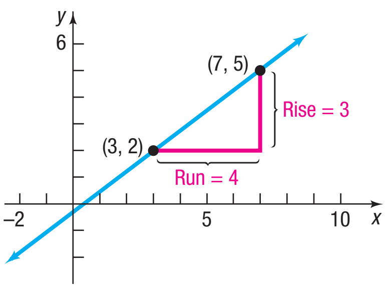


Figure 32

(b) A slope of

$$-\frac{4}{5} = \frac{-4}{5} = \frac{\text{Rise}}{\text{Run}}$$

means that for every horizontal change of 5 units to the right, there is a corresponding vertical change of -4 units (a downward movement). Start at the point (3, 2) and move 5 units to the right and then 4 units down, arriving at the point $(8, -2)$. Drawing the line through these points gives the graph. See Figure 33.

Alternatively, consider that

$$-\frac{4}{5} = \frac{4}{-5} = \frac{\text{Rise}}{\text{Run}}$$

so for every horizontal change of -5 units (a movement to the left), there is a corresponding vertical change of 4 units (upward). This approach leads to the point $(-2, 6)$, which is also on the graph of the line in Figure 33.

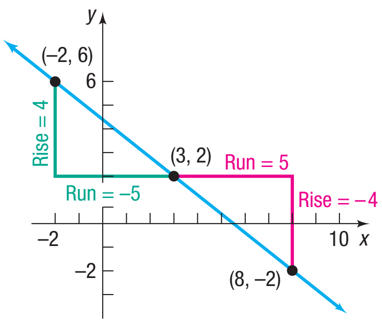


Figure 33

Now Work PROBLEM 25

3 Find the Equation of a Vertical Line

EXAMPLE 4 **Graphing a Line**

Graph the equation: $x = 3$

Solution To graph $x = 3$, we find all points (x, y) in the plane for which $x = 3$. No matter what y-coordinate is used, the corresponding x-coordinate always equals 3. Consequently, the graph of the equation $x = 3$ is a vertical line with x-intercept 3 and an undefined slope. See Figure 34 .

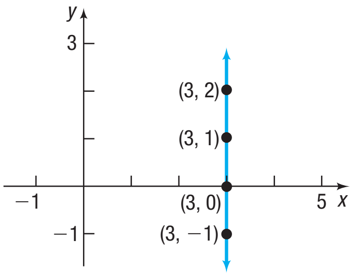


Figure 34 $x = 3$

Example 4 suggests the following result:

THEOREM Equation of a Vertical Line

A vertical line is given by an equation of the form

$$\boxed{x = a}$$

where a is the x-intercept.

COMMENT To graph an equation using most graphing utilities, we need to express the equation in the form $y = \{expression\ in\ x\}$. But $x = 3$ cannot be put in this form. To overcome this, most graphing utilities have special commands for drawing vertical lines. DRAW, LINE, PLOT, and VERT are among the more common ones. Consult your manual to determine the correct methodology for your graphing utility. ■

4 Use the Point-Slope Form of a Line; Identify Horizontal Lines

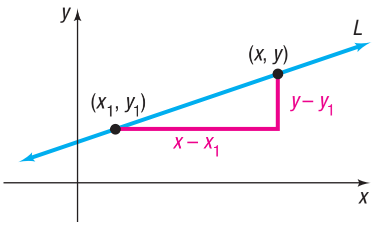


Figure 35

Let L be a nonvertical line with slope m that contains the point (x_1, y_1). See Figure 35. For any other point (x, y) on L, we have

$$m = \frac{y - y_1}{x - x_1} \quad \text{or} \quad y - y_1 = m(x - x_1)$$

THEOREM Point-Slope Form of an Equation of a Line

An equation of a nonvertical line with slope m that contains the point (x_1, y_1) is

$$\boxed{y - y_1 = m(x - x_1) \qquad \textbf{(2)}}$$

EXAMPLE 5 Finding the Point-Slope Form of an Equation of a Line

Find the point-slope form of an equation of the line with slope 4, containing the point $(1, 2)$.

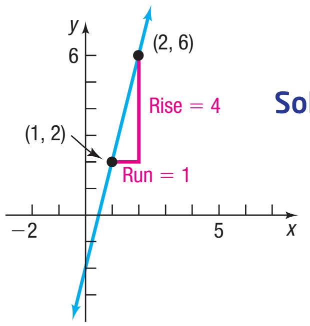


Figure 36 $y - 2 = 4(x - 1)$

Solution An equation of the line with slope 4 that contains the point $(1, 2)$ can be found by using the point-slope form with $m = 4$, $x_1 = 1$, and $y_1 = 2$.

$$y - y_1 = m(x - x_1)$$

$$y - 2 = 4(x - 1) \qquad m = 4, x_1 = 1, y_1 = 2$$

See Figure 36 for the graph.

Now Work PROBLEM 33

EXAMPLE 6 Finding the Equation of a Horizontal Line

Find an equation of the horizontal line containing the point $(3, 2)$.

Solution Because all the y-values are equal on a horizontal line, the slope of a horizontal line is 0. To get an equation, use the point-slope form with $m = 0$, $x_1 = 3$, and $y_1 = 2$.

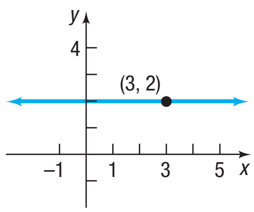


Figure 37 $y = 2$

$$y - y_1 = m(x - x_1)$$

$$y - 2 = 0 \cdot (x - 3) \qquad m = 0, x_1 = 3, y_1 = 2$$

$$y - 2 = 0$$

$$y = 2$$

See Figure 37 for the graph.

Example 6 suggests the following result:

THEOREM Equation of a Horizontal Line

A horizontal line is given by an equation of the form

$$\boxed{y = b}$$

where b is the y-intercept.

5 Use the Slope-Intercept Form of a Line

Another useful equation of a line is obtained when the slope m and y-intercept b are known. Then the point $(0, b)$ is on the line. Using the point-slope form, equation (2), we obtain

$$y - b = m(x - 0) \quad \text{or} \quad y = mx + b$$

THEOREM Slope-Intercept Form of an Equation of a Line

An equation of a line with slope m and y-intercept b is

$$\boxed{y = mx + b} \qquad \textbf{(3)}$$

For example, if a line has slope 5 and y-intercept 2, we can write the equation in slope-intercept form as

$$y = \underbrace{5}_{\text{slope}}x + \underbrace{2}_{y\text{-intercept}}$$

Now Work PROBLEMS 53 AND 59 (express answer in slope-intercept form)

Seeing the Concept

To see the role that the slope m plays, graph the following lines on the same screen.

$$
\begin{aligned}
Y_1 &= 2 & m &= 0\\
Y_2 &= x + 2 & m &= 1\\
Y_3 &= -x + 2 & m &= -1\\
Y_4 &= 3x + 2 & m &= 3\\
Y_5 &= -3x + 2 & m &= -3
\end{aligned}
$$

Figure 38 displays the graphs using Desmos. What do you conclude about lines of the form $y = mx + 2$?

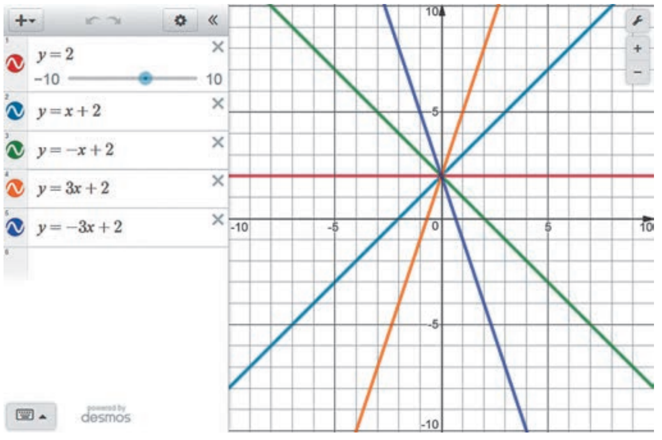


Figure 38 $y = mx + 2$

To see the role of the y-intercept b, graph the following lines on the same screen.

$$
\begin{aligned}
Y_1 &= 2x & b &= 0\\
Y_2 &= 2x + 1 & b &= 1\\
Y_3 &= 2x - 1 & b &= -1\\
Y_4 &= 2x + 4 & b &= 4\\
Y_5 &= 2x - 4 & b &= -4
\end{aligned}
$$

Figure 39 displays the graphs using Desmos. What do you conclude about lines of the form $y = 2x + b$?

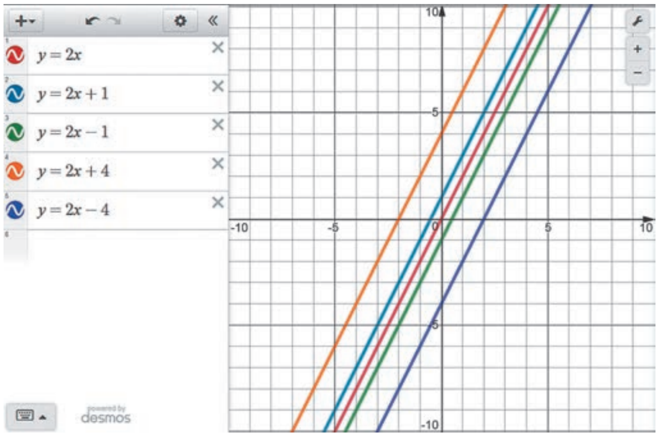


Figure 39 $y = 2x + b$

When the equation of a line is written in slope-intercept form, it is easy to find the slope m and y-intercept b of the line. For example, suppose that the equation of a line is

$$y = -2x + 7$$

Compare this equation to $y = mx + b$.

$$\underset{y \,=\, mx \,+\, b}{y = -2x + 7}$$

The slope of this line is -2 and its y-intercept is 7.

Now Work PROBLEM 79

EXAMPLE 7 Finding the Slope and y-Intercept of a Line

Find the slope m and y-intercept b of the equation $2x + 4y = 8$. Graph the equation.

Solution To find the slope and y-intercept, write the equation in slope-intercept form by solving for y.

$$
\begin{aligned}
2x + 4y &= 8\\
4y &= -2x + 8\\
y &= -\frac{1}{2}x + 2 \qquad y = mx + b
\end{aligned}
$$

The coefficient of x, $-\frac{1}{2}$, is the slope, and the constant, 2, is the y-intercept. To graph the line with y-intercept 2 and with slope $-\frac{1}{2}$, start at the point $(0, 2)$ and move to the right 2 units and then down 1 unit to the point $(2, 1)$. Draw the line through these points. See Figure 40.

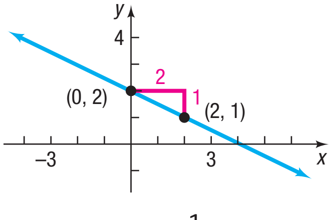


Figure 40 $y = -\frac{1}{2}x + 2$

Now Work PROBLEM 85

6 Find the Equation of a Line Given Two Points

EXAMPLE 8 **Finding an Equation of a Line Given Two Points**

Find an equation of the line containing the points $(2, 3)$ and $(-4, 5)$. Graph the line.

Solution First compute the slope of the line.

$$m = \frac{5-3}{-4-2} = \frac{2}{-6} = -\frac{1}{3} \qquad m = \frac{y_2 - y_1}{x_2 - x_1}$$

Use the point (2, 3) and the slope $m = -\frac{1}{3}$ to get the point-slope form of the equation of the line.

$$y - 3 = -\frac{1}{3}(x - 2) \qquad y - y_1 = m(x - x_1)$$

Continue to get the slope-intercept form.

$$y - 3 = -\frac{1}{3}x + \frac{2}{3}$$

$$y = -\frac{1}{3}x + \frac{11}{3} \qquad y = mx + b$$

See Figure 41 for the graph.

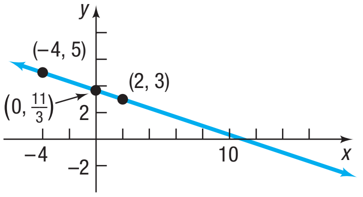


Figure 41 $y = -\frac{1}{3}x + \frac{11}{3}$

In the solution to Example 8, we could have used the other point, $(-4, 5)$, instead of the point $(2, 3)$. The equation that results, when written in slope-intercept form, is the equation that we obtained in the example. (Try it for yourself.)

Now Work PROBLEM 45

7 Graph Lines Written in General Form Using Intercepts

DEFINITION General Form

The equation of a line is in **general form*** when it is written as

$$Ax + By = C \qquad \textbf{(4)}$$

where A, B, and C are real numbers and A and B are not both 0.

If $B = 0$ in equation (4), then $A \neq 0$ and the graph of the equation is a vertical line: $x = \frac{C}{A}$.

If $B \neq 0$ in equation (4), then we can solve the equation for y and write the equation in slope-intercept form as we did in Example 7.

One way to graph a line given in general form, equation (4), is to find its intercepts. Remember, the intercepts of the graph of an equation are the points where the graph crosses or touches a coordinate axis.

EXAMPLE 9 **Graphing an Equation in General Form Using Its Intercepts**

Graph the equation $2x + 4y = 8$ by finding its intercepts.

Solution To obtain the x-intercept, let $y = 0$ in the equation and solve for x.

$$\begin{aligned} 2x + 4y &= 8 \\ 2x + 4 \cdot 0 &= 8 \qquad \text{Let } y = 0. \\ 2x &= 8 \\ x &= 4 \qquad \text{Divide both sides by 2.} \end{aligned}$$

The x-intercept is 4, and the point $(4, 0)$ is on the graph of the equation.

(continued)

*Some texts use the term **standard form**.

To obtain the y-intercept, let $x = 0$ in the equation and solve for y.

$$\begin{aligned} 2x + 4y &= 8 \\ 2 \cdot 0 + 4y &= 8 \quad \text{Let } x = 0. \\ 4y &= 8 \\ y &= 2 \quad \text{Divide both sides by 4.} \end{aligned}$$

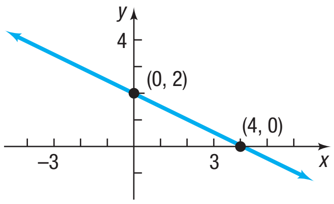


Figure 42 $2x + 4y = 8$

The y-intercept is 2, and the point $(0, 2)$ is on the graph of the equation.

Plot the points (4, 0) and (0, 2) and draw the line through the points. See Figure 42.

Now Work PROBLEM 99

The equation of every line can be written in general form. For example, a vertical line whose equation is

$$x = a$$

can be written in the general form

$$1 \cdot x + 0 \cdot y = a \quad A = 1, B = 0, C = a$$

A horizontal line whose equation is

$$y = b$$

can be written in the general form

$$0 \cdot x + 1 \cdot y = b \quad A = 0, B = 1, C = b$$

Lines that are neither vertical nor horizontal have general equations of the form

$$Ax + By = C \quad A \neq 0 \text{ and } B \neq 0$$

Because the equation of every line can be written in general form, any equation equivalent to equation (4) is called a **linear equation**.

8 Find Equations of Parallel Lines

When two lines (in the plane) do not intersect (that is, they have no points in common), they are **parallel**. Look at Figure 43. We have drawn two parallel lines and have constructed two right triangles by drawing sides parallel to the coordinate axes. The right triangles are similar. (Do you see why? Two angles are equal.) Because the triangles are similar, the ratios of corresponding sides are equal.

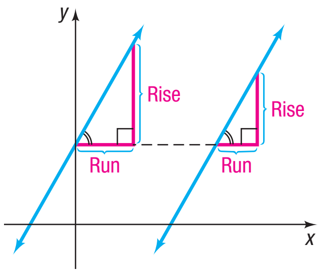


Figure 43 Parallel lines

THEOREM Criteria for Parallel Lines

Two nonvertical lines are parallel if and only if their slopes are equal and they have different y-intercepts.

The use of the phrase "if and only if" in the preceding theorem means that actually two statements are being made, one the converse of the other.

- If two nonvertical lines are parallel, then their slopes are equal and they have different y-intercepts.
- If two nonvertical lines have equal slopes and they have different y-intercepts, then they are parallel.

EXAMPLE 10 Showing That Two Lines Are Parallel

Show that the lines given by the following equations are parallel.

$$L_1: \quad 2x + 3y = 6 \qquad L_2: \quad 4x + 6y = 0$$

Solution To determine whether these lines have equal slopes and different y-intercepts, write each equation in slope-intercept form.

$$L_1: \quad 2x + 3y = 6 \qquad\qquad L_2: \quad 4x + 6y = 0$$
$$3y = -2x + 6 \qquad\qquad 6y = -4x$$
$$y = -\frac{2}{3}x + 2 \qquad\qquad y = -\frac{2}{3}x$$
$$\text{Slope} = -\frac{2}{3}; \quad y\text{-intercept} = 2 \qquad\qquad \text{Slope} = -\frac{2}{3}; \quad y\text{-intercept} = 0$$

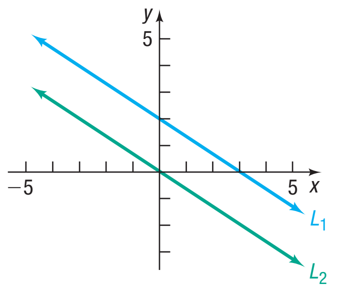


Figure 44 Parallel lines

Because these lines have the same slope, $-\frac{2}{3}$, but different y-intercepts, the lines are parallel. See Figure 44.

EXAMPLE 11 Finding a Line That Is Parallel to a Given Line

Find an equation for the line that contains the point $(2, -3)$ and is parallel to the line $2x + y = 6$.

Solution Since the two lines are to be parallel, the slope of the line containing the point $(2, -3)$ equals the slope of the line $2x + y = 6$. Begin by writing the equation of the line $2x + y = 6$ in slope-intercept form.

$$2x + y = 6$$
$$y = -2x + 6$$

The slope is -2. Since the line containing the point $(2, -3)$ also has slope -2, use the point-slope form to obtain its equation.

$$y - y_1 = m(x - x_1) \qquad \text{Point-slope form}$$
$$y - (-3) = -2(x - 2) \qquad m = -2, x_1 = 2, y_1 = -3$$
$$y + 3 = -2x + 4 \qquad \text{Simplify.}$$
$$y = -2x + 1 \qquad \text{Slope-intercept form}$$
$$2x + y = 1 \qquad \text{General form}$$

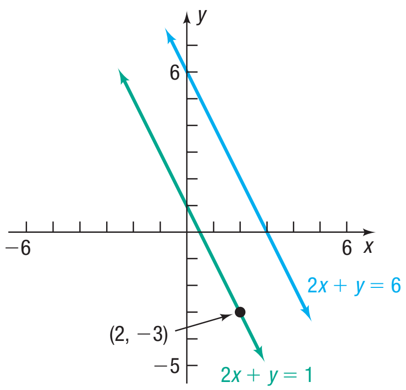


Figure 45

This line is parallel to the line $2x + y = 6$ and contains the point $(2, -3)$. See Figure 45.

Now Work PROBLEM 67

9 Find Equations of Perpendicular Lines

When two lines intersect at a right angle (90°), they are **perpendicular**. See Figure 46.

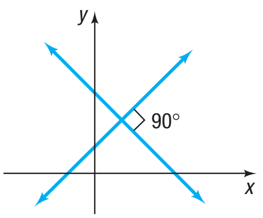


Figure 46 Perpendicular lines

The following result gives a condition, in terms of their slopes, for two lines to be perpendicular.

THEOREM Criterion for Perpendicular Lines

Two nonvertical lines are perpendicular if and only if the product of their slopes is -1.

Here we prove the "only if" part of the statement:

If two nonvertical lines are perpendicular, then the product of their slopes is -1.

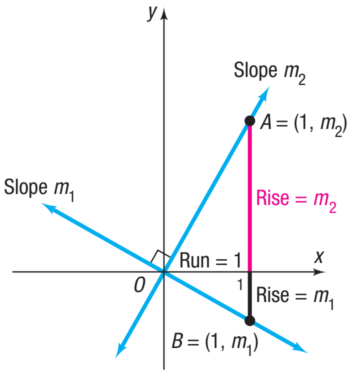


Figure 47

Proof Let m_1 and m_2 denote the slopes of the two lines. There is no loss in generality (that is, neither the angle nor the slopes are affected) if we situate the lines so that they meet at the origin. See Figure 47. The point $A = (1, m_2)$ is on the line having slope m_2, and the point $B = (1, m_1)$ is on the line having slope m_1. (Do you see why this must be true?)

Suppose that the lines are perpendicular. Then triangle OAB is a right triangle. As a result of the Pythagorean Theorem, it follows that

$$[d(O, A)]^2 + [d(O, B)]^2 = [d(A, B)]^2 \tag{5}$$

Using the distance formula, the squares of these distances are

$$[d(O, A)]^2 = (1-0)^2 + (m_2 - 0)^2 = 1 + m_2^2$$
$$[d(O, B)]^2 = (1-0)^2 + (m_1 - 0)^2 = 1 + m_1^2$$
$$[d(A, B)]^2 = (1-1)^2 + (m_2 - m_1)^2 = m_2^2 - 2m_1m_2 + m_1^2$$

Using these facts in equation (5), we get

$$(1 + m_2^2) + (1 + m_1^2) = m_2^2 - 2m_1m_2 + m_1^2$$

which, upon simplification, can be written as

$$m_1m_2 = -1$$

If the lines are perpendicular, the product of their slopes is -1. ■

In Problem 138, you are asked to prove the "if" part of the theorem:

> If two nonvertical lines have slopes whose product is -1, then the lines are perpendicular.

You may find it easier to remember the condition for two nonvertical lines to be perpendicular by observing that the equality $m_1m_2 = -1$ means that m_1 and m_2 are negative reciprocals of each other; that is,

$$m_1 = -\frac{1}{m_2} \quad \text{and} \quad m_2 = -\frac{1}{m_1}$$

EXAMPLE 12 Finding the Slope of a Line Perpendicular to Another Line

If a line has slope $\frac{3}{2}$, any line having slope $-\frac{2}{3}$ is perpendicular to it.

EXAMPLE 13 Finding an Equation of a Line Perpendicular to a Given Line

Find an equation of the line that contains the point $(1, -2)$ and is perpendicular to the line $x + 3y = 6$. Graph the two lines.

Solution First write the equation of the line $x + 3y = 6$ in slope-intercept form to find its slope.

$$x + 3y = 6$$

$$3y = -x + 6 \qquad \text{Proceed to solve for } y.$$

$$y = -\frac{1}{3}x + 2 \qquad \text{Place in the form } y = mx + b.$$

The slope of the line $x + 3y = 6$ is $-\frac{1}{3}$. Any line perpendicular to this line has slope 3.

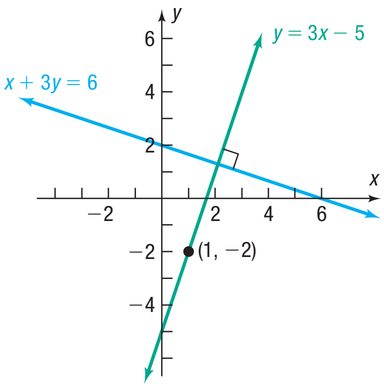


Figure 48

Because the point $(1, -2)$ is on the line with slope 3, use the point-slope form of the equation of a line.

$$y - y_1 = m(x - x_1) \qquad \text{Point-slope form}$$

$$y - (-2) = 3(x - 1) \qquad m = 3, x_1 = 1, y_1 = -2$$

$$y + 2 = 3(x - 1)$$

To obtain other forms of the equation, proceed as follows:

$$y + 2 = 3x - 3 \qquad \text{Simplify.}$$

$$y = 3x - 5 \qquad \text{Slope-intercept form}$$

$$3x - y = 5 \qquad \text{General form}$$

Figure 48 shows the graphs.

Now Work PROBLEM 73

WARNING Be sure to use a square screen when you use a graphing calculator to graph perpendicular lines. Otherwise, the angle between the two lines will appear distorted. A discussion of square screens is given in Section B.5.

1.3 Assess Your Understanding

Concepts and Vocabulary

1. The slope of a vertical line is ____________; the slope of a horizontal line is _____.

2. For the line $2x + 3y = 6$, the x-intercept is _____ and the y-intercept is _____.

3. ***True or False*** The equation $3x + 4y = 6$ is written in general form.

4. ***True or False*** The slope of the line $2y = 3x + 5$ is 3.

5. ***True or False*** The point $(1, 2)$ is on the line $2x + y = 4$.

6. Two nonvertical lines have slopes m_1 and m_2, respectively. The lines are parallel if __________ and the ____________ are unequal; the lines are perpendicular if ____________.

7. The lines $y = 2x + 3$ and $y = ax + 5$ are parallel if $a =$ _____.

8. The lines $y = 2x - 1$ and $y = ax + 2$ are perpendicular if $a =$ _______.

9. ***Multiple Choice*** If a line slants downward from left to right, then which of the following describes its slope?
(a) positive **(b)** zero
(c) negative **(d)** undefined

10. ***Multiple Choice*** Choose the formula for finding the slope m of a nonvertical line that contains the two distinct points (x_1, y_1) and (x_2, y_2).

(a) $m = \dfrac{y_2 - x_2}{y_1 - x_1} \quad x_1 \neq y_1$ **(b)** $m = \dfrac{y_2 - x_1}{x_2 - y_1} \quad y_1 \neq x_2$

(c) $m = \dfrac{x_2 - x_1}{y_2 - y_1} \quad y_1 \neq y_2$ **(d)** $m = \dfrac{y_2 - y_1}{x_2 - x_1} \quad x_1 \neq x_2$

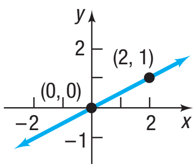
11. ***Multiple Choice*** Choose the correct statement about the graph of the line $y = -3$.
(a) The graph is vertical with x-intercept -3.
(b) The graph is horizontal with y-intercept -3.
(c) The graph is vertical with y-intercept -3.
(d) The graph is horizontal with x-intercept -3.

12. ***Multiple Choice*** Choose the point-slope equation of a nonvertical line with slope m that passes through the point (x_1, y_1).
(a) $y + y_1 = mx + x_1$ **(b)** $y - y_1 = mx - x_1$
(c) $y + y_1 = m(x + x_1)$ **(d)** $y - y_1 = m(x - x_1)$

Skill Building

In Problems 13–16, (a) find the slope of the line and (b) interpret the slope.

13.



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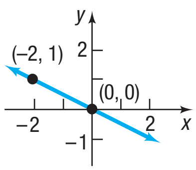

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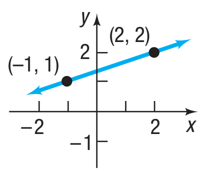

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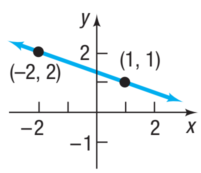


In Problems 17–24, plot each pair of points and determine the slope of the line containing the points. Graph the line.

17. $(4, 2); (3, 4)$ **18.** $(2, 3); (4, 0)$ **19.** $(-2, 3); (2, 1)$ **20.** $(-1, 1); (2, 3)$
21. $(4, 2); (-5, 2)$ **22.** $(-3, -1); (2, -1)$ **23.** $(2, 0); (2, 2)$ **24.** $(-1, 2); (-1, -2)$

In Problems 25–32, graph the line that contains the point P and has slope m.

25. $P = (1, 2); m = 3$ **26.** $P = (2, 1); m = 4$ **27.** $P = (1, 3); m = -\frac{2}{5}$ **28.** $P = (2, 4); m = -\frac{3}{4}$
29. $P = (2, -4); m = 0$ **30.** $P = (-1, 3); m = 0$ **31.** $P = (-2, 0)$; slope undefined **32.** $P = (0, 3)$; slope undefined

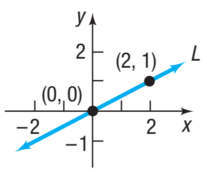
In Problems 33–38, a point on a line and its slope are given. Find the point-slope form of the equation of the line.

33. $P = (1, 2); m = 3$ **34.** $P = (2, 1); m = 4$ **35.** $P = (1, 3); m = -\frac{2}{5}$
36. $P = (2, 4); m = -\frac{3}{4}$ **37.** $P = (2, -4); m = 0$ **38.** $P = (-1, 3); m = 0$

In Problems 39–44, the slope and a point on a line are given. Use this information to locate three additional points on the line. Answers may vary. [**Hint**: *It is not necessary to find the equation of the line. See Example 3.*]

39. Slope 2; point $(-2, 3)$ **40.** Slope 4; point $(1, 2)$ **41.** Slope $\frac{4}{3}$; point $(-3, 2)$
42. Slope $-\frac{3}{2}$; point $(2, -4)$ **43.** Slope -1; point $(4, 1)$ **44.** Slope -2; point $(-2, -3)$

In Problems 45–52, find an equation of the line L.

45.


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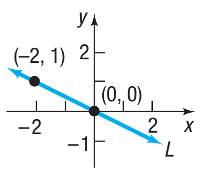

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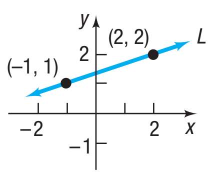

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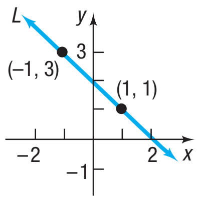

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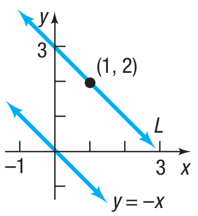
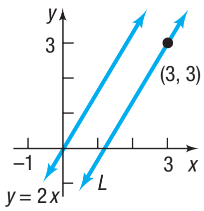
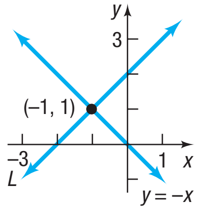


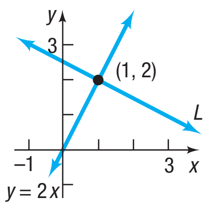
L is parallel to $y = -x$

50.


L is parallel to $y = 2x$

51.


L is perpendicular to $y = -x$

52.


L is perpendicular to $y = 2x$

In Problems 53–78, find an equation for the line with the given properties. Express your answer using either the general form or the slope-intercept form of the equation of a line, whichever you prefer.

53. Slope $= 3$; containing the point $(-2, 3)$
54. Slope $= 2$; containing the point $(4, -3)$
55. Slope $= -\frac{2}{3}$; containing the point $(1, -1)$
56. Slope $= \frac{1}{2}$; containing the point $(3, 1)$
57. Containing the points $(-3, 4)$ and $(2, 5)$
58. Containing the points $(1, 3)$ and $(-1, 2)$
59. Slope $= -3$; y-intercept $= 3$
60. Slope $= -2$; y-intercept $= -2$
61. x-intercept $= 2$; y-intercept $= -1$
62. x-intercept $= -4$; y-intercept $= 4$
63. Slope undefined; containing the point $(3, 8)$
64. Slope undefined; containing the point $(2, 4)$
65. Vertical; containing the point $(4, -5)$
66. Horizontal; containing the point $(-3, 2)$
67. Parallel to the line $y = 2x$; containing the point $(-1, 2)$
68. Parallel to the line $y = -3x$; containing the point $(-1, 2)$
69. Parallel to the line $2x - y = -2$; containing the point $(0, 0)$
70. Parallel to the line $x - 2y = -5$; containing the point $(0, 0)$
71. Parallel to the line $y = 5$; containing the point $(4, 2)$
72. Parallel to the line $x = 5$; containing the point $(4, 2)$
73. Perpendicular to the line $y = \frac{1}{2}x + 4$; containing the point $(1, -2)$
74. Perpendicular to the line $y = 2x - 3$; containing the point $(1, -2)$

75. Perpendicular to the line $2x + y = 2$; containing the point $(-3, 0)$

76. Perpendicular to the line $x - 2y = -5$; containing the point $(0, 4)$

77. Perpendicular to the line $y = 8$; containing the point $(3, 4)$

78. Perpendicular to the line $x = 8$; containing the point $(3, 4)$

In Problems 79–98, find the slope and y-intercept of each line. Graph the line.

79. $y = 2x + 3$ **80.** $y = -3x + 4$ **81.** $\frac{1}{3}x + y = 2$ **82.** $\frac{1}{2}y = x - 1$ **83.** $y = 2x + \frac{1}{2}$

84. $y = \frac{1}{2}x + 2$ **85.** $x + 2y = 4$ **86.** $-x + 3y = 6$ **87.** $3x + 2y = 6$ **88.** $2x - 3y = 6$

89. $x - y = 2$ **90.** $x + y = 1$ **91.** $y = -1$ **92.** $x = -4$ **93.** $x = 2$

94. $y = 5$ **95.** $x + y = 0$ **96.** $y - x = 0$ **97.** $3x + 2y = 0$ **98.** $2y - 3x = 0$

In Problems 99–108, (a) find the intercepts of the graph of each equation and (b) graph the equation.

99. $2x + 3y = 6$ **100.** $3x - 2y = 6$ **101.** $6x - 4y = 24$ **102.** $-4x + 5y = 40$ **103.** $5x + 3y = 18$

104. $7x + 2y = 21$ **105.** $x - \frac{2}{3}y = 4$ **106.** $\frac{1}{2}x + \frac{1}{3}y = 1$ **107.** $-0.3x + 0.4y = 1.2$ **108.** $0.2x - 0.5y = 1$

109. Find an equation of the y-axis.

110. Find an equation of the x-axis.

In Problems 111–114, the equations of two lines are given. Determine whether the lines are parallel, perpendicular, or neither.

111. $y = \frac{1}{2}x - 3$
$y = -2x + 4$

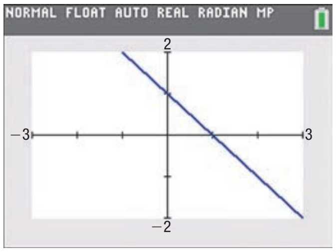
112. $y = 2x - 3$
$y = 2x + 4$

113. $y = -2x + 3$
$y = -\frac{1}{2}x + 2$

114. $y = 4x + 5$
$y = -4x + 2$

In Problems 115–118, write an equation of each line. Express your answer using either the general form or the slope-intercept form of the equation of a line, whichever you prefer.

115.



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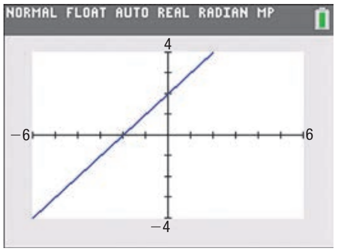

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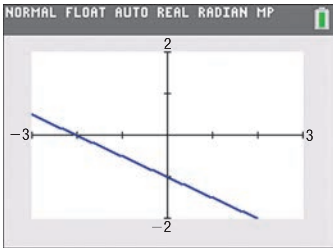
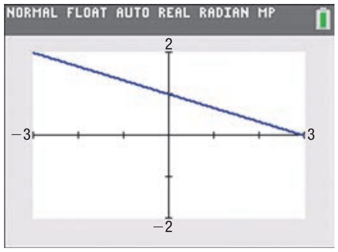


118.



Applications and Extensions

119. Geometry Use slopes to show that the quadrilateral whose vertices are $(1, -1)$, $(4, 1)$, $(2, 2)$, and $(5, 4)$ is a parallelogram.

120. Geometry Use slopes to determine if the triangle whose vertices are $(-4, 4)$, $(1, 5)$, and $(2, 0)$ is a right triangle.

121. Geometry Use slopes and the distance formula to show that the quadrilateral whose vertices are $(0, 0)$, $(1, 3)$, $(4, 2)$, and $(3, -1)$ is a square.

122. Geometry Use slopes to determine if the quadrilateral whose vertices are $(-2, 1)$, $(-3, 2)$, $(-4, -1)$ and $(-5, 0)$ is a rectangle.

123. Truck Rentals A truck rental company rents a moving truck for one day by charging \$33 plus \$0.09 per mile. Write a linear equation that relates the cost C, in dollars, of renting the truck to the number x of miles driven. What is the cost of renting the truck if the truck is driven 175 miles? 403 miles?

124. Cost Equation The **fixed costs** of operating a business are the costs incurred regardless of the level of production. Fixed costs include rent, fixed salaries, and costs of leasing machinery. The **variable costs** of operating a business are the costs that change with the level of output. Variable costs include raw materials, hourly wages, and electricity. Suppose that a manufacturer of jeans has fixed daily costs of \$1200 and variable costs of \$20 for each pair of jeans manufactured. Write a linear equation that relates the daily cost C, in dollars, of manufacturing the jeans to the number x of jeans manufactured. What is the cost of manufacturing 400 pairs of jeans? 740 pairs?

125. Cost of Driving a Car The annual fixed costs of owning a small sedan are \$4436, assuming the car is completely paid for. The cost to drive the car is approximately \$0.16 per mile. Write a linear equation that relates the cost C and the number x of miles driven annually.

126. Wages of a Car Salesperson Dan receives \$525 per week for selling new and used cars. He also receives 5% of the profit on any sales. Write a linear equation that represents Dan's weekly salary S when he has sales that generate a profit of x dollars.

127. Electricity Rates in Florida Florida Power & Light Company supplies electricity to residential customers for a monthly customer charge of \$8.01 plus 8.89 cents per kilowatt hour (kWh) for up to 1000 kilowatt hours.

(a) Write a linear equation that relates the monthly charge C, in dollars, to the number x of kilowatt hours used in a month, $0 \le x \le 1000$.

(b) Graph this equation.

(c) What is the monthly charge for using 200 kilowatt hours?

(d) What is the monthly charge for using 500 kilowatt hours?

(e) Interpret the slope of the line.

***Source**: Florida Power & Light Company, March 2018*

128. Natural Gas Rates in Illinois Ameren Illinois supplies natural gas to residential customers for a monthly customer charge of \$21.82 plus 64.9 cents per therm of heat energy.

(a) Write a linear equation that relates the monthly charge C, in dollars, to the number x of therms used in a month.

(b) What is the monthly charge for using 90 therms?

(c) What is the monthly charge for using 150 therms?

(d) Graph the equation.

(e) Interpret the slope of the line.

***Source**: Ameren Illinois, March 2018*

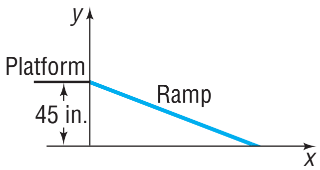
129. Measuring Temperature The relationship between Celsius (°C) and Fahrenheit (°F) degrees of measuring temperature is linear. Find a linear equation relating °C and °F if 0°C corresponds to 32°F and 100°C corresponds to 212°F. Use the equation to find the Celsius measure of 70°F.

130. Measuring Temperature The Kelvin (K) scale for measuring temperature is obtained by adding 273 to the Celsius temperature.

(a) Write a linear equation relating K and °C.

(b) Write a linear equation relating K and °F (see Problem 129).

131. Access Ramp A wooden access ramp is being built to reach a platform that sits 45 inches above the floor. The ramp drops 3 inches for every 29-inch run.



(a) Write a linear equation that relates the height y of the ramp above the floor to the horizontal distance x from the platform.

(b) Find and interpret the x-intercept of the graph of your equation.

(c) Design requirements stipulate that the maximum run be 30 feet (360 inches) and that the maximum slope be a drop of 1 inch for each 8 inches of run. Will this ramp meet the requirements? Explain.

(d) What slopes could be used to obtain the 45-inch rise and still meet design requirements?

***Source**: www.adaptiveaccess.com/wood_ramps.php*

132. U.S. Advertising Share A report showed that Internet ads accounted for 19% of all U.S. advertisement spending when print ads (magazines and newspapers) accounted for 26% of the spending. The report further showed that Internet ads accounted for 35% of all advertisement spending when print ads accounted for 16% of the spending.

(a) Write a linear equation that relates that percent y of print ad spending to the percent x of Internet ad spending.

(b) Find the intercepts of the graph of your equation.

(c) Do the intercepts have any meaningful interpretation?

(d) Predict the percent of print ad spending if Internet ads account for 39% of all advertisement spending in the United States.

***Source**: Marketing Fact Pack 2018. Ad Age Datacenter, December 18, 2017*

133. Product Promotion A cereal company finds that the number of people who will buy one of its products in the first month that it is introduced is linearly related to the amount of money it spends on advertising. If it spends \$10,000 on advertising, then 100,000 boxes of cereal will be sold, and if it spends \$50,000 on advertising, then 200,000 boxes of cereal will be sold.

(a) Write an equation that relates the amount A spent on advertising to the number x of boxes the company aims to sell.

(b) How much advertising is needed to sell 700,000 boxes of cereal?

(c) Interpret the slope.

134. The equation $2x - y = C$ defines a **family of lines**, one line for each value of C. On one set of coordinate axes, graph the members of the family when $C = -4$, $C = 0$, and $C = 2$. Can you draw a conclusion from the graph about each member of the family?

135. *Challenge Problem* Show that the line containing the points (a, b) and (b, a), $a \neq b$, is perpendicular to the line $y = x$. Also show that the midpoint of (a, b) and (b, a) lies on the line $y = x$.

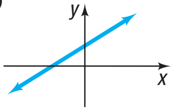
136. *Challenge Problem* Find three numbers a for which the lines $x + 2y = 5$, $2x - 3y + 4 = 0$, and $ax + y = 0$ do not form a triangle.

137. *Challenge Problem* Form a triangle using the points $(0, 0)$, $(a, 0)$, and (b, c), where $a > 0$, $b > 0$, and $c > 0$. Find the point of intersection of the three lines joining the midpoint of a side of the triangle to the opposite vertex.

138. *Challenge Problem* Prove that if two nonvertical lines have slopes whose product is -1, then the lines are perpendicular. [**Hint**: Refer to Figure 47 and use the converse of the Pythagorean Theorem.]

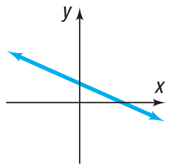
Explaining Concepts: Discussion and Writing

139. Which of the following equations might have the graph shown? (More than one answer is possible.)



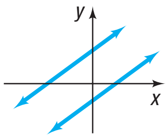
(a) $2x + 3y = 6$ **(b)** $-2x + 3y = 6$
(c) $3x - 4y = -12$ **(d)** $x - y = 1$
(e) $x - y = -1$ **(f)** $y = 3x - 5$
(g) $y = 2x + 3$ **(h)** $y = -3x + 3$

140. Which of the following equations might have the graph shown? (More than one answer is possible.)



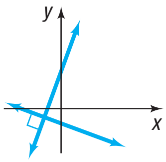
(a) $2x + 3y = 6$ **(b)** $2x - 3y = 6$
(c) $3x + 4y = 12$ **(d)** $x - y = 1$
(e) $x - y = -1$ **(f)** $y = -2x - 1$
(g) $y = -\frac{1}{2}x + 10$ **(h)** $y = x + 4$

141. The figure shows the graph of two parallel lines. Which of the following pairs of equations might have such a graph?



(a) $x - 2y = 3$
$x + 2y = 7$
(b) $x + y = 2$
$x + y = -1$
(c) $x - y = -2$
$x - y = 1$
(d) $x - y = -2$
$2x - 2y = -4$
(e) $x + 2y = 2$
$x + 2y = -1$

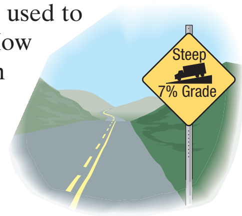
142. The figure shows the graph of two perpendicular lines. Which of the following pairs of equations might have such a graph?



(a) $y - 2x = 2$
$y + 2x = -1$
(b) $y - 2x = 0$
$2y + x = 0$
(c) $2y - x = 2$
$2y + x = -2$
(d) $y - 2x = 2$
$x + 2y = -1$
(e) $2x + y = -2$
$2y + x = -2$

143. ***m* is for Slope** The accepted symbol used to denote the slope of a line is the letter m. Investigate the origin of this practice. Begin by consulting a French dictionary and looking up the French word *monter*. Write a brief essay on your findings.

144. **Grade of a Road** The term *grade* is used to describe the inclination of a road. How is this term related to the notion of slope of a line? Is a 4% grade very steep? Investigate the grades of some mountainous roads and determine their slopes. Write a brief essay on your findings.



145. **Carpentry** Carpenters use the term *pitch* to describe the steepness of staircases and roofs. How is pitch related to slope? Investigate typical pitches used for stairs and for roofs. Write a brief essay on your findings.

146. Can the equation of every line be written in slope-intercept form? Why?

147. Does every line have exactly one x-intercept and one y-intercept? Are there any lines that have no intercepts?

148. What can you say about two lines that have equal slopes and equal y-intercepts?

149. What can you say about two lines with the same x-intercept and the same y-intercept? Assume that the x-intercept is not 0.

150. If two distinct lines have the same slope but different x-intercepts, can they have the same y-intercept?

151. If two distinct lines have the same y-intercept but different slopes, can they have the same x-intercept?

152. Which form of the equation of a line do you prefer to use? Justify your position with an example that shows that your choice is better than another. Have reasons.

153. **What Went Wrong?** A student is asked to find the slope of the line joining $(-3, 2)$ and $(1, -4)$. He states that the slope is $\frac{3}{2}$. Is he correct? If not, what went wrong?

1.4 Circles

PREPARING FOR THIS SECTION *Before getting started, review the following:*

- Completing the Square (Section A.3, p. A29)
- Square Root Method (Section A.6, p. A48)

Now Work the **'Are You Prepared?'** problems on page 75.

OBJECTIVES
1 Write the Standard Form of the Equation of a Circle (p. 72)
2 Graph a Circle (p. 73)
3 Work with the General Form of the Equation of a Circle (p. 74)

1 Write the Standard Form of the Equation of a Circle

One advantage of a coordinate system is that it enables us to translate a geometric statement into an algebraic statement, and vice versa. Consider, for example, the following geometric statement that defines a circle.

DEFINITION Circle

A **circle** is a set of points in the xy-plane that are a fixed distance r from a fixed point (h, k). The fixed distance r is called the **radius**, and the fixed point (h, k) is called the **center** of the circle.

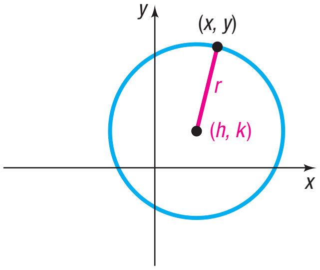


Figure 49
$(x - h)^2 + (y - k)^2 = r^2$

Need to Review?
The distance formula is discussed in Section 1.1, pp. 39–40.

Figure 49 shows the graph of a circle. To find the equation, let (x, y) represent the coordinates of any point on a circle with radius r and center (h, k). Then the distance between the points (x, y) and (h, k) must always equal r. That is, by the distance formula,

$$\sqrt{(x-h)^2 + (y-k)^2} = r$$

or, equivalently,

$$(x-h)^2 + (y-k)^2 = r^2$$

DEFINITION Standard Form of an Equation of a Circle

The **standard form of an equation of a circle** with radius r and center (h, k) is

$$(x-h)^2 + (y-k)^2 = r^2 \qquad \textbf{(1)}$$

THEOREM

The standard form of an equation of a circle of radius r with center at the origin $(0, 0)$ is

$$x^2 + y^2 = r^2$$

DEFINITION Unit Circle

If the radius $r = 1$, the circle whose center is at the origin is called the **unit circle** and has the equation

$$x^2 + y^2 = 1$$

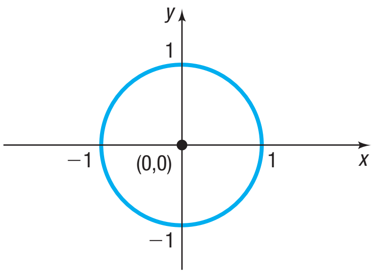


Figure 50
Unit circle $x^2 + y^2 = 1$

See Figure 50. Notice that the graph of the unit circle is symmetric with respect to the x-axis, the y-axis, and the origin.

EXAMPLE 1 Writing the Standard Form of the Equation of a Circle

Write the standard form of the equation of the circle with radius 5 and center $(-3, 6)$.

Solution Substitute the values $r = 5$, $h = -3$, and $k = 6$ into equation (1).

$$(x-h)^2 + (y-k)^2 = r^2 \qquad \text{Equation (1)}$$
$$[x - (-3)]^2 + (y-6)^2 = 5^2$$
$$(x+3)^2 + (y-6)^2 = 25$$

Now Work PROBLEM 9

2 Graph a Circle

EXAMPLE 2 **Graphing a Circle**

Graph the equation: $(x+3)^2 + (y-2)^2 = 16$

Solution The given equation is in the standard form of an equation of a circle. To graph the circle, compare the equation to equation (1). The comparison gives information about the circle.

$$\begin{aligned}(x+3)^2 + (y-2)^2 &= 16\\ (x-(-3))^2 + (y-2)^2 &= 4^2\\ (x-h)^2 + (y-k)^2 &= r^2\end{aligned}$$

We see that $h = -3$, $k = 2$, and $r = 4$. The circle has center $(-3, 2)$ and radius 4. To graph this circle, first plot the center $(-3, 2)$. Since the radius is 4, locate four points on the circle by plotting points 4 units to the left, to the right, up, and down from the center. These four points are then used as guides to obtain the graph. See Figure 51.

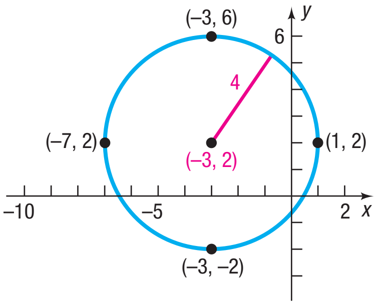


Figure 51 $(x+3)^2 + (y-2)^2 = 16$

Now Work PROBLEMS 27(a) AND (b)

EXAMPLE 3 **Finding the Intercepts of a Circle**

Find the intercepts, if any, of the graph of the circle $(x+3)^2 + (y-2)^2 = 16$.

Solution This is the equation graphed in Example 2. To find the x-intercepts, if any, let $y = 0$. Then

> **In Words**
> The symbol $\pm$ is read "plus or minus." It means to add and subtract the quantity following the $\pm$ symbol. For example, 5 ± 2 means "$5 - 2 = 3$ or $5 + 2 = 7$."

$$\begin{aligned}(x+3)^2 + (y-2)^2 &= 16\\ (x+3)^2 + (0-2)^2 &= 16 && y = 0\\ (x+3)^2 + 4 &= 16 && \text{Simplify.}\\ (x+3)^2 &= 12 && \text{Simplify.}\\ x+3 &= \pm\sqrt{12} && \text{Use the Square Root Method.}\\ x &= -3 \pm 2\sqrt{3} && \text{Solve for } x.\end{aligned}$$

The x-intercepts are $-3 - 2\sqrt{3} \approx -6.46$ and $-3 + 2\sqrt{3} \approx 0.46$.

To find the y-intercepts, if any, let $x = 0$. Then

$$\begin{aligned}(x+3)^2 + (y-2)^2 &= 16\\ (0+3)^2 + (y-2)^2 &= 16\\ 9 + (y-2)^2 &= 16\\ (y-2)^2 &= 7\\ y-2 &= \pm\sqrt{7}\\ y &= 2 \pm \sqrt{7}\end{aligned}$$

The y-intercepts are $2 - \sqrt{7} \approx -0.65$ and $2 + \sqrt{7} \approx 4.65$.

Look back at Figure 51 to verify the approximate locations of the intercepts.

Now Work PROBLEM 27(c)

3 Work with the General Form of the Equation of a Circle

If we eliminate the parentheses from the standard form of the equation of the circle given in Example 2, we get

$$(x + 3)^2 + (y - 2)^2 = 16$$
$$x^2 + 6x + 9 + y^2 - 4y + 4 = 16$$

which simplifies to

$$x^2 + y^2 + 6x - 4y - 3 = 0$$

It can be shown that any equation of the form

$$x^2 + y^2 + ax + by + c = 0$$

has a graph that is a circle, is a point, or has no graph at all. For example, the graph of the equation $x^2 + y^2 = 0$ is the single point $(0, 0)$. The equation $x^2 + y^2 + 5 = 0$, or $x^2 + y^2 = -5$, has no graph, because sums of squares of real numbers are never negative.

DEFINITION General Form of the Equation of a Circle

When its graph is a circle, the equation

$$\boxed{x^2 + y^2 + ax + by + c = 0}$$

is the **general form of the equation of a circle**.

Now Work PROBLEM 15

Need to Review?
Completing the square is discussed in Section A.3, p. A29.

If an equation of a circle is in general form, we use the method of completing the square to put the equation in standard form so that we can identify its center and radius.

EXAMPLE 4 Graphing a Circle Whose Equation Is in General Form

Graph the equation: $x^2 + y^2 + 4x - 6y + 12 = 0$

Solution Group the terms involving x, group the terms involving y, and put the constant on the right side of the equation. The result is

$$(x^2 + 4x) + (y^2 - 6y) = -12$$

Next, complete the square of each expression in parentheses. Remember that any number added on the left side of the equation must also be added on the right.

$$(x^2 + 4x + \underbrace{4}_{\left(\frac{4}{2}\right)^2 = 4}) + (y^2 - 6y + \underbrace{9}_{\left(\frac{-6}{2}\right)^2 = 9}) = -12 + 4 + 9$$

$$(x + 2)^2 + (y - 3)^2 = 1 \quad \text{Factor.}$$

This equation is the standard form of the equation of a circle with radius 1 and center $(-2, 3)$. To graph the equation, use the center $(-2, 3)$ and the radius 1. See Figure 52.

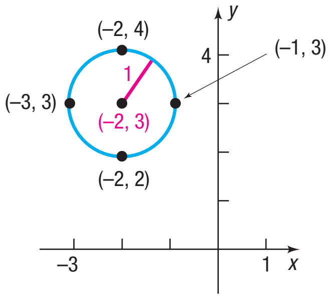


Figure 52 $(x + 2)^2 + (y - 3)^2 = 1$

Now Work PROBLEM 31

EXAMPLE 5 Using a Graphing Utility to Graph a Circle

Graph the equation: $x^2 + y^2 = 4$

Solution This is the equation of a circle with center at the origin and radius 2. To graph this equation using a graphing utility, begin by solving for y.

$$x^2 + y^2 = 4$$

$$y^2 = 4 - x^2 \qquad \text{Subtract } x^2 \text{ from both sides.}$$

$$y = \pm\sqrt{4 - x^2} \qquad \text{Use the Square Root Method to solve for } y.$$

There are two equations to graph: first graph $Y_1 = \sqrt{4 - x^2}$ and then graph $Y_2 = -\sqrt{4 - x^2}$ on the same square screen. (Your circle will appear oval if you do not use a square screen.*) See Figure 53.

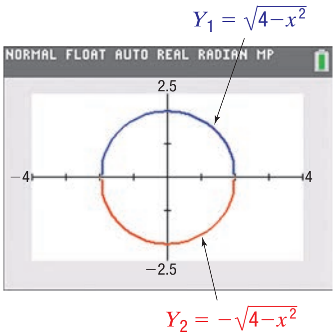


Figure 53 $x^2 + y^2 = 4$

Overview

The discussion in Sections 1.3 and 1.4 about lines and circles deals with two problems that can be generalized as follows:

- Given an equation, classify it and graph it.
- Given a graph, or information about a graph, find its equation.

This text deals with both problems. We shall study various equations, classify them, and graph them.

*The square screen ratio for the TI-84 Plus C graphing calculator is 8:5.

1.4 Assess Your Understanding

'Are You Prepared?' *Answers are given at the end of these exercises. If you get a wrong answer, read the pages listed in red.*

1. To complete the square of $x^2 + 10x$, you would _______ (*add/subtract*) the number ______. (p. A29)

2. Use the Square Root Method to solve the equation $(x - 2)^2 = 9$. (p. A48)

Concepts and Vocabulary

3. ***True or False*** Every equation of the form

$$x^2 + y^2 + ax + by + c = 0$$

has a circle as its graph.

4. For a circle, the __________ is the distance from the center to any point on the circle.

5. ***True or False*** The radius of the circle $x^2 + y^2 = 9$ is 3.

6. ***True or False*** The center of the circle

$$(x + 3)^2 + (y - 2)^2 = 13$$

is $(3, -2)$.

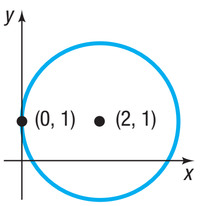
7. ***Multiple Choice*** Choose the equation of a circle with radius 6 and center $(3, -5)$.
(a) $(x - 3)^2 + (y + 5)^2 = 6$
(b) $(x + 3)^2 + (y - 5)^2 = 36$
(c) $(x + 3)^2 + (y - 5)^2 = 6$
(d) $(x - 3)^2 + (y + 5)^2 = 36$

8. ***Multiple Choice*** The equation of a circle can be changed from general form to standard from by doing which of the following?
(a) completing the squares **(b)** solving for x
(c) solving for y **(d)** squaring both sides

Skill Building

In Problems 9–12, find the center and radius of each circle. Write the standard form of the equation.

9.



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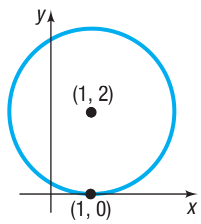

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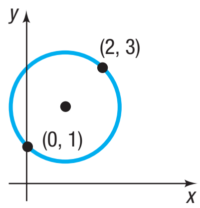

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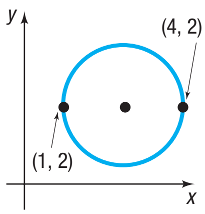


In Problems 13–24, write the standard form of the equation and the general form of the equation of each circle of radius r and center (h, k). Graph each circle.

13. $r = 3;\ (h, k) = (0, 0)$ **14.** $r = 2;\ (h, k) = (0, 0)$ **15.** $r = 2;\ (h, k) = (0, 2)$ **16.** $r = 3;\ (h, k) = (1, 0)$

17. $r = 4;\ (h, k) = (2, -3)$ **18.** $r = 5;\ (h, k) = (4, -3)$ **19.** $r = 7;\ (h, k) = (-5, -2)$ **20.** $r = 4;\ (h, k) = (-2, 1)$

21. $r = \frac{1}{2};\ (h, k) = \left(0, -\frac{1}{2}\right)$ **22.** $r = \frac{1}{2};\ (h, k) = \left(\frac{1}{2}, 0\right)$ **23.** $r = 2\sqrt{5};\ (h, k) = (-3, 2)$ **24.** $r = \sqrt{13};\ (h, k) = (5, -1)$

In Problems 25–38, (a) find the center (h, k) and radius r of each circle; (b) graph each circle; (c) find the intercepts, if any.

25. $x^2 + (y-1)^2 = 1$ **26.** $x^2 + y^2 = 4$ **27.** $2(x-3)^2 + 2y^2 = 8$

28. $3(x+1)^2 + 3(y-1)^2 = 6$ **29.** $x^2 + y^2 + 4x + 2y - 20 = 0$ **30.** $x^2 + y^2 - 2x - 4y - 4 = 0$

31. $x^2 + y^2 + 4x - 4y - 1 = 0$ **32.** $x^2 + y^2 - 6x + 2y + 9 = 0$ **33.** $x^2 + y^2 + x + y - \frac{1}{2} = 0$

34. $x^2 + y^2 - x + 2y + 1 = 0$ **35.** $2x^2 + 2y^2 - 12x + 8y - 24 = 0$ **36.** $2x^2 + 2y^2 + 8x + 7 = 0$

37. $3x^2 + 3y^2 - 12y = 0$ **38.** $2x^2 + 8x + 2y^2 = 0$

In Problems 39–44, find the standard form of the equation of each circle.

39. Center $(1, 0)$ and containing the point $(-3, 2)$

40. Center at the origin and containing the point $(-2, 3)$

41. With endpoints of a diameter at $(4, 3)$ and $(0, 1)$

42. With endpoints of a diameter at $(1, 4)$ and $(-3, 2)$

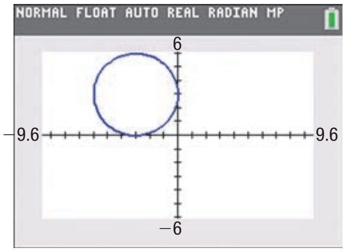
43. Center $(-5, 6)$ and area 49π

44. Center $(2, -4)$ and circumference 16π

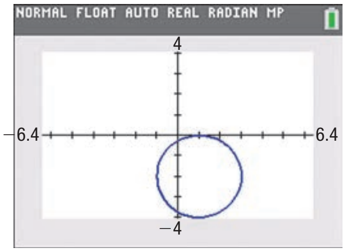
In Problems 45–48, match each graph with the correct equation.

(a) $(x-3)^2 + (y+3)^2 = 9$ **(b)** $(x+1)^2 + (y-2)^2 = 4$ **(c)** $(x-1)^2 + (y+2)^2 = 4$ **(d)** $(x+3)^2 + (y-3)^2 = 9$

45.



46.



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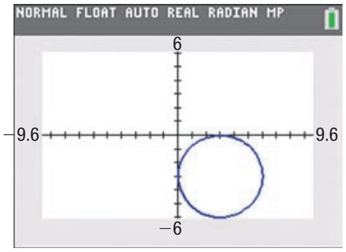
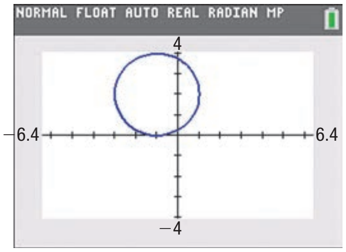


48.



Applications and Extensions

49. Find an equation of the line containing the centers of the two circles

$$(x-4)^2 + (y+2)^2 = 25$$

and

$$(x+1)^2 + (y-5)^2 = 16$$

50. Find an equation of the line containing the centers of the two circles

$$x^2 + y^2 - 4x + 6y + 4 = 0$$

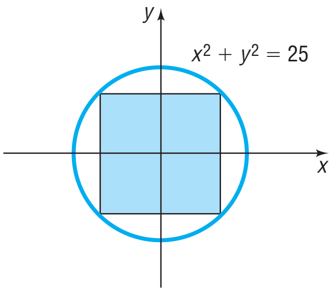
and

$$x^2 + y^2 + 6x + 4y + 9 = 0$$

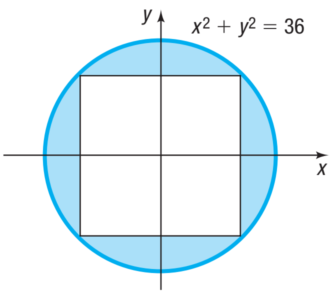
51. If a circle of radius 2 rolls along the x-axis in quadrants I and II, what is an equation for the path of the center of the circle?

52. A circle of radius 7 rolls along the y-axis in quadrants I and IV. Find an equation for the path of the center of the circle.

53. Find the area of the square in the figure.



54. Find the area of the blue shaded region in the figure, assuming the quadrilateral inside the circle is a square.



55. Ferris Wheel The High Roller observation wheel in Las Vegas has a maximum height of 550 feet and a diameter of 520 feet, with one full rotation taking approximately 30 minutes. Find an equation for the wheel if the center of the wheel is on the y-axis.

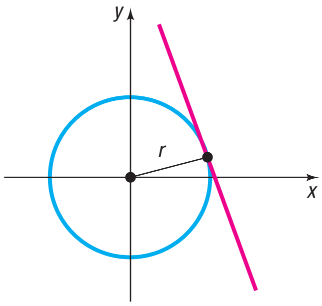
Source: *Las Vegas Review Journal*

56. Ferris Wheel A Ferris wheel has a maximum height of 270 feet and a wheel diameter of 260 feet. Find an equation for the wheel if the center of the wheel is on the y-axis and y represents the height above the ground.

57. Vertically Circular Building The Sunrise Kempinski Hotel in Beijing, China, is a vertically circular building whose outline is described by the equation $x^2 + y^2 - 78y - 1843 = 0$ if the center of the building is on the y-axis. If x and y are in meters, what is the height of the building?

Problems 59–66 require the following discussion.

The **tangent line** to a circle may be defined as the line that intersects the circle in a single point, called the **point of tangency**. See the figure.



58. Vertically Circular Building Located in Al Raha, Abu Dhabi, the headquarters of property developing company Aldar is a vertically circular building with a diameter of 121 meters. The tip of the building is 110 meters aboveground. Find an equation for the building's outline if the center of the building is on the y-axis.

In Problems 59–62, find the standard form of the equation of each circle. (Refer to the preceding discussion).

59. Center $(-3, 1)$ and tangent to the y-axis

60. Center $(2, 3)$ and tangent to the x-axis

61. Center $(4, -2)$ and tangent to the line $x = 1$

62. Center $(-1, 3)$ and tangent to the line $y = 2$

63. ***Challenge Problem*** If the equation of a circle is $x^2 + y^2 = r^2$ and the equation of a tangent line is $y = mx + b$, show that:

(a) $r^2(1 + m^2) = b^2$

[**Hint**: The quadratic equation $x^2 + (mx + b)^2 = r^2$ has exactly one solution.]

(b) The point of tangency is $\left(\dfrac{-r^2 m}{b}, \dfrac{r^2}{b}\right)$.

(c) The tangent line is perpendicular to the line containing the center of the circle and the point of tangency.

64. The line $x - 2y + 16 = 0$ is tangent to a circle at $(0, 8)$. The line $y = 2x - 1$ is tangent to the same circle at $(3, 5)$. Find the center of the circle.

65. ***Challenge Problem*** **The Greek Method** The Greek method for finding the equation of the tangent line to a circle uses the fact that at any point on a circle, the line containing the center and the tangent line are perpendicular. Use this method to find an equation of the tangent line to the circle $x^2 + y^2 = 9$ at the point $(1, 2\sqrt{2})$.

66. ***Challenge Problem*** Use the Greek method described in Problem 65 to find an equation of the tangent line to the circle

$$x^2 + y^2 - 4x + 6y + 4 = 0$$

at the point $(3, 2\sqrt{2} - 3)$.

67. ***Challenge Problem*** If $x^2 + y^2 + dx + ey + f = 0$ is the equation of a circle, show that

$$x_0 x + y_0 y + d\left(\frac{x + x_0}{2}\right) + e\left(\frac{y + y_0}{2}\right) + f = 0$$

is an equation of the tangent line to the circle at the point (x_0, y_0).

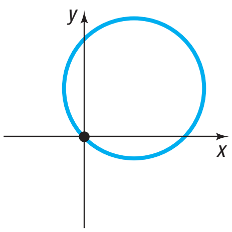
68. ***Challenge Problem*** If (x_1, y_1) and (x_2, y_2) are the endpoints of a diameter of a circle, show that an equation of the circle is

$$(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$$

Explaining Concepts: Discussion and Writing

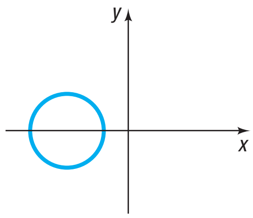
69. Which of the following equations might have the graph shown? (More than one answer is possible.)

(a) $(x - 2)^2 + (y + 3)^2 = 13$
(b) $(x - 2)^2 + (y - 2)^2 = 8$
(c) $(x - 2)^2 + (y - 3)^2 = 13$
(d) $(x + 2)^2 + (y - 2)^2 = 8$
(e) $x^2 + y^2 - 4x - 9y = 0$
(f) $x^2 + y^2 + 4x - 2y = 0$
(g) $x^2 + y^2 - 9x - 4y = 0$
(h) $x^2 + y^2 - 4x - 4y = 4$



70. Which of the following equations might have the graph shown? (More than one answer is possible.)

(a) $(x - 2)^2 + y^2 = 3$
(b) $(x + 2)^2 + y^2 = 3$
(c) $x^2 + (y - 2)^2 = 3$
(d) $(x + 2)^2 + y^2 = 4$
(e) $x^2 + y^2 + 10x + 16 = 0$
(f) $x^2 + y^2 + 10x - 2y = 1$
(g) $x^2 + y^2 + 9x + 10 = 0$
(h) $x^2 + y^2 - 9x - 10 = 0$



71. Explain how the center and radius of a circle can be used to graph the circle.

72. **What Went Wrong?** A student stated that the center and radius of the graph whose equation is $(x+3)^2+(y-2)^2=16$ are $(3,-2)$ and 4, respectively. Why is this incorrect?

'Are You Prepared?' Answers

1. add; 25 **2.** $\{-1, 5\}$

Chapter Review

Things to Know

Formulas

Distance formula (p. 39)	$d=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$
Midpoint formula (p. 42)	$(x, y)=\left(\dfrac{x_1+x_2}{2}, \dfrac{y_1+y_2}{2}\right)$
Slope of a line (p. 56)	$m=\dfrac{y_2-y_1}{x_2-x_1}$ if $x_1 \neq x_2$; undefined if $x_1=x_2$
Parallel lines (p. 64)	Equal slopes $(m_1=m_2)$ and different y-intercepts $(b_1 \neq b_2)$
Perpendicular lines (p. 65)	Product of slopes is -1 $(m_1 \cdot m_2=-1)$

Equations of Lines and Circles

Vertical line (p. 60)	$x=a$; a is the x-intercept
Point-slope form of the equation of a line (p. 60)	$y-y_1=m(x-x_1)$; m is the slope of the line, (x_1, y_1) is a point on the line
Horizontal line (p. 61)	$y=b$; b is the y-intercept
Slope-intercept form of the equation of a line (p. 61)	$y=mx+b$; m is the slope of the line, b is the y-intercept
General form of the equation of a line (p. 63)	$Ax+By=C$; A, B not both 0
Standard form of the equation of a circle (p. 72)	$(x-h)^2+(y-k)^2=r^2$; r is the radius of the circle, (h, k) is the center of the circle
Equation of the unit circle (p. 72)	$x^2+y^2=1$
General form of the equation of a circle (p. 74)	$x^2+y^2+ax+by+c=0$

Objectives

Section	You should be able to . . .	Examples	Review Exercises
1.1	1 Use the distance formula (p. 39)	1–3	1(a)–3(a), 29, 30(a), 31
	2 Use the midpoint formula (p. 41)	4	1(b)–3(b), 31
1.2	1 Graph equations by plotting points (p. 46)	1–3	4
	2 Find intercepts from a graph (p. 48)	4	5
	3 Find intercepts from an equation (p. 48)	5	6–10
	4 Test an equation for symmetry with respect to the x-axis, the y-axis, and the origin (p. 49)	6–10	6–10
	5 Know how to graph key equations (p. 51)	11–13	26, 27

Section	You should be able to . . .	Examples	Review Exercises
1.3	1 Calculate and interpret the slope of a line (p. 56)	1, 2	1(c)–3(c), 1(d)–3(d), 32
	2 Graph lines given a point and the slope (p. 59)	3	28
	3 Find the equation of a vertical line (p. 60)	4	17
	4 Use the point-slope form of a line; identify horizontal lines (p. 60)	5, 6	16
	5 Use the slope-intercept form of a line (p. 61)	7	16, 18–23
	6 Find an equation of a line given two points (p. 63)	8	18, 19
	7 Graph lines written in general form using intercepts (p. 63)	9	24, 25
	8 Find equations of parallel lines (p. 64)	10, 11	20
	9 Find equations of perpendicular lines (p. 65)	12, 13	21, 30(b)
1.4	1 Write the standard form of the equation of a circle (p. 72)	1	11, 12, 31
	2 Graph a circle (p. 73)	2, 3, 5	13–15
	3 Work with the general form of the equation of a circle (p. 74)	4	14, 15

Review Exercises

In Problems 1–3, find the following for each pair of points:

(a) The distance between the points

(b) The midpoint of the line segment connecting the points

(c) The slope of the line containing the points

(d) Interpret the slope found in part (c)

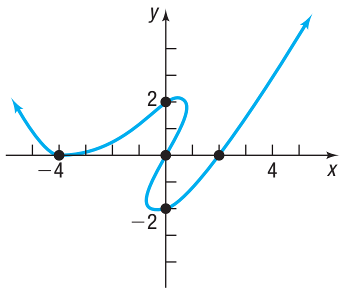
1. $(0, 0); (4, 2)$

2. $(1, -1); (-2, 3)$

3. $(4, -4); (4, 8)$

4. Graph $y = x^2 + 4$ by plotting points.

5. List the intercepts of the graph below.



In Problems 6–10, list the intercepts and test for symmetry with respect to the x-axis, the y-axis, and the origin.

6. $3x^2 = 2y$

7. $4x^2 + y^2 = 16$

8. $y = x^4 + 2x^2 + 1$

9. $y^5 = x^2 - 4x$

10. $x^2 + x + y^2 + 2y = 0$

In Problems 11 and 12, find the standard form of the equation of the circle whose center and radius are given.

11. $(h, k) = (-3, 4); r = 5$

12. $(h, k) = (-1, -2); r = 1$

In Problems 13–15, find the center and radius of each circle. Graph each circle. Find the intercepts, if any, of each circle.

13. $x^2 + (y - 1)^2 = 4$

14. $x^2 + y^2 - 2x + 4y - 4 = 0$

15. $3x^2 + 3y^2 - 6x + 12y = 0$

In Problems 16–21, find an equation of the line having the given characteristics. Express your answer using either the general form or the slope-intercept form of the equation of a line, whichever you prefer.

16. Slope $= -2$; containing the point $(3, -1)$

17. Horizontal; containing the point $(1, -4)$

18. y-intercept $= -2$; containing the point $(5, -3)$

19. Containing the points $(2, -3)$ and $(4, 1)$

20. Parallel to the line $2x - 3y = -4$; containing the point $(-5, 3)$

21. Perpendicular to the line $x - y = 3$; containing the point $(-3, 5)$

In Problems 22 and 23, find the slope and y-intercept of each line. Graph the line, labeling any intercepts.

22. $4x - 5y = -20$

23. $\frac{1}{2}x - \frac{1}{3}y = -\frac{1}{6}$

In Problems 24 and 25, find the intercepts and graph each line.

24. $2x - 3y = 12$

25. $\frac{1}{2}x + \frac{1}{3}y = 2$

26. Graph $y = x^3$.

27. Graph $y = \sqrt{x}$.

28. Graph the line with slope $\frac{2}{3}$ containing the point $(1, 2)$.

29. Show that the points $A = (3, 4)$, $B = (1, 1)$, and $C = (-2, 3)$ are the vertices of an isosceles triangle.

30. Show that the points $A = (-2, 0)$, $B = (-4, 4)$, and $C = (8, 5)$ are the vertices of a right triangle in two ways:

(a) By using the converse of the Pythagorean Theorem

(b) By using the slopes of the lines joining the vertices

31. The endpoints of the diameter of a circle are $(-3, 2)$ and $(5, -6)$. Find the center and radius of the circle. Write the standard equation of this circle.

32. Show that the points $A = (2, 5)$, $B = (6, 1)$, and $C = (8, -1)$ lie on a line by using slopes.

Chapter Test

CHAPTER Test Prep VIDEOS

The Chapter Test Prep Videos include step-by-step solutions to all chapter test exercises. These videos are available in MyLab™ Math, or on this text's YouTube Channel. Refer to the Preface for a link to the YouTube channel.

In Problems 1–3, use $P_1 = (-1, 3)$ and $P_2 = (5, -1)$.

1. Find the distance from P_1 to P_2.

2. Find the midpoint of the line segment joining P_1 and P_2.

3. (a) Find the slope of the line containing P_1 and P_2.

(b) Interpret this slope.

4. Graph $y = x^2 - 9$ by plotting points.

5. Graph $y^2 = x$.

6. List the intercepts and test for symmetry: $x^2 + y = 9$

7. Write the slope-intercept form of the line with slope -2 containing the point $(3, -4)$. Graph the line.

8. Write the general form of the circle with center $(4, -3)$ and radius 5.

9. Find the center and radius of the circle

$$x^2 + y^2 + 4x - 2y - 4 = 0$$

Graph this circle.

10. For the line $2x + 3y = 6$, find a line parallel to the given line containing the point $(1, -1)$. Also find a line perpendicular to the given line containing the point $(0, 3)$.

Chapter Project

Internet-based Project

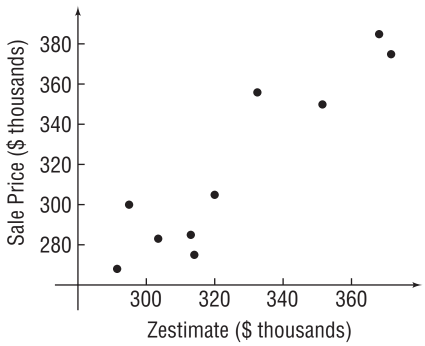
I. Determining the Selling Price of a Home Determining how much to pay for a home is one of the more difficult decisions that must be made when purchasing a home. There are many factors that play a role in a home's value. Location, size, number of bedrooms, number of bathrooms, lot size, and building materials are just a few. Fortunately, the website Zillow.com has developed its own formula for predicting the selling price of a home. This information is a great tool for predicting the actual sale price. For example, the data to the right show the "zestimate"—the selling price of a home as predicted by the folks at Zillow—and the actual selling price of the home, for homes in Oak Park, Illinois.

Zestimate ($ thousands)	Sale Price ($ thousands)
291.5	268
320	305
371.5	375
303.5	283
351.5	350
314	275
332.5	356
295	300
313	285
368	385

The graph below, called a scatter plot, shows the points

$$(291.5, 268), (320, 305), \ldots, (368, 385)$$

in a Cartesian plane. From the graph, it appears that the data follow a linear relation.



1. Imagine drawing a line through the data that appears to fit the data well. Do you believe the slope of the line would be positive, negative, or close to zero? Why?

2. Pick two points from the scatter plot. Treat the zestimate as the value of x, and treat the sale price as the corresponding value of y. Find the equation of the line through the two points you selected.

3. Interpret the slope of the line.

4. Use your equation to predict the selling price of a home whose zestimate is $335,000.

5. Do you believe it would be a good idea to use the equation you found in part 2 if the zestimate were $950,000? Why or why not?

6. Choose a location in which you would like to live. Go to zillow.com and randomly select at least ten homes that have recently sold.

(a) Draw a scatter plot of your data.

(b) Select two points from the scatter plot and find the equation of the line through the points.

(c) Interpret the slope.

(d) Find a home from the Zillow website that interests you under the "Make Me Move" option for which a zestimate is available. Use your equation to predict the sale price based on the zestimate.

2 Functions and Their Graphs

Choosing a Data Plan

When selecting a data plan for a device, most consumers choose a service provider first and then select an appropriate data plan from that provider. The choice as to the type of plan selected depends on your use of the device. For example, is online gaming important? Do you want to stream audio or video? The mathematics learned in this chapter can help you decide what plan is best suited to your particular needs.

—See the Internet-based Chapter Project—

Outline

A Look Back

So far, our discussion has focused on techniques for graphing equations containing two variables.

A Look Ahead

In this chapter, we look at a special type of equation involving two variables called a *function*. This chapter deals with what a function is, how to graph functions, properties of functions, and how functions are used in applications. The word *function* apparently was introduced by René Descartes in 1637. For him, a function was simply any positive integral power of a variable x. Gottfried Wilhelm Leibniz (1646–1716), who always emphasized the geometric side of mathematics, used the word *function* to denote any quantity associated with a curve, such as the coordinates of a point on the curve. Leonhard Euler (1707–1783) used the word to mean any equation or formula involving variables and constants. His idea of a function is similar to the one most often seen in courses that precede calculus. Later, the use of functions in investigating heat flow equations led to a very broad definition that originated with Lejeune Dirichlet (1805–1859), which describes a function as a correspondence between two sets. That is the definition used in this text.

2.1 Functions

PREPARING FOR THIS SECTION *Before getting started, review the following:*

- Intervals (Section A.9, pp. A72–A74)
- Solving Inequalities (Section A.9, pp. A76–A78)
- Evaluating Algebraic Expressions, Domain of a Variable (Section A.1, pp. A6–A7)
- Rationalizing Denominators and Numerators (Section A.10, pp. A85–A86)

Now Work the **'Are You Prepared?'** problems on page 95.

OBJECTIVES
1. Describe a Relation (p. 83)
2. Determine Whether a Relation Represents a Function (p. 85)
3. Use Function Notation; Find the Value of a Function (p. 87)
4. Find the Difference Quotient of a Function (p. 90)
5. Find the Domain of a Function Defined by an Equation (p. 91)
6. Form the Sum, Difference, Product, and Quotient of Two Functions (p. 93)

1 Describe a Relation

Often there are situations where one variable is somehow linked to another variable. For example, the price of a gallon of gas is linked to the price of a barrel of oil. A person can be associated to her telephone number(s). The volume V of a sphere depends on its radius R. The force F exerted by an object corresponds to its acceleration a. These are examples of a *relation*, a correspondence between two sets called the *domain* and the *range*.

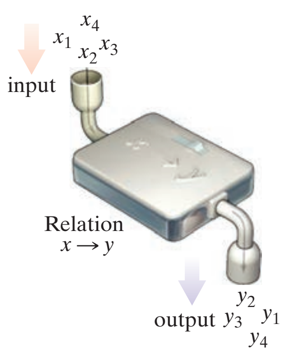


Figure 1

DEFINITION Relation

A **relation** is a correspondence between two sets: a set X, called the **domain**, and a set Y, called the **range**. In a relation, each element from the domain corresponds to at least one element from the range.

If x is an element of the domain and y is an element of the range, and if a relation exists from x to y, then we say that y **corresponds** to x or that y **depends on** x, and we write $x \rightarrow y$. It is often helpful to think of x as the **input** and y as the **output** of the relation. See Figure 1.

Suppose an astronaut standing on the Moon throws a rock 20 meters up and starts a stopwatch as the rock begins to fall back down. The astronaut measures the height of the rock at 1, 2, 2.5, 3, 4, and 5 seconds and obtains heights of 19.2, 16.8, 15, 12.8, 7.2, and 0 meters, respectively. This is an example of a relation expressed **verbally**. The domain of the relation is the set $\{0, 1, 2, 2.5, 3, 4, 5\}$ and the range of the relation is the set $\{20, 19.2, 16.8, 15, 12.8, 7.2, 0\}$.

Table 1

Time (in seconds)	Height (in meters)
0	20
1	19.2
2	16.8
2.5	15
3	12.8
4	7.2
5	0

The astronaut could also express this relation *numerically*, *graphically*, or *algebraically*.

The relation can be expressed **numerically** using a table of numbers, as in Table 1, or by using a **set of ordered pairs**. Using ordered pairs, the relation is

$$\{(0, 20), (1, 19.2), (2, 16.8), (2.5, 15), (3, 12.8), (4, 7.2), (5, 0)\}$$

where the first element of each pair denotes the time and the second element denotes the height.

Suppose x represents the number of seconds on the stopwatch and y represents the height of the rock in meters. Then the relation can be expressed **graphically** by plotting the points (x, y). See Figure 2 on the next page.

The relation can be represented as a **mapping** by drawing an arrow from an element in the domain to the corresponding element in the range. See Figure 3 on the next page.

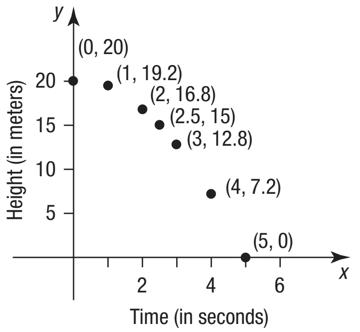


Figure 2

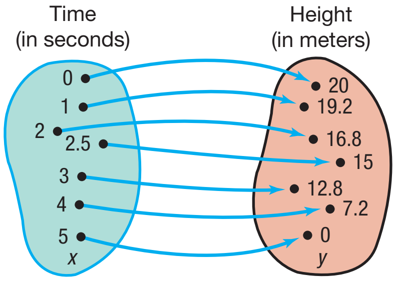


Figure 3

Finally, from physics, the relation can be expressed **algebraically** using the equation

$$y = 20 - 0.8x^2$$

EXAMPLE 1 Describing a Relation

A verbal description of a relation is given below.

The price of First Class U.S. postage stamps has changed over the years. To mail a letter in 2015 cost \$0.49. In 2016 it cost \$0.49 for part of the year and \$0.47 for the rest of the year. In 2017 it cost \$0.47 for part of the year and \$0.49 for the rest of the year. In 2018 it cost \$0.49 for part of the year and \$0.50 for the rest of the year.

Using year as input and price as output,

(a) What is the domain and the range of the relation?

(b) Express the relation as a set of ordered pairs.

(c) Express the relation as a mapping.

(d) Express the relation as a graph.

Solution The relation establishes a correspondence between the input, year, and the output, price of a First Class U.S. postage stamp.

(a) The domain of the relation is $\{2015, 2016, 2017, 2018\}$. The range of the relation is $\{\$0.47, \$0.49, \$0.50\}$.

(b) The relation expressed as a set of ordered pairs is $\{(2015, \$0.49), (2016, \$0.47), (2016, \$0.49), (2017, \$0.47), (2017, \$0.49), (2018, \$0.49), (2018, \$0.50)\}$

(c) See Figure 4 for the relation expressed as a mapping, using *t* for year and *p* for price.

(d) Figure 5 shows a graph of the relation.

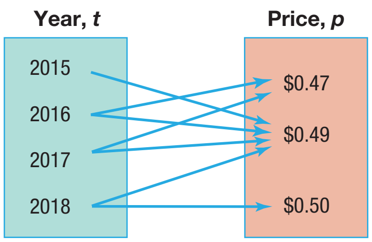


Figure 4 First Class stamp price

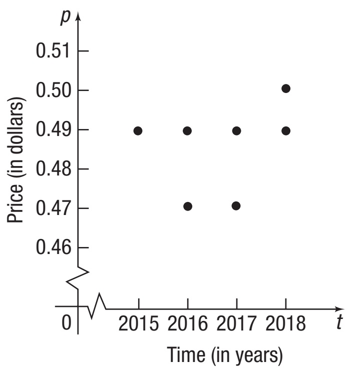


Figure 5 First Class stamp price (2015–2018)

Now Work PROBLEM 17

2 Determine Whether a Relation Represents a Function

Look back at the relation involving the height of a rock on the Moon described at the beginning of the section. Notice that each input, time, corresponds to exactly one output, height. Given a time, you could tell the exact height of the rock. But that is not the case with the price of stamps. Given the year 2018, you cannot determine the price of a stamp with certainty. It could be \$0.49, or it could be \$0.50.

Consider the mapping of the relation in Figure 6. It shows a correspondence between a substance and its specific heat. Notice that for each substance you can tell its specific heat with certainty.

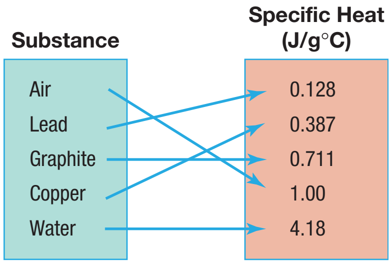


Figure 6 Specific heat of some common substances

The relation associating the time to the height of the rock is a *function*, and the relation associating a given substance to its specific heat is a function. But the relation associating the year to the price of a First Class postage stamp is *not* a function. To be a function, each input must correspond to *exactly one* output.

> **DEFINITION Function**
>
> Let X and Y be two nonempty sets.* A **function** from X into Y is a relation that associates with each element of X exactly one element of Y.

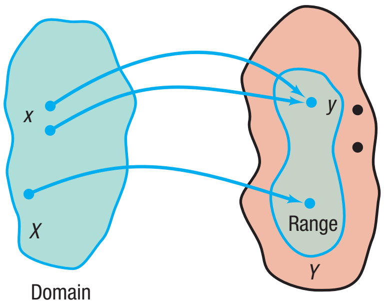


Figure 7

The set X is called the **domain** of the function. For each element x in X, the corresponding element y in Y is called the **value** of the function at x, or the **image** of x. The set of all images of the elements in the domain is called the **range** of the function. See Figure 7.

Since there may be some elements in Y that are not the image of some x in X, it follows that the range of a function may be a proper subset of Y, as shown in Figure 7.

The idea behind a function is its certainty. If an input is given, we can use the function to determine the output. This is not possible if a relation is not a function. The requirement of "one output" provides a predictable behavior that is important when using mathematics to model or analyze the real world. It allows doctors to know exactly how much medicine to give a patient, an engineer to determine the material to use in construction, a store manager to choose how many units to keep in stock, etc.

EXAMPLE 2 Determining Whether a Relation Given by a Mapping Is a Function

For each relation, state the domain and range. Then determine whether the relation is a function.

(a) See Figure 8. This relation shows a correspondence between an item on McDonald's \$1-\$2-\$3 menu and its price.
***Source**: McDonald's Corporation, 2018*

(b) See Figure 9. This relation shows a correspondence between activity level and daily calories needed for an individual with a basal metabolic rate (BMR) of 1975 kilocalories per day (kcal/day).

(c) See Figure 10. This relation shows a correspondence between gestation period (in days) and life expectancy (in years) for five animal species.

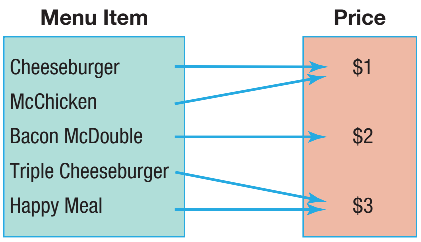


Figure 8

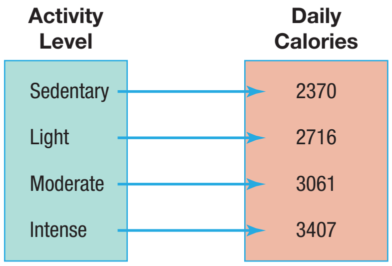


Figure 9

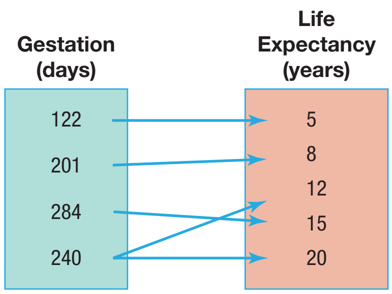


Figure 10

*The sets X and Y will usually be sets of real numbers, in which case a (real) function results. The two sets can also be sets of complex numbers, and then we have defined a complex function. In the broad definition (proposed by Lejeune Dirichlet), X and Y can be any two sets.

Solution (a) The domain of the relation is {Cheeseburger, McChicken, Bacon McDouble, Triple Cheeseburger, Happy Meal}, and the range of the relation is $\{\$1, \$2, \$3\}$. The relation in Figure 8 is a function because each element in the domain corresponds to exactly one element in the range.

(b) The domain of the relation is {Sedentary, Light, Moderate, Intense}, and the range of the relation is $\{2370, 2716, 3061, 3407\}$. The relation in Figure 9 is a function because each element in the domain corresponds to exactly one element in the range.

(c) The domain of the relation is $\{122, 201, 240, 284\}$, and the range of the relation is $\{5, 8, 12, 15, 20\}$. The relation in Figure 10 is not a function because there is an element in the domain, 240, that corresponds to two elements in the range, 12 and 20. If a gestation period of 240 days is selected from the domain, a single life expectancy cannot be associated with it.

Now Work PROBLEM 19

We may also think of a function as a set of ordered pairs (x, y) in which no ordered pairs have the same first element and different second elements. The set of all first elements x is the domain of the function, and the set of all second elements y is its range. Each element x in the domain corresponds to exactly one element y in the range.

EXAMPLE 3 Determining Whether a Relation Given by a Set of Ordered Pairs Is a Function

For each relation, state the domain and range. Then determine whether the relation is a function.

(a) $\{(1, 4), (2, 5), (3, 6), (4, 7)\}$

(b) $\{(1, 4), (2, 4), (3, 5), (6, 10)\}$

(c) $\{(-3, 9), (-2, 4), (0, 0), (1, 1), (-3, 8)\}$

Solution (a) The domain of this relation is $\{1, 2, 3, 4\}$, and its range is $\{4, 5, 6, 7\}$. This relation is a function because there are no ordered pairs with the same first element and different second elements.

(b) The domain of this relation is $\{1, 2, 3, 6\}$, and its range is $\{4, 5, 10\}$. This relation is a function because there are no ordered pairs with the same first element and different second elements.

(c) The domain of this relation is $\{-3, -2, 0, 1\}$, and its range is $\{0, 1, 4, 8, 9\}$. This relation is not a function because there are two ordered pairs, $(-3, 9)$ and $(-3, 8)$, that have the same first element and different second elements.

In Example 3(b), notice that 1 and 2 in the domain both have the same image in the range. This does not violate the definition of a function; two different first elements can have the same second element. The definition is violated when two ordered pairs have the same first element and different second elements, as in Example 3(c).

Now Work PROBLEM 23

Up to now we have shown how to identify when a relation is a function for relations defined by mappings (Example 2) and ordered pairs (Example 3). But relations can also be expressed as equations.

It is usually easiest to determine whether an equation, where y depends on x, is a function, when the equation is solved for y. If any value of x in the domain corresponds to more than one y, the equation does not define a function; otherwise, it does define a function.

EXAMPLE 4 Determining Whether an Equation Is a Function

Determine whether the equation $y = 2x - 5$ defines y as a function of x.

Solution The equation tells us to take an input x, multiply it by 2, and then subtract 5. For any input x, these operations yield only one output y, so the equation is a function. For example, if $x = 1$, then $y = 2 \cdot 1 - 5 = -3$. If $x = 3$, then $y = 2 \cdot 3 - 5 = 1$.

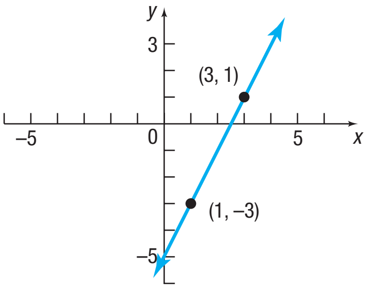


Figure 11 $y = 2x - 5$

The graph of the equation $y = 2x - 5$ is a line with slope 2 and y-intercept -5. The function is called a *linear function*. See Figure 11.

EXAMPLE 5 Determining Whether an Equation Is a Function

Determine whether the equation $x^2 + y^2 = 1$ defines y as a function of x.

Solution To determine whether the equation $x^2 + y^2 = 1$, which defines the unit circle, is a function, solve the equation for y.

$$x^2 + y^2 = 1$$
$$y^2 = 1 - x^2$$
$$y = \pm\sqrt{1 - x^2}$$

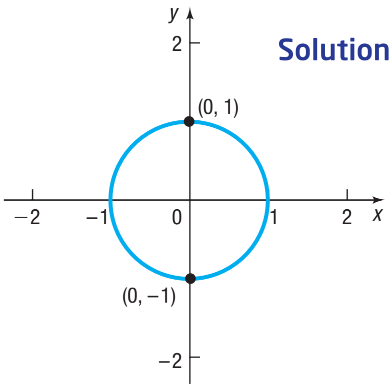


Figure 12 $x^2 + y^2 = 1$

For values of x for which $-1 < x < 1$, two values of y result. For example, if $x = 0$, then $y = \pm 1$, so two different outputs result from the same input. This means that the equation $x^2 + y^2 = 1$ does not define a function. See Figure 12.

Now Work PROBLEM 37

3 Use Function Notation; Find the Value of a Function

It is common practice to denote functions by letters such as f, g, F, G, and others. If f is a function, then for each number x in the domain, the corresponding number y in the range is designated by the symbol $f(x)$, read as "f of x," and we write $y = f(x)$. When a function is expressed in this way, we are using **function notation**.

We refer to $f(x)$ as the **value** of the function f at the number x. For example, the function in Example 4 may be written using function notation as $y = f(x) = 2x - 5$. Then $f(1) = -3$ and $f(3) = 1$.

Sometimes it is helpful to think of a function f as a machine that receives as input a number from the domain, manipulates it, and outputs a value in the range. See Figure 13.

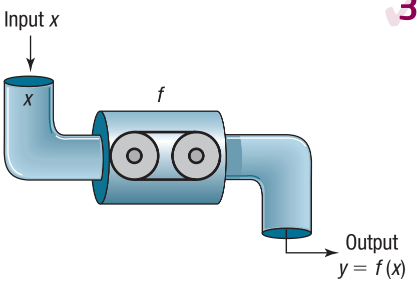


Figure 13 Input/output machine

In Words
If $y = f(x)$, then x is the input and y is the output corresponding to x.

The restrictions on this input/output machine are as follows:

- It accepts only numbers from the domain of the function.
- For each input, there is exactly one output (which may be repeated for different inputs).

WARNING The notation $y = f(x)$ denotes a function f. It does NOT mean "f times x." ■

Figure 14 illustrates some other functions. Notice that in every function, for each x in the domain, there is one corresponding value in the range.

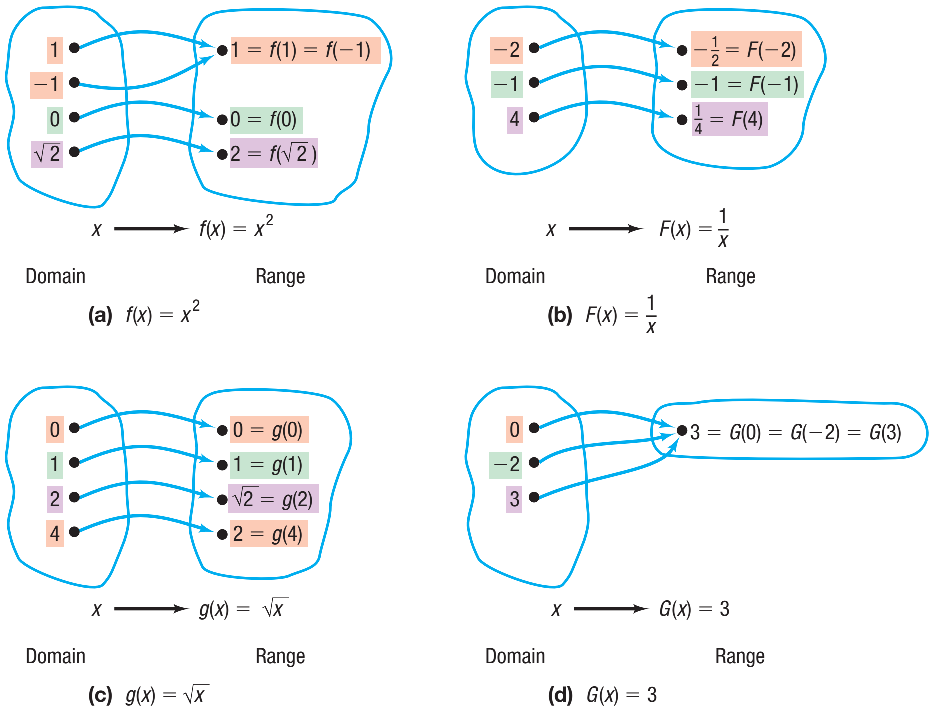


Figure 14

For a function $y = f(x)$, the variable x is called the **independent variable** because it can be assigned any number from the domain. The variable y is called the **dependent variable**, because its value depends on x.

Any symbols can be used to represent the independent and dependent variables. For example, if f is the *cube function*, then f can be given by $f(x) = x^3$ or $f(t) = t^3$ or $f(z) = z^3$. All three functions are the same. Each says to cube the independent variable to get the output. In practice, the symbols used for the independent and dependent variables are based on common usage, such as using t for time and a for acceleration.

The independent variable is also called the **argument** of the function. Thinking of the independent variable as an argument can sometimes make it easier to find the value of a function. For example, if f is the function defined by $f(x) = x^3$, then f tells us to cube the argument. Then $f(2)$ means to cube 2, $f(a)$ means to cube the number a, and $f(x + h)$ means to cube the quantity $x + h$.

EXAMPLE 6 Finding Values of a Function

For the function f defined by $f(x) = 2x^2 - 3x$, evaluate

(a) $f(3)$ (b) $f(x) + f(3)$ (c) $3f(x)$ (d) $f(-x)$

(e) $-f(x)$ (f) $f(3x)$ (g) $f(x + 3)$ (h) $f(x + h)$

Solution

(a) Substitute 3 for x in the equation for f, $f(x) = 2x^2 - 3x$, to get

$$f(3) = 2 \cdot 3^2 - 3 \cdot 3 = 18 - 9 = 9$$

The image of 3 is 9.

(b) $f(x) + f(3) = (2x^2 - 3x) + 9 = 2x^2 - 3x + 9$

(c) Multiply the equation for f by 3.

$$3f(x) = 3(2x^2 - 3x) = 6x^2 - 9x$$

(d) Substitute $-x$ for x in the equation for f and simplify.

$$f(-x) = 2(-x)^2 - 3(-x) = 2x^2 + 3x$$ Notice the use of parentheses here.

(e) $-f(x) = -(2x^2 - 3x) = -2x^2 + 3x$

(f) Substitute $3x$ for x in the equation for f and simplify.

$$f(3x) = 2(3x)^2 - 3 \cdot 3x = 2 \cdot 9x^2 - 9x = 18x^2 - 9x$$

(g) Substitute $x + 3$ for x in the equation for f and simplify.

$$\begin{aligned} f(x+3) &= 2(x+3)^2 - 3(x+3) \\ &= 2(x^2 + 6x + 9) - 3x - 9 \\ &= 2x^2 + 12x + 18 - 3x - 9 \\ &= 2x^2 + 9x + 9 \end{aligned}$$

(h) Substitute $x + h$ for x in the equation for f and simplify.

$$\begin{aligned} f(x+h) &= 2(x+h)^2 - 3(x+h) \\ &= 2(x^2 + 2xh + h^2) - 3x - 3h \\ &= 2x^2 + 4xh + 2h^2 - 3x - 3h \end{aligned}$$

Notice in this example that $f(x+3) \neq f(x) + f(3)$, $f(-x) \neq -f(x)$, and $3f(x) \neq f(3x)$.

Now Work PROBLEM 43

Most calculators have special keys that allow you to find the value of certain commonly used functions. Examples are keys for the square function $f(x) = x^2$, the square root function $f(x) = \sqrt{x}$, and the reciprocal function $f(x) = \dfrac{1}{x} = x^{-1}$.

EXAMPLE 7 Finding Values of a Function on a Calculator

Verify the following results on your calculator.

(a) $f(x) = x^2$ $\quad f(1.234) = 1.234^2 = 1.522756$

(b) $F(x) = \dfrac{1}{x}$ $\quad F(1.234) = \dfrac{1}{1.234} \approx 0.8103727715$

(c) $g(x) = \sqrt{x}$ $\quad g(1.234) = \sqrt{1.234} \approx 1.110855526$

COMMENT Graphing calculators can be used to evaluate a function. Figure 15 shows the function $Y_1 = f(x) = 2x^2 - 3x$ evaluated at 3 on a TI-84 Plus C graphing calculator. Compare this result with the solution to Example 6(a).

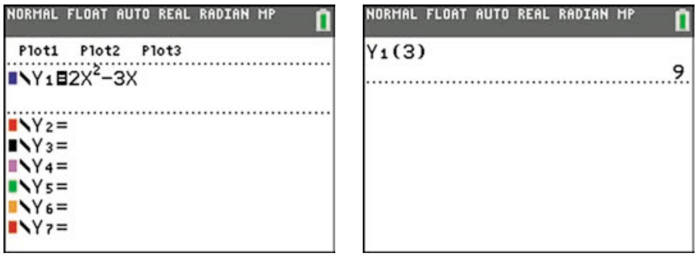


Figure 15 Evaluating $f(x) = 2x^2 - 3x$ for $x = 3$

Implicit Form of a Function

COMMENT A graphing calculator requires the explicit form of a function. ■

In general, when a function f is defined by an equation in x and y, we say that the function f is given **implicitly**. If it is possible to solve the equation for y in terms of x, then we write $y = f(x)$ and say that the function is given **explicitly**. For example,

Implicit Form	**Explicit Form**
$3x + y = 5$	$y = f(x) = -3x + 5$
$x^2 - y = 6$	$y = f(x) = x^2 - 6$
$xy = 4$	$y = f(x) = \dfrac{4}{x}$

SUMMARY

Important Facts about Functions

- For each x in the domain of a function f, there is exactly one image $f(x)$ in the range; however, more than one x in the domain can have the same image in the range.
- f is the symbol that we use to denote the function. It is symbolic of the equation (rule) that we use to get from x in the domain to $f(x)$ in the range.
- If $y = f(x)$, then x is the independent variable, or the argument of f, and y, or $f(x)$, is the dependent variable, or the value of f at x.

4 Find the Difference Quotient of a Function

An important concept in calculus involves using a certain quotient. For a function f, the inputs x and $x + h$, $h \neq 0$, result in the values $f(x)$ and $f(x + h)$. The quotient of their differences

$$\frac{f(x+h) - f(x)}{(x+h) - x} = \frac{f(x+h) - f(x)}{h}$$

with $h \neq 0$, is called the *difference quotient of f at x*.

DEFINITION Difference Quotient

The **difference quotient** of a function f at x is given by

$$\frac{f(x+h) - f(x)}{h} \qquad h \neq 0 \qquad \textbf{(1)}$$

The difference quotient is used in calculus to define the *derivative*, which leads to applications such as the velocity of an object and optimization of resources.

When finding a difference quotient, it is necessary to simplify expression (1) so that the h in the denominator can be cancelled.

EXAMPLE 8 Finding the Difference Quotient of a Function

Find the difference quotient of each function.

(a) $f(x) = 2x^2 - 3x$

(b) $f(x) = \dfrac{4}{x}$

(c) $f(x) = \sqrt{x}$

Solution

(a) $$\frac{f(x+h)-f(x)}{h} \underset{\substack{\uparrow \\ f(x+h)=2(x+h)^2-3(x+h)}}{=} \frac{[2(x+h)^2-3(x+h)]-[2x^2-3x]}{h}$$

$$= \frac{2(x^2+2xh+h^2)-3x-3h-2x^2+3x}{h} \qquad \text{Simplify.}$$

$$= \frac{2x^2+4xh+2h^2-3h-2x^2}{h} \qquad \text{Distribute and combine like terms.}$$

$$= \frac{4xh+2h^2-3h}{h} \qquad \text{Combine like terms.}$$

$$= \frac{h(4x+2h-3)}{h} \qquad \text{Factor out } h.$$

$$= 4x+2h-3 \qquad \text{Cancel the } h\text{'s.}$$

(b) $$\frac{f(x+h)-f(x)}{h} = \frac{\frac{4}{x+h}-\frac{4}{x}}{h} \qquad f(x+h)=\frac{4}{x+h}$$

$$= \frac{\frac{4x-4(x+h)}{x(x+h)}}{h} \qquad \text{Subtract.}$$

$$= \frac{4x-4x-4h}{x(x+h)h} \qquad \text{Divide and distribute.}$$

$$= \frac{-4h}{x(x+h)h} \qquad \text{Simplify.}$$

$$= -\frac{4}{x(x+h)} \qquad \text{Cancel the } h\text{'s.}$$

Need to Review?
Rationalizing Numerators is discussed in Section A.10, pp. A85–A86.

(c) When a function involves a square root, we rationalize the numerator of expression (1). After simplifying, the h's cancel.

$$\frac{f(x+h)-f(x)}{h} = \frac{\sqrt{x+h}-\sqrt{x}}{h} \qquad f(x+h)=\sqrt{x+h}$$

$$= \frac{\sqrt{x+h}-\sqrt{x}}{h}\cdot\frac{\sqrt{x+h}+\sqrt{x}}{\sqrt{x+h}+\sqrt{x}} \qquad \text{Rationalize the numerator.}$$

$$= \frac{(\sqrt{x+h})^2-(\sqrt{x})^2}{h(\sqrt{x+h}+\sqrt{x})} \qquad (A-B)(A+B)=A^2-B^2$$

$$= \frac{h}{h(\sqrt{x+h}+\sqrt{x})} \qquad (\sqrt{x+h})^2-(\sqrt{x})^2 = x+h-x=h$$

$$= \frac{1}{\sqrt{x+h}+\sqrt{x}} \qquad \text{Cancel the } h\text{'s.}$$

Now Work PROBLEM 83

5 Find the Domain of a Function Defined by an Equation

Often the domain of a function f is not specified; instead, only the equation defining the function is given. In such cases, the **domain of f** is the largest set of real numbers for which the value $f(x)$ is a real number. The domain of a function f is the same as the domain of the variable x in the expression $f(x)$.

EXAMPLE 9 Finding the Domain of a Function

Find the domain of each function.

(a) $f(x) = x^2 + 5x$ (b) $g(x) = \dfrac{3x}{x^2 - 4}$

(c) $h(t) = \sqrt{4 - 3t}$ (d) $F(x) = \dfrac{\sqrt{3x + 12}}{x - 5}$

Solution

(a) The function $f(x) = x^2 + 5x$ says to sum the square of a number and five times the number. Since these operations can be performed on any real number, the domain of f is the set of all real numbers.

In Words
The domain of g found in Example 9(b) is $\{x \mid x \neq -2, x \neq 2\}$. This notation is read, "The domain of the function g is the set of all real numbers x such that x does not equal -2 and x does not equal 2."

(b) The function $g(x) = \dfrac{3x}{x^2 - 4}$ says to divide $3x$ by $x^2 - 4$. Since division by 0 is not defined, the denominator $x^2 - 4$ cannot be 0, so x cannot equal -2 or 2. The domain of the function g is $\{x \mid x \neq -2, x \neq 2\}$.

(c) The function $h(t) = \sqrt{4 - 3t}$ says to take the square root of $4 - 3t$. Since only nonnegative numbers have real square roots, the expression under the square root (the radicand) must be nonnegative (greater than or equal to zero). That is,

$$\begin{aligned} 4 - 3t &\geq 0 \\ -3t &\geq -4 \\ t &\leq \frac{4}{3} \end{aligned}$$

The domain of h is $\left\{t \,\middle|\, t \leq \dfrac{4}{3}\right\}$, or the interval $\left(-\infty, \dfrac{4}{3}\right]$.

(d) The function $F(x) = \dfrac{\sqrt{3x + 12}}{x - 5}$ says to take the square root of $3x + 12$ and divide the result by $x - 5$. This requires that $3x + 12 \geq 0$, so $x \geq -4$, and also that $x - 5 \neq 0$, so $x \neq 5$. Combining these two restrictions, the domain of F is

$$\{x \mid x \geq -4, x \neq 5\}$$

The following steps may prove helpful for finding the domain of a function that is defined by an equation and whose domain is a subset of the real numbers.

Finding the Domain of a Function Defined by an Equation

- Start with the domain as the set of all real numbers.
- If the equation has a denominator, exclude any numbers for which the denominator is zero.
- If the equation has a radical with an even index, exclude any numbers for which the expression inside the radical (the radicand) is negative.

Now Work PROBLEM 55

We express the domain of a function using interval notation, set notation, or words, whichever is most convenient. If x is in the domain of a function f, we say that **f is defined at x**, or **$f(x)$ exists**. If x is not in the domain of f, we say that **f is not defined at x**, or **$f(x)$ does not exist**. For example, if $f(x) = \dfrac{x}{x^2 - 1}$, then $f(0)$ exists, but $f(1)$ and $f(-1)$ do not exist. (Do you see why?)

When a function is defined by an equation, it can be difficult to find its range unless we also have a graph of the function. So we are usually content to find only the domain.

When we use functions in applications, the domain may be restricted by physical or geometric considerations. For example, the domain of the function f defined by $f(x) = x^2$ is the set of all real numbers. However, if f represents the area of a square whose sides are of length x, the domain of f is restricted to the positive real numbers, since the length of a side can never be 0 or negative.

EXAMPLE 10 **Finding the Domain of a Function Used in an Application**

Express the area of a circle as a function of its radius. Find the domain.

Solution See Figure 16. The formula for the area A of a circle of radius r is $A = \pi r^2$. Using r to represent the independent variable and A to represent the dependent variable, the function expressing this relationship is

$$A = A(r) = \pi r^2$$

In this application, the domain is $\{r \mid r > 0\}$. (Do you see why?)

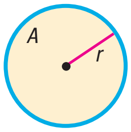


Figure 16 Circle of radius r

Now Work PROBLEM 105

6 Form the Sum, Difference, Product, and Quotient of Two Functions

Functions, like numbers, can be added, subtracted, multiplied, and divided. For example, if $f(x) = x^2 + 9$ and $g(x) = 3x + 5$, then

$$f(x) + g(x) = (x^2 + 9) + (3x + 5) = x^2 + 3x + 14$$

The new function $y = x^2 + 3x + 14$ is called the *sum function* $f + g$. Similarly,

$$f(x) \cdot g(x) = (x^2 + 9)(3x + 5) = 3x^3 + 5x^2 + 27x + 45$$

The new function $y = 3x^3 + 5x^2 + 27x + 45$ is called the *product function* $f \cdot g$.

The general definitions are given next.

DEFINITION Sum Function

Given functions f and g, the **sum function** is defined by

$$\boxed{(f + g)(x) = f(x) + g(x)}$$

RECALL
The symbol $\cap$ stands for intersection. It means the set of elements that are common to both sets.

The domain of $f + g$ consists of all real numbers x that are in the domains of both f and g. That is, domain of $f + g$ = domain of $f \cap$ domain of g.

DEFINITION Difference Function

Given functions f and g, the **difference function** is defined by

$$\boxed{(f - g)(x) = f(x) - g(x)}$$

The domain of $f - g$ consists of all real numbers x that are in the domains of both f and g. That is, domain of $f - g$ = domain of $f \cap$ domain of g.

DEFINITION Product Function

Given functions f and g, the **product function** is defined by

$$\boxed{(f \cdot g)(x) = f(x) \cdot g(x)}$$

The domain of $f \cdot g$ consists of all real numbers x that are in the domains of both f and g. That is, domain of $f \cdot g$ = domain of $f \cap$ domain of g.

DEFINITION Quotient Function

Given functions f and g, the **quotient function** is defined by

$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)} \qquad g(x) \neq 0$$

The domain of $\frac{f}{g}$ consists of all real numbers x for which $g(x) \neq 0$ that are also in the domains of both f and g. That is,

$$\text{domain of } \frac{f}{g} = \{x \mid g(x) \neq 0\} \cap \text{domain of } f \cap \text{domain of } g$$

EXAMPLE 11 Operations on Functions

Let f and g be two functions defined as

$$f(x) = \frac{1}{x+2} \quad \text{and} \quad g(x) = \frac{x}{x-1}$$

Find the following functions, and determine the domain.

(a) $(f+g)(x)$ (b) $(f-g)(x)$ (c) $(f \cdot g)(x)$ (d) $\left(\frac{f}{g}\right)(x)$

Solution The domain of f is $\{x \mid x \neq -2\}$ and the domain of g is $\{x \mid x \neq 1\}$.

(a) $$(f+g)(x) = f(x) + g(x) = \frac{1}{x+2} + \frac{x}{x-1}$$
$$= \frac{x-1}{(x+2)(x-1)} + \frac{x(x+2)}{(x+2)(x-1)}$$
$$= \frac{x - 1 + x^2 + 2x}{(x+2)(x-1)} = \frac{x^2 + 3x - 1}{(x+2)(x-1)}$$

The domain of $f + g$ consists of all real numbers x that are in the domains of both f and g. The domain of $f + g$ is $\{x \mid x \neq -2, x \neq 1\}$.

(b) $$(f-g)(x) = f(x) - g(x) = \frac{1}{x+2} - \frac{x}{x-1}$$
$$= \frac{x-1}{(x+2)(x-1)} - \frac{x(x+2)}{(x+2)(x-1)}$$
$$= \frac{x - 1 - x^2 - 2x}{(x+2)(x-1)} = \frac{-x^2 - x - 1}{(x+2)(x-1)} = -\frac{x^2 + x + 1}{(x+2)(x-1)}$$

The domain of $f - g$ consists of all real numbers x that are in the domains of both f and g. The domain of $f - g$ is $\{x \mid x \neq -2, x \neq 1\}$.

(c) $$(f \cdot g)(x) = f(x) \cdot g(x) = \frac{1}{x+2} \cdot \frac{x}{x-1} = \frac{x}{(x+2)(x-1)}$$

The domain of $f \cdot g$ consists of all real numbers x that are in the domains of both f and g. The domain of $f \cdot g$ is $\{x \mid x \neq -2, x \neq 1\}$.

(d) $$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)} = \frac{\frac{1}{x+2}}{\frac{x}{x-1}} = \frac{1}{x+2} \cdot \frac{x-1}{x} = \frac{x-1}{x(x+2)}$$

The domain of $\dfrac{f}{g}$ consists of all real numbers x for which $g(x) \neq 0$ that are also in the domains of both f and g. Since $g(x) = 0$ when $x = 0$, exclude 0 as well as -2 and 1 from the domain. The domain of $\dfrac{f}{g}$ is $\{x \mid x \neq -2, x \neq 0, x \neq 1\}$.

Now Work PROBLEM 71

In calculus, it is sometimes helpful to view a complicated function as the sum, difference, product, or quotient of simpler functions. For example,

$$F(x) = x^2 + \sqrt{x} \text{ is the sum of } f(x) = x^2 \text{ and } g(x) = \sqrt{x}.$$

$$H(x) = \frac{x^2 - 1}{x^2 + 1} \quad \text{is the quotient of } f(x) = x^2 - 1 \text{ and } g(x) = x^2 + 1.$$

SUMMARY

Function
- A relation between two sets of real numbers so that each number x in the first set, the domain, corresponds to exactly one number y in the second set.
- A set of ordered pairs (x, y) or $(x, f(x))$ in which no first element is paired with two different second elements.
- The range is the set of y-values of the function that are the images of the x-values in the domain.
- A function f may be defined implicitly by an equation involving x and y or explicitly by writing $y = f(x)$.

Unspecified domain
If a function f is defined by an equation and no domain is specified, then the domain is the largest set of real numbers for which $f(x)$ is a real number.

Function notation
- $y = f(x)$
- f is a symbol for the function.
- x is the independent variable, or argument.
- y is the dependent variable.
- $f(x)$ is the value of the function at x.

2.1 Assess Your Understanding

'Are You Prepared?'

Answers are given at the end of these exercises. If you get a wrong answer, read the pages listed in red.

1. The inequality $-1 < x < 3$ can be written in interval notation as __________. (pp. A72–A74)
2. If $x = -2$, the value of the expression $3x^2 - 5x + \dfrac{1}{x}$ is __________. (pp. A6–A7)
3. The domain of the variable in the expression $\dfrac{x-3}{x+4}$ is ______________. (p. A7)
4. Solve the inequality: $3 - 2x > 5$. Graph the solution set. (pp. A76–A78)
5. To rationalize the denominator of $\dfrac{3}{\sqrt{5} - 2}$, multiply the numerator and denominator by ____________. (pp. A85–A86)
6. A quotient is considered rationalized if its denominator has no __________. (pp. A85–A86)

Concepts and Vocabulary

7. For a function $y = f(x)$, the variable x is the ________________ variable, and the variable y is the _____________ variable.
8. ***Multiple Choice*** The set of all images of the elements in the domain of a function is called the _____.
   (a) range **(b)** domain **(c)** solution set **(d)** function
9. ***Multiple Choice*** The independent variable is sometimes referred to as the _____ of the function.
   (a) range **(b)** value **(c)** argument **(d)** definition
10. ***True or False*** The domain of $\dfrac{f}{g}$ consists of the numbers x that are in the domains of both f and g.
11. ***True or False*** Every relation is a function.
12. Four ways of expressing a relation are ___________, ______________, ______________, and ______________.
13. ***True or False*** If no domain is specified for a function f, then the domain of f is the set of real numbers.
14. ***True or False*** If x is in the domain of a function f, we say that f is not defined at x, or $f(x)$ does not exist.

15. The expression $\dfrac{f(x+h)-f(x)}{h}$ is called the ____________ ____________ of f.

16. When written as $y = f(x)$, a function is said to be defined ____________.

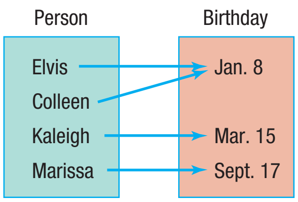
Skill Building

In Problems 17 and 18, a relation expressed verbally is given.
(a) What is the domain and the range of the relation? (b) Express the relation using a mapping. (c) Express the relation as a set of ordered pairs.

17. The density of a gas under constant pressure depends on temperature. Holding pressure constant at 14.5 pounds per square inch, a chemist measures the density of an oxygen sample at temperatures of 0, 22, 40, 70, and 100ºC and obtains densities of 1.411, 1.305, 1.229, 1.121, and 1.031 kg/m^3, respectively.

18. A researcher wants to investigate how weight depends on height among adult males in Europe. She visits five regions in Europe and determines the average heights in those regions to be 1.80, 1.78, 1.77, 1.77, and 1.80 meters. The corresponding average weights are 87.1, 86.9, 83.0, 84.1, and 86.4 kg, respectively.

In Problems 19–30, find the domain and range of each relation. Then determine whether the relation represents a function.

19.


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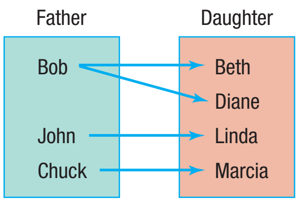

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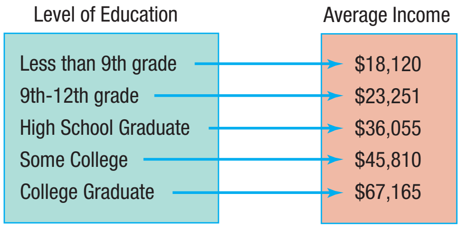

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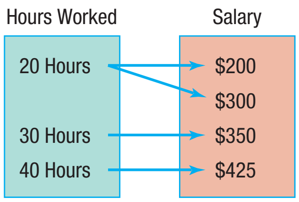


23. $\{(2,6), (-3,6), (4,9), (2,10)\}$ **24.** $\{(-2,5), (-1,3), (3,7), (4,12)\}$ **25.** $\{(0,-2), (1,3), (2,3), (3,7)\}$
26. $\{(1,3), (2,3), (3,3), (4,3)\}$ **27.** $\{(-4,4), (-3,3), (-2,2), (-1,1), (-4,0)\}$ **28.** $\{(3,3), (3,5), (0,1), (-4,6)\}$
29. $\{(-2,16), (-1,4), (0,3), (1,4)\}$ **30.** $\{(-1,8), (0,3), (2,-1), (4,3)\}$

In Problems 31–42, determine whether the equation defines y as a function of x.

31. $y = x^3$ **32.** $y = 2x^2 - 3x + 4$ **33.** $y = |x|$ **34.** $y = \dfrac{1}{x}$

35. $y = \pm\sqrt{1-2x}$ **36.** $x^2 = 8 - y^2$ **37.** $x = y^2$ **38.** $x + y^2 = 1$

39. $y = \dfrac{3x-1}{x+2}$ **40.** $y = \sqrt[3]{x}$ **41.** $x^2 - 4y^2 = 1$ **42.** $|y| = 2x + 3$

In Problems 43–50, find the following for each function:
(a) $f(0)$ *(b)* $f(1)$ *(c)* $f(-1)$ *(d)* $f(-x)$ *(e)* $-f(x)$ *(f)* $f(x+1)$ *(g)* $f(2x)$ *(h)* $f(x+h)$

43. $f(x) = 3x^2 + 2x - 4$ **44.** $f(x) = -2x^2 + x - 1$ **45.** $f(x) = \dfrac{x^2-1}{x+4}$ **46.** $f(x) = \dfrac{x}{x^2+1}$

47. $f(x) = \sqrt{x^2 + x}$ **48.** $f(x) = |x| + 4$ **49.** $f(x) = 1 - \dfrac{1}{(x+2)^2}$ **50.** $f(x) = \dfrac{2x+1}{3x-5}$

In Problems 51–70, find the domain of each function.

51. $f(x) = x^2 + 2$ **52.** $f(x) = -5x + 4$ **53.** $f(x) = \dfrac{x^2}{x^2+1}$ **54.** $f(x) = \dfrac{x+1}{2x^2+8}$

55. $g(x) = \dfrac{x}{x^2-16}$ **56.** $h(x) = \dfrac{2x}{x^2-4}$ **57.** $G(x) = \dfrac{x+4}{x^3-4x}$ **58.** $F(x) = \dfrac{x-2}{x^3+x}$

59. $G(x) = \sqrt{1 - x}$

60. $h(x) = \sqrt{3x - 12}$

61. $f(x) = \dfrac{x - 1}{|3x - 1| - 4}$

62. $p(x) = \dfrac{x}{|2x + 3| - 1}$

63. $f(x) = \dfrac{-x}{\sqrt{-x - 2}}$

64. $f(x) = \dfrac{x}{\sqrt{x - 4}}$

65. $h(z) = \dfrac{\sqrt{z + 3}}{z - 2}$

66. $P(t) = \dfrac{\sqrt{t - 4}}{3t - 21}$

67. $g(t) = -t^2 + \sqrt[3]{t^2 + 7t}$

68. $f(x) = \sqrt[3]{5x - 4}$

69. $N(p) = \sqrt[5]{\dfrac{p}{2p^2 - 98}}$

70. $M(t) = \sqrt[5]{\dfrac{t + 1}{t^2 - 5t - 14}}$

In Problems 71–80, for the given functions f and g, find the following. For parts (a)–(d), also find the domain.

(a) $(f + g)(x)$ *(b)* $(f - g)(x)$ *(c)* $(f \cdot g)(x)$ *(d)* $\left(\dfrac{f}{g}\right)(x)$

(e) $(f + g)(3)$ *(f)* $(f - g)(4)$ *(g)* $(f \cdot g)(2)$ *(h)* $\left(\dfrac{f}{g}\right)(1)$

71. $f(x) = 3x + 4;\quad g(x) = 2x - 3$

72. $f(x) = 2x + 1;\quad g(x) = 3x - 2$

73. $f(x) = 2x^2 + 3;\quad g(x) = 4x^3 + 1$

74. $f(x) = x - 1;\quad g(x) = 2x^2$

75. $f(x) = |x|;\quad g(x) = x$

76. $f(x) = \sqrt{x};\quad g(x) = 3x - 5$

77. $f(x) = \sqrt{x - 1};\quad g(x) = \sqrt{4 - x}$

78. $f(x) = 1 + \dfrac{1}{x};\quad g(x) = \dfrac{1}{x}$

79. $f(x) = \sqrt{x + 1};\quad g(x) = \dfrac{2}{x}$

80. $f(x) = \dfrac{2x + 3}{3x - 2};\quad g(x) = \dfrac{4x}{3x - 2}$

81. Given $f(x) = \dfrac{1}{x}$ and $\left(\dfrac{f}{g}\right)(x) = \dfrac{x + 1}{x^2 - x}$, find the function g.

82. Given $f(x) = 3x + 1$ and $(f + g)(x) = 6 - \dfrac{1}{2}x$, find the function g.

In Problems 83–98, find the difference quotient of f; that is, find $\dfrac{f(x + h) - f(x)}{h}$, $h \neq 0$, *for each function. Be sure to simplify.*

83. $f(x) = 4x + 3$

84. $f(x) = -3x + 1$

85. $f(x) = 3x^2 + 2$

86. $f(x) = x^2 - 4$

87. $f(x) = 3x^2 - 2x + 6$

88. $f(x) = x^2 - x + 4$

89. $f(x) = \dfrac{1}{x + 3}$

90. $f(x) = \dfrac{5}{4x - 3}$

91. $f(x) = \dfrac{5x}{x - 4}$

92. $f(x) = \dfrac{2x}{x + 3}$

93. $f(x) = \sqrt{x + 1}$

94. $f(x) = \sqrt{x - 2}$

95. $f(x) = \dfrac{1}{x^2 + 1}$

96. $f(x) = \dfrac{1}{x^2}$

97. $f(x) = \dfrac{1}{\sqrt{x + 2}}$

98. $f(x) = \sqrt{4 - x^2}$

Applications and Extensions

99. Given $f(x) = x^2 - 3x + 3$, find the value(s) for x such that $f(x) = 31$.

100. If $f(x) = \dfrac{5}{6}x - \dfrac{3}{4}$, find the value(s) of x so that $f(x) = -\dfrac{7}{16}$.

101. If $f(x) = 2x^3 + Ax^2 + 7x - 5$ and $f(2) = 5$, what is the value of A?

102. If $f(x) = 3x^2 - Bx + 4$ and $f(-1) = 12$, what is the value of B ?

103. If $f(b) = \dfrac{4b + 4}{b - A}$ and $f(-3) = 2$, what is the value of A?

104. If $f(x) = \dfrac{2x - B}{3x + 4}$ and $f(2) = \dfrac{1}{2}$, what is the value of B?

105. Geometry Express the perimeter P of a rectangle as a function of the length L if the width of the rectangle is twice its length.

106. Geometry Express the area A of an isosceles right triangle as a function of the length x of one of the two equal sides.

107. Constructing Functions Ann, a commissioned salesperson, earns \$100 base pay plus \$10 per item sold. Express her gross salary G as a function of the number x of items sold.

108. Constructing Functions Express the gross salary G of a person who earns \$16 per hour as a function of the number x of hours worked.

109. Effect of Gravity on Jupiter If a rock falls from a height of 20 meters on the planet Jupiter, its height H (in meters) after x seconds is approximately

$$H(x) = 20 - 13x^2$$

(a) What is the height of the rock when $x = 1$ second? When $x = 1.1$ seconds? When $x = 1.2$ seconds?
(b) When is the height of the rock 15 meters? When is it 10 meters? When is it 5 meters?
(c) When does the rock strike the ground?

110. Effect of Gravity on Earth If a rock falls from a height of 31 meters on Earth, the height H (in meters) after x seconds 31 is approximately

$$H(x) = 31 - 4.9x^2.$$

(a) What is the height of the rock when $x = 1.3$ seconds?
(b) When is the height of the rock 14 meters?
(c) When does the rock strike the ground?

111. Cost of Trans-Atlantic Travel An airplane crosses the Atlantic Ocean (3000 miles) with an airspeed of 550 miles per hour. The cost C (in dollars) per passenger is given by

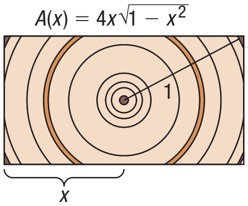
$$C(x) = 150 + \frac{x}{15} + \frac{36{,}000}{x}$$

where x is the ground speed (airspeed $\pm$ wind).

(a) What is the cost per passenger for quiescent (no wind) conditions?
(b) What is the cost per passenger with a head wind of 50 miles per hour?
(c) What is the cost per passenger with a tail wind of 100 miles per hour?
(d) What is the cost per passenger with a head wind of 100 miles per hour?

112. Cross-sectional Area The cross-sectional area of a beam cut from a log with radius 1 foot is given by the function $A(x) = 4x\sqrt{1 - x^2}$, where x represents the length, in feet, of half the base of the beam. See the figure. Determine the cross-sectional area of the beam if the length of half the base of the beam is as follows:

(a) One-third of a foot
(b) One-half of a foot
(c) Two-thirds of a foot



113. Economics The **participation rate** is the number of people in the labor force divided by the civilian population (excludes military). Let $L(x)$ represent the size of the labor force in year x, and $P(x)$ represent the civilian population in year x. Determine a function that represents the participation rate R as a function of x.

114. Crimes Suppose that $V(x)$ represents the number of violent crimes committed in year x and $P(x)$ represents the number of property crimes committed in year x. Determine a function T that represents the combined total of violent crimes and property crimes in year x.

115. Health Care Suppose that $P(x)$ represents the percentage of income spent on health care in year x and $I(x)$ represents income in year x. Find a function H that represents total health care expenditures in year x.

116. Income Tax Suppose that $I(x)$ represents the income of an individual in year x before taxes and $T(x)$ represents the individual's tax bill in year x. Find a function N that represents the individual's net income (income after taxes) in year x.

117. Profit Function Suppose that the revenue R, in dollars, from selling x cell phones, in hundreds, is $R(x) = -1.7x^2 + 320x$. The cost C, in dollars, from selling x cell phones, in hundreds, is $C(x) = 0.06x^3 - 3x^2 + 85x + 400$.
(a) Find the profit function, $P(x) = R(x) - C(x)$.
(b) Find the profit if $x = 12$ hundred cell phones are sold.
(c) Interpret $P(12)$.

118. Population as a Function of Age The function

$$P = P(a) = 0.027a^2 - 6.530a + 363.804$$

represents the population P (in millions) of Americans who are at least a years old in 2015.
(a) Identify the dependent and independent variables.
(b) Evaluate $P(20)$. Explain the meaning of $P(20)$.
(c) Evaluate $P(0)$. Explain the meaning of $P(0)$.

***Source:** U.S. Census Bureau*

119. Stopping Distance When the driver of a vehicle observes an impediment, the total stopping distance involves both the reaction distance R (the distance the vehicle travels while the driver moves his or her foot to the brake pedal) and the braking distance B (the distance the vehicle travels once the brakes are applied). For a car traveling at a speed of v miles per hour, the reaction distance R, in feet, can be estimated by $R(v) = 2.2v$. Suppose that the braking distance B, in feet, for a car is given by $B(v) = 0.05v^2 + 0.4v - 15$.
(a) Find the stopping distance function

$$D(v) = R(v) + B(v)$$

(b) Find the stopping distance if the car is traveling at a speed of 60 mph.
(c) Interpret $D(60)$.

120. Some functions f have the property that

$$f(a + b) = f(a) + f(b)$$

for all real numbers a and b. Which of the following functions have this property?

(a) $h(x) = 2x$ **(b)** $g(x) = x^2$
(c) $F(x) = 5x - 2$ **(d)** $G(x) = \frac{1}{x}$

121. *Challenge Problem* If $f\left(\frac{x+4}{5x-4}\right) = 3x^2 - 2$, find $f(1)$.

122. *Challenge Problem* Find the difference quotient of the function $f(x) = \sqrt[3]{x}$.
(**Hint**: Factor using $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$ with $a = \sqrt[3]{x + h}$ and $b = \sqrt[3]{x}$.)

123. *Challenge Problem* Find the domain of $f(x) = \sqrt{\frac{x^2+1}{7-|3x-1|}}$.

Explaining Concepts: Discussion and Writing

124. Are the functions $f(x) = x - 1$ and $g(x) = \dfrac{x^2 - 1}{x + 1}$ the same? Explain.

125. Find a function H that multiplies a number x by 3 and then subtracts the cube of x and divides the result by your age.

126. Investigate when, historically, the use of the function notation $y = f(x)$ first appeared.

Retain Your Knowledge

Problems 127–135 are based on material learned earlier in the course. The purpose of these problems is to keep the material fresh in your mind so that you are better prepared for the final exam.

127. List the intercepts and test for symmetry the graph of
$$(x + 12)^2 + y^2 = 16$$

128. Determine which of the given points are on the graph of the equation $y = 3x^2 - 8\sqrt{x}$.
Points: $(-1, -5)$, $(4, 32)$, $(9, 171)$

129. How many pounds of lean hamburger that is 7% fat must be mixed with 12 pounds of ground chuck that is 20% fat to have a hamburger mixture that is 15% fat?

130. Solve $x^3 - 9x = 2x^2 - 18$.

131. Given $a + bx = ac + d$, solve for a.

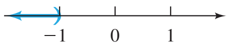
132. **Rotational Inertia** The rotational inertia I of an object varies with the square of the perpendicular distance d from the object to the axis of rotation according to the model
$$I = \frac{10}{9}d^2 \text{ (in kg} \cdot \text{m}^2\text{)}$$
What is the rotational inertia of the object if the perpendicular distance is 1.5 m?

133. Find the slope of a line perpendicular to the line
$$3x - 10y = 12$$

134. Simplify $\dfrac{(4x^2 - 7) \cdot 3 - (3x + 5) \cdot 8x}{(4x^2 - 7)^2}$.

135. Determine the degree of the polynomial
$$9x^2(3x - 5)(5x + 1)^4$$

'Are You Prepared?' Answers

1. $(-1, 3)$ **2.** 21.5 **3.** $\{x \mid x \neq -4\}$ **4.** $\{x \mid x < -1\}$ 

5. $\sqrt{5} + 2$ **6.** radicals

2.2 The Graph of a Function

PREPARING FOR THIS SECTION *Before getting started, review the following:*

- Graphs of Equations (Section 1.2, pp. 46–47)
- Intercepts (Section 1.2, pp. 48–49)

Now Work the **'Are You Prepared?'** problems on page 103.

OBJECTIVES
1 Identify the Graph of a Function (p. 100)
2 Obtain Information from or about the Graph of a Function (p. 100)

In Section 1.2 we saw how a graph can more clearly demonstrate the relationship between two variables. Consider the average gasoline price data and corresponding graph provided again for convenience in Table 2 and Figure 17.

Table 2

Year	Price	Year	Price	Year	Price
1991	1.98	2000	2.11	2009	2.68
1992	1.90	2001	1.97	2010	3.12
1993	1.81	2002	1.83	2011	3.84
1994	1.78	2003	2.07	2012	3.87
1995	1.78	2004	2.40	2013	3.68
1996	1.87	2005	2.85	2014	3.48
1997	1.83	2006	3.13	2015	2.51
1998	1.55	2007	3.31	2016	2.19
1999	1.67	2008	3.70	2017	2.38

Source: *U.S. Energy Information Administration*

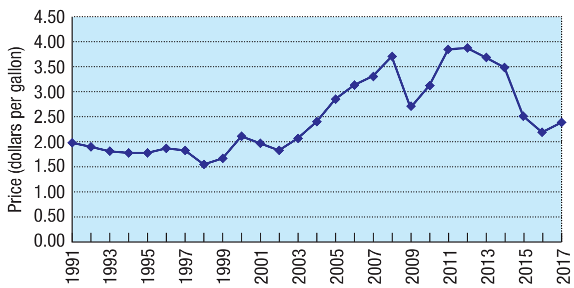


Figure 17 Average retail price of gasoline (2017 dollars)

We can see from the graph that the price of gasoline (adjusted for inflation) stayed roughly the same from 1993 to 1997 and was increasing from 2002 to 2008. The graph also shows that the lowest price occurred in 1998.

Look again at Figure 17. The graph shows that for each date on the horizontal axis, there is only one price on the vertical axis. The graph represents a function, although a rule for getting from date to price is not given.

When a function is defined by an equation in x and y, the **graph of the function** is the graph of the equation; that is, it is the set of points (x, y) in the xy-plane that satisfy the equation.

1 Identify the Graph of a Function

Not every collection of points in the xy-plane represents the graph of a function. Remember, for a function, each number x in the domain has exactly one image y in the range. This means that the graph of a function cannot contain two points with the same x-coordinate and different y-coordinates. Therefore, the graph of a function must satisfy the following *vertical-line test*.

In Words
If any vertical line intersects a graph at more than one point, the graph is not the graph of a function.

THEOREM Vertical-Line Test

A set of points in the xy-plane is the graph of a function if and only if every vertical line intersects the graph in at most one point.

EXAMPLE 1 Identifying the Graph of a Function

Which of the graphs in Figure 18 are graphs of functions?

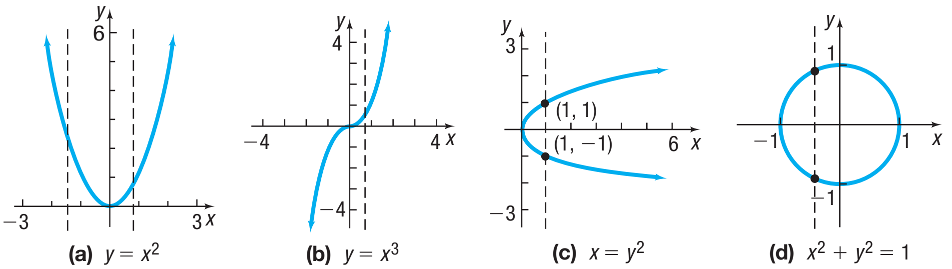


Figure 18

Solution The graphs in Figures 18(a) and 18(b) are graphs of functions, because every vertical line intersects each graph in at most one point. The graphs in Figures 18(c) and 18(d) are not graphs of functions, because there is a vertical line that intersects each graph in more than one point. Notice in Figure 18(c) that the input 1 corresponds to two outputs, -1 and 1. This is why the graph does not represent a function.

Now Work PROBLEMS 15 AND 17

2 Obtain Information from or about the Graph of a Function

If (x, y) is a point on the graph of a function f, then y is the value of f at x; that is, $y = f(x)$. Also if $y = f(x)$, then (x, y) is a point on the graph of f. For example, if $(-2, 7)$ is on the graph of f, then $f(-2) = 7$, and if $f(5) = 8$, then the point $(5, 8)$ is on the graph of $y = f(x)$.

EXAMPLE 2 Obtaining Information from the Graph of a Function

Let f be the function whose graph is given in Figure 19. (The graph of f might represent the distance y that the bob of a pendulum is from its *at-rest* position at time x. Negative values of y mean that the pendulum is to the left of the at-rest position, and positive values of y mean that the pendulum is to the right of the at-rest position.)

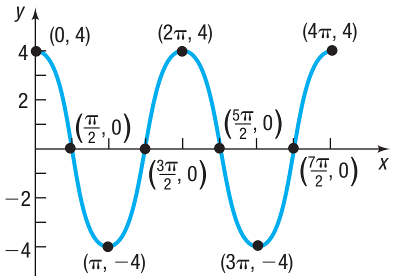


Figure 19

(a) What are $f(0)$, $f\left(\frac{3\pi}{2}\right)$, and $f(3\pi)$?

(b) What is the domain of f?

(c) What is the range of f?

(d) List the intercepts. (Recall that these are the points, if any, where the graph crosses or touches the coordinate axes.)

(e) How many times does the line $y = 2$ intersect the graph?

(f) For what values of x does $f(x) = -4$?

(g) For what values of x is $f(x) > 0$?

Solution

(a) Since $(0, 4)$ is on the graph of f, the y-coordinate 4 is the value of f at the x-coordinate 0; that is, $f(0) = 4$. In a similar way, when $x = \frac{3\pi}{2}$, then $y = 0$, so $f\left(\frac{3\pi}{2}\right) = 0$. When $x = 3\pi$, then $y = -4$, so $f(3\pi) = -4$.

(b) To determine the domain of f, notice that the points on the graph of f have x-coordinates between 0 and 4π, inclusive; and for each number x between 0 and 4π, there is a point $(x, f(x))$ on the graph. The domain of f is $\{x \mid 0 \le x \le 4\pi\}$ or the interval $[0, 4\pi]$.

(c) The points on the graph all have y-coordinates between -4 and 4, inclusive; and for each such number y, there is at least one number x in the domain. The range of f is $\{y \mid -4 \le y \le 4\}$ or the interval $[-4, 4]$.

(d) The intercepts are the points

$$(0, 4),\ \left(\frac{\pi}{2}, 0\right),\ \left(\frac{3\pi}{2}, 0\right),\ \left(\frac{5\pi}{2}, 0\right),\quad \text{and} \quad \left(\frac{7\pi}{2}, 0\right)$$

(e) Draw the horizontal line $y = 2$ on the graph in Figure 19. Notice that the line intersects the graph four times.

(f) Since $(\pi, -4)$ and $(3\pi, -4)$ are the only points on the graph for which $y = f(x) = -4$, we have $f(x) = -4$ when $x = \pi$ and $x = 3\pi$.

(g) To determine where $f(x) > 0$, look at Figure 19 and determine the x-values from 0 to 4π for which the y-coordinate is positive. This occurs on $\left[0, \frac{\pi}{2}\right) \cup \left(\frac{3\pi}{2}, \frac{5\pi}{2}\right) \cup \left(\frac{7\pi}{2}, 4\pi\right]$. Using inequality notation, $f(x) > 0$ for $0 \le x < \frac{\pi}{2}$ or $\frac{3\pi}{2} < x < \frac{5\pi}{2}$ or $\frac{7\pi}{2} < x \le 4\pi$.

RECALL
The symbol $\cup$ stands for union. It means the set of elements that are in either of two sets.

When the graph of a function is given, its domain may be viewed as the shadow created by the graph on the x-axis by vertical beams of light. Its range can be viewed as the shadow created by the graph on the y-axis by horizontal beams of light. Try this technique with the graph given in Figure 19.

Now Work PROBLEM 11

EXAMPLE 3 Obtaining Information about the Graph of a Function

Consider the function: $f(x) = \dfrac{x + 1}{x + 2}$

(a) Find the domain of f.

(b) Is the point $\left(1, \frac{1}{2}\right)$ on the graph of f?

(c) If $x = 2$, what is $f(x)$? What point is on the graph of f?

(d) If $f(x) = 2$, what is x? What point is on the graph of f?

(e) What are the x-intercepts of the graph of f (if any)? What corresponding point(s) are on the graph of f?

Solution (a) The domain of f is $\{x \mid x \neq -2\}$.

(b) When $x = 1$, then

$$f(1) = \frac{1+1}{1+2} = \frac{2}{3} \qquad f(x) = \frac{x+1}{x+2}$$

The point $\left(1, \frac{2}{3}\right)$ is on the graph of f; the point $\left(1, \frac{1}{2}\right)$ is not.

(c) If $x = 2$, then

$$f(2) = \frac{2+1}{2+2} = \frac{3}{4}$$

The point $\left(2, \frac{3}{4}\right)$ is on the graph of f.

(d) If $f(x) = 2$, then

$$\begin{aligned} \frac{x+1}{x+2} &= 2 && f(x) = 2 \\ x + 1 &= 2(x+2) && \text{Multiply both sides by } x + 2. \\ x + 1 &= 2x + 4 && \text{Distribute.} \\ x &= -3 && \text{Solve for } x. \end{aligned}$$

NOTE In Example 3, -2 is not in the domain of f, so $x + 2$ is not zero and we can multiply both sides of the equation by $x + 2$. ■

If $f(x) = 2$, then $x = -3$. The point $(-3, 2)$ is on the graph of f.

(e) The x-intercepts of the graph of f are the real solutions of the equation $f(x) = 0$ that are in the domain of f.

$$\begin{aligned} \frac{x+1}{x+2} &= 0 && \\ x + 1 &= 0 && \text{Multiply both sides by } x + 2. \\ x &= -1 && \text{Subtract 1 from both sides.} \end{aligned}$$

The only real solution of the equation $f(x) = \frac{x+1}{x+2} = 0$ is $x = -1$, so -1 is the only x-intercept. Since $f(-1) = 0$, the point $(-1, 0)$ is on the graph of f.

Now Work PROBLEM 27

EXAMPLE 4 Energy Expended

For an individual walking, the energy expended E in terms of speed v can be approximated by

$$E(v) = \frac{29}{v} + 0.0053v$$

where E has units of cal/min/kg and v has units of m/min.

(a) Find the energy expended for a speed of $v = 40$ m/min.
(b) Find the energy expended for a speed of $v = 70$ m/min.
(c) Find the energy expended for a speed of $v = 100$ m/min.
(d) Graph the function $E = E(v), 0 < v \leq 200$
(e) Create a TABLE with TblStart $= 1$ and ΔTbl $= 1$. Which value of v minimizes the energy expended?

Source: Ralston, H.J. Int. Z. Angew. Physiol. Einschl. Arbeitsphysiol. *(1958) 17:277.*

Solution (a) The energy expended for a walking speed of $v = 40$ meters per minute is

$$E(40) = \frac{29}{40} + 0.0053 \cdot 40 \approx 0.94 \text{ cal/min/kg}$$

(b) The energy expended for a walking speed of $v = 70$ meters per minute is

$$E(70) = \frac{29}{70} + 0.0053 \cdot 70 \approx 0.79 \text{ cal/min/kg}$$

(c) The energy expended for a walking speed of $v = 100$ meters per minute is

$$E(100) = \frac{29}{100} + 0.0053 \cdot 100 = 0.82 \text{ cal/min/kg}$$

(d) See Figure 20 for the graph of $E = E(v)$ on a TI-84 Plus C.

(e) With the function $E = E(v)$ in Y_1, we create Table 3. Scroll down to find a value of v for which Y_1 is smallest. Table 4 shows that a walking speed of $v = 74$ meters per minute minimizes the expended energy at about 0.784 cal/min/kg.

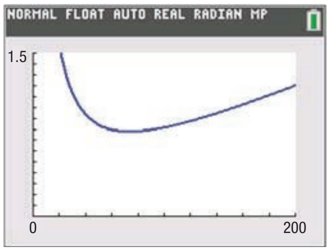


Figure 20 $E(v) = \dfrac{29}{v} + 0.0053v$

Table 3

NORMAL FLOAT AUTO REAL RADIAN MP
PRESS ENTER TO EDIT

X	Y1
1	29.005
2	14.511
3	9.6826
4	7.2712
5	5.8265
6	4.8651
7	4.18
8	3.6674
9	3.2699
10	2.953
11	2.6947

Y1=29/X+.0053X

Table 4

NORMAL FLOAT AUTO REAL RADIAN MP
PRESS ◆ TO EDIT FUNCTION

X	Y1
70	.78529
71	.78475
72	.78438
73	.78416
74	.78409
75	.78417
76	.78438
77	.78472
78	.78519
79	.78579
80	.7865

Y1=.784091891892

Now Work PROBLEM 35

SUMMARY

- **Graph of a Function** The set of points (x, y) in the xy-plane that satisfy the equation $y = f(x)$.
- **Vertical-Line Test** A set of points in the xy-plane is the graph of a function if and only if every vertical line intersects the graph in at most one point.

2.2 Assess Your Understanding

'Are You Prepared?' *Answers are given at the end of these exercises. If you get a wrong answer, read the pages listed in red.*

1. The intercepts of the equation $x^2 + 4y^2 = 16$ are ________. (pp. 48–49)

2. ***True or False*** The point $(-2, -6)$ is on the graph of the equation $x = 2y - 2$. (pp. 46–47)

Concepts and Vocabulary

3. A set of points in the xy-plane is the graph of a function if and only if every __________ line intersects the graph in at most one point.

4. If the point $(5, -3)$ is a point on the graph of f, then $f(\_\_\_\_) = \_\_\_\_$.

5. Find a so that the point $(-1, 2)$ is on the graph of $f(x) = ax^2 + 4$.

6. ***True or False*** Every graph represents a function.

7. ***True or False*** The graph of a function $y = f(x)$ always crosses the y-axis.

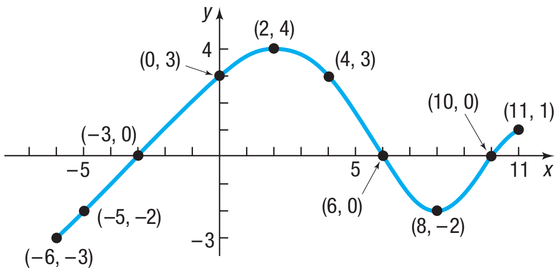
8. ***True or False*** The y-intercept of the graph of the function $y = f(x)$, whose domain is all real numbers, is $f(0)$.

9. ***Multiple Choice*** If a function is defined by an equation in x and y, then the set of points (x, y) in the xy-plane that satisfy the equation is called the __________.
   (a) domain of the function **(b)** range of the function
   (c) graph of the function **(d)** relation of the function

10. ***Multiple Choice*** The graph of a function $y = f(x)$ can have more than one of which type of intercept?
    (a) x-intercept **(b)** y-intercept **(c)** both **(d)** neither

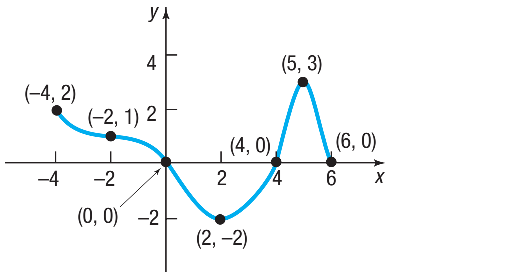
Skill Building

11. Use the given graph of the function f to answer parts (a)–(n).



(a) Find $f(0)$ and $f(-6)$.
(b) Find $f(6)$ and $f(11)$.
(c) Is $f(3)$ positive or negative?
(d) Is $f(-4)$ positive or negative?
(e) For what values of x is $f(x) = 0$?
(f) For what values of x is $f(x) > 0$?
(g) What is the domain of f?
(h) What is the range of f?
(i) What are the x-intercepts?
(j) What is the y-intercept?
(k) How often does the line $y = \frac{1}{2}$ intersect the graph?
(l) How often does the line $x = 5$ intersect the graph?
(m) For what values of x does $f(x) = 3$?
(n) For what values of x does $f(x) = -2$?

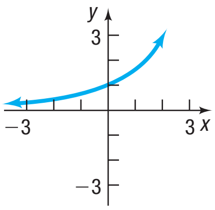
12. Use the given graph of the function f to answer parts (a)–(n).



(a) Find $f(0)$ and $f(6)$.
(b) Find $f(2)$ and $f(-2)$.
(c) Is $f(3)$ positive or negative?
(d) Is $f(-1)$ positive or negative?
(e) For what values of x is $f(x) = 0$?
(f) For what values of x is $f(x) < 0$?
(g) What is the domain of f?
(h) What is the range of f?
(i) What are the x-intercepts?
(j) What is the y-intercept?
(k) How often does the line $y = -1$ intersect the graph?
(l) How often does the line $x = 1$ intersect the graph?
(m) For what value of x does $f(x) = 3$?
(n) For what value of x does $f(x) = -2$?

In Problems 13–24, determine whether or not the graph is that of a function by using the vertical-line test. In either case, use the graph to find:
(a) The domain and range *(b) The intercepts, if any* *(c) Any symmetry with respect to the x-axis, the y-axis, or the origin*

13.



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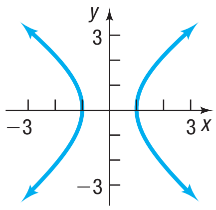

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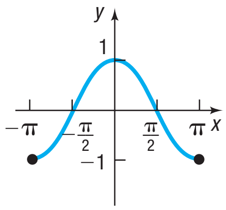

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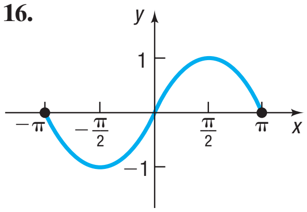

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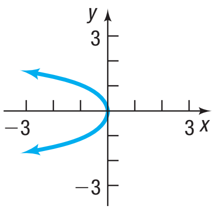

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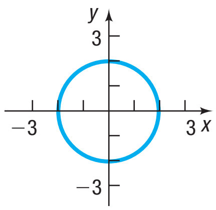
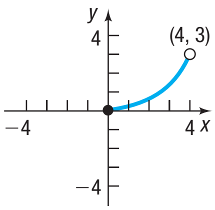


19.



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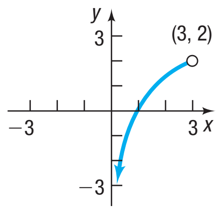
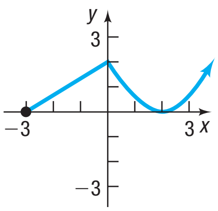


21.



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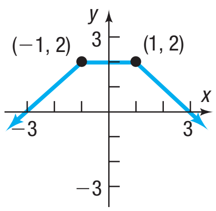

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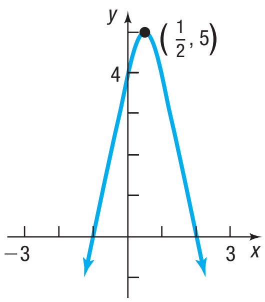

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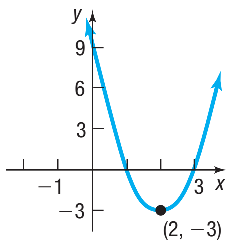


In Problems 25–30, answer the questions about each function.

25. $f(x) = -3x^2 + 5x$

(a) Is the point $(-1, 2)$ on the graph of f?

(b) If $x = -2$, what is $f(x)$? What point is on the graph of f?

(c) If $f(x) = -2$, what is x? What point(s) are on the graph of f?

(d) What is the domain of f?

(e) List the x-intercepts, if any, of the graph of f.

(f) List the y-intercept, if there is one, of the graph of f.

26. $f(x) = 3x^2 + x - 2$

(a) Is the point $(1, 2)$ on the graph of f?

(b) If $x = -2$, what is $f(x)$? What point is on the graph of f?

(c) If $f(x) = -2$, what is x? What point(s) are on the graph of f?

(d) What is the domain of f?

(e) List the x-intercepts, if any, of the graph of f.

(f) List the y-intercept, if there is one, of the graph of f.

27. $f(x) = \dfrac{x+2}{x-6}$

(a) Is the point $(3, 14)$ on the graph of f?

(b) If $x = 4$, what is $f(x)$? What point is on the graph of f?

(c) If $f(x) = 2$, what is x? What point(s) are on the graph of f?

(d) What is the domain of f?

(e) List the x-intercepts, if any, of the graph of f.

(f) List the y-intercept, if there is one, of the graph of f.

28. $f(x) = \dfrac{x^2+2}{x+4}$

(a) Is the point $\left(1, \dfrac{3}{5}\right)$ on the graph of f?

(b) If $x = 0$, what is $f(x)$? What point is on the graph of f?

(c) If $f(x) = \dfrac{1}{2}$, what is x? What point(s) are on the graph of f?

(d) What is the domain of f?

(e) List the x-intercepts, if any, of the graph of f.

(f) List the y-intercept, if there is one, of the graph of f.

29. $f(x) = \dfrac{2x}{x-2}$

(a) Is the point $\left(\dfrac{1}{2}, -\dfrac{2}{3}\right)$ on the graph of f?

(b) If $x = 4$, what is $f(x)$? What point is on the graph of f?

(c) If $f(x) = 1$, what is x? What point(s) are on the graph of f?

(d) What is the domain of f?

(e) List the x-intercepts, if any, of the graph of f.

(f) List the y-intercept, if there is one, of the graph of f.

30. $f(x) = \dfrac{12x^4}{x^2+1}$

(a) Is the point $(-1, 6)$ on the graph of f?

(b) If $x = 3$, what is $f(x)$? What point is on the graph of f?

(c) If $f(x) = 1$, what is x? What point(s) are on the graph of f?

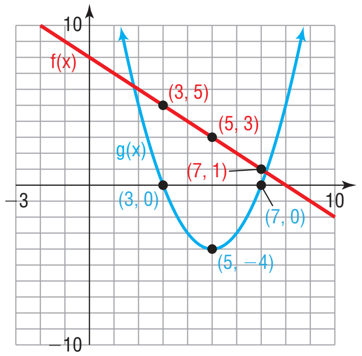
(d) What is the domain of f?

(e) List the x-intercepts, if any, of the graph of f.

(f) List the y-intercept, if there is one, of the graph of f.

Applications and Extensions

31. The graphs of two functions, f and g, is illustrated below. Use the graph to answer parts (a) through (f).



(a) $(f+g)(3) = ?$ **(b)** $(f+g)(5) = ?$

(c) $(f-g)(7) = ?$ **(d)** $(g-f)(7) = ?$

(e) $(f \cdot g)(3) = ?$ **(f)** $\left(\dfrac{f}{g}\right)(5) = ?$

32. If a player launched a shot at a 30-degree angle from a point 5.5 feet above the floor, then the function $h(x) = \dfrac{-21x^2}{v^2} + 0.6x + 5.5$ approximates its height. In the function, h is the initial height of the shot's point of launch, x is the shot's horizontal distance traveled, and v is the magnitude of the shot's initial velocity in feet per second.

(a) Assuming that the height of the shot is approximately 9.9 feet when the horizontal distance traveled by the shot is 15 feet, determine the magnitude of the shot's initial velocity.

(b) Write a function for the path of the shot using the velocity found in part (a).

(c) Determine the height of the shot after it has traveled a horizontal distance of 20 feet.

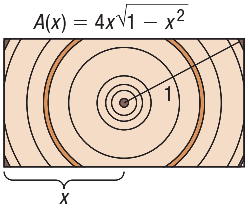
(d) Find some additional points, and graph the path of the shot.

33. If a football player kicks a ball at a 60-degree angle with respect to the ground from a position 1 foot above the ground, then the function $h(x) = \dfrac{-64x^2}{v^2} + 1.7x + 1$ models the ball's trajectory. In this equation, h is the height of the ball above the ground, x is the forward distance traversed by the ball, and v is the magnitude of the initial velocity the ball was kicked with measured in feet per second. Suppose a player kicks a ball with an initial velocity having a magnitude of 28 feet per second.

(continued)

(a) Find the height of the ball after it has traversed 5 feet in front of the goalpost.
(b) Find the height of the ball after it has traversed 10 feet in front of the goalpost.
(c) Find some additional points, and graph the path of the ball.
(d) The goalpost's crossbar is 8 feet high, and the goal is 20 feet away from the initial point of contact. Will the ball hit the goalpost's crossbar? Why or why not? If not, then what initial velocity must the ball have to hit the goalpost's crossbar, assuming the same point of contact, magnitude, and direction of initial velocity?

34. **Cross-sectional Area** The cross-sectional area of a beam cut from a log with radius 1 foot is given by the function $A(x) = 4x\sqrt{1 - x^2}$, where x represents the length, in feet, of half the base of the beam. See the figure.
(a) Find the domain of A.
(b) Use a graphing utility to graph the function $A = A(x)$.
(c) Create a TABLE with TblStart $= 0$ and ΔTbl $= 0.1$ for $0 \le x \le 1$. Which value of x maximizes the cross-sectional area? What should be the length of the base of the beam to maximize the cross-sectional area?



35. **Motion of a Golf Ball** A golf ball is hit with an initial velocity of 130 feet per second at an inclination of 45° to the horizontal. In physics, it is established that the height h of the golf ball is given by the function

$$h(x) = \frac{-32x^2}{130^2} + x$$

where x is the horizontal distance that the golf ball has traveled.
(a) Determine the height of the golf ball after it has traveled 100 feet.
(b) What is the height after it has traveled 300 feet?
(c) What is $h(500)$? Interpret this value.
(d) How far was the golf ball hit?
(e) Use a graphing utility to graph the function $h = h(x)$.
(f) Use a graphing utility to determine the distance that the ball has traveled when the height of the ball is 90 feet.
(g) Create a TABLE with TblStart $= 0$ and ΔTbl $= 25$. To the nearest 25 feet, how far does the ball travel before it reaches a maximum height? What is the maximum height?
(h) Adjust the value of ΔTbl until you determine the distance, to within 1 foot, that the ball travels before it reaches its maximum height.

36. **Effect of Elevation on Weight** If an object weighs m pounds at sea level, then its weight W (in pounds) at a height of h miles above sea level is given approximately by

$$W(h) = m\left(\frac{4000}{4000 + h}\right)^2$$

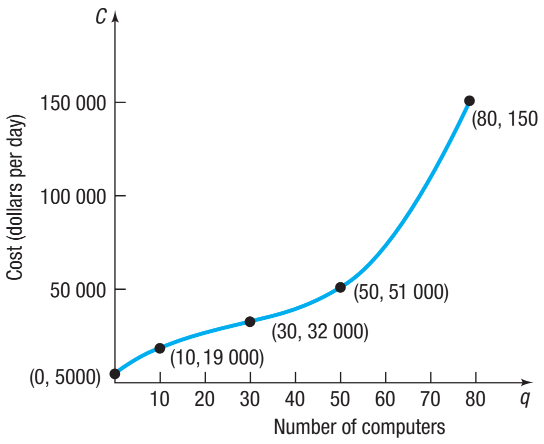
(a) If Amy weighs 120 pounds at sea level, how much will she weigh on Pikes Peak, which is 14,110 feet above sea level?
(b) Use a graphing utility to graph the function $W = W(h)$. Use $m = 120$ pounds.
(c) Create a TABLE with TblStart $= 0$ and ΔTbl $= 0.5$ to see how the weight W varies as h changes from 0 to 5 miles.
(d) At what height will Amy weigh 119.95 pounds?
(e) Does your answer to part (d) seem reasonable? Explain.

37. **Cost of Transatlantic Travel** A Boeing 747 crosses the Atlantic Ocean (3000 miles) with an airspeed of 500 miles per hour. The cost C (in dollars) per passenger is given by

$$C(x) = 100 + \frac{x}{10} + \frac{36{,}000}{x}$$

where x is the groundspeed (airspeed $\pm$ wind).
(a) What is the cost when the groundspeed is 480 miles per hour? 600 miles per hour?
(b) Find the domain of C.
(c) Use a graphing utility to graph the function $C = C(x)$.
(d) Create a TABLE with TblStart $= 0$ and ΔTbl $= 50$.
(e) To the nearest 50 miles per hour, what groundspeed minimizes the cost per passenger?

38. **Reading and Interpreting Graphs** Let C be the function whose graph is given below. This graph represents the cost C of manufacturing q computers in a day.
(a) Find $C(0)$. Interpret this value.
(b) Find $C(10)$. Interpret this value.
(c) Find $C(50)$. Interpret this value.
(d) What is the domain of C? What does this domain imply in terms of daily production?
(e) Describe the shape of the graph.
(f) The point (30, 32 000) is called an *inflection point*. Describe the behavior of the graph around the inflection point.



39. Reading and Interpreting Graphs Let C be the function whose graph is given below. This graph represents the cost C of using g gigabytes of data in a month for a data-only plan.

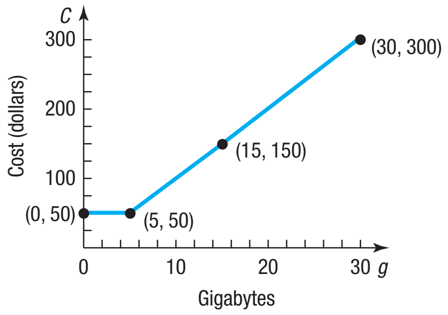
(a) Find $C(0)$. Interpret this value.

(b) Find $C(5)$. Interpret this value.

(c) Find $C(15)$. Interpret this value.

(d) What is the domain of C? What does this domain imply in terms of the number of gigabytes?

(e) Describe the shape of the graph.



40. *Challenge Problem* Suppose $f(x) = x^2 - 4x + c$ and $g(x) = \dfrac{f(x)}{3} - 4$. Find $f(3)$ if $g(-2) = 5$.

41. *Challenge Problem* Suppose $f(x) = \sqrt{x} + 2$ and $g(x) = x^2 + n$. If $f(g(5)) = 4$, what is the value of $g(n)$?

Explaining Concepts: Discussion and Writing

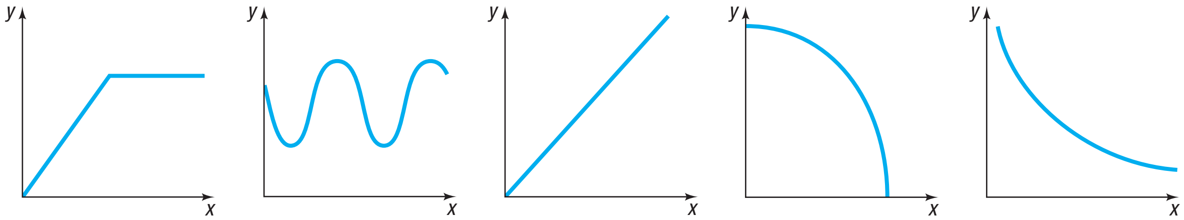
42. Describe how you would find the domain and range of a function if you were given its graph. How would your strategy change if you were given the equation defining the function instead of its graph?

43. How many x-intercepts can the graph of a function have? How many y-intercepts can the graph of a function have? Explain why.

44. Is a graph that consists of a single point the graph of a function? Can you write the equation of such a function?

45. Match each of the following functions with the graph that best describes the situation.

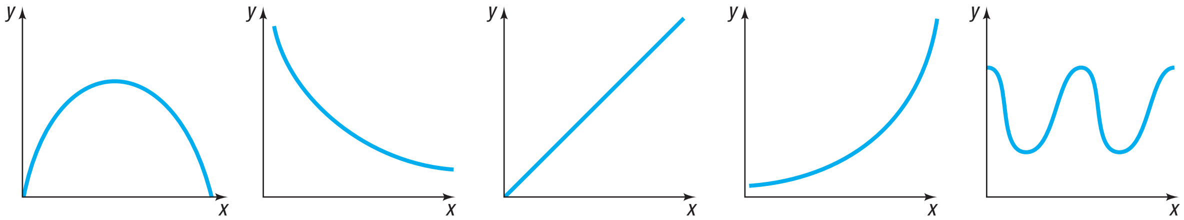
(a) The cost of building a house as a function of its square footage
(b) The height of an egg dropped from a 300-foot building as a function of time
(c) The height of a human as a function of time
(d) The demand for Big Macs as a function of price
(e) The height of a child on a swing as a function of time



(I) (II) (III) (IV) (V)

46. Match each of the following functions with the graph that best describes the situation.

(a) The temperature of a bowl of soup as a function of time
(b) The number of hours of daylight per day over a 2-year period
(c) The population of Florida as a function of time
(d) The distance traveled by a car going at a constant velocity as a function of time
(e) The height of a golf ball hit with a 7-iron as a function of time

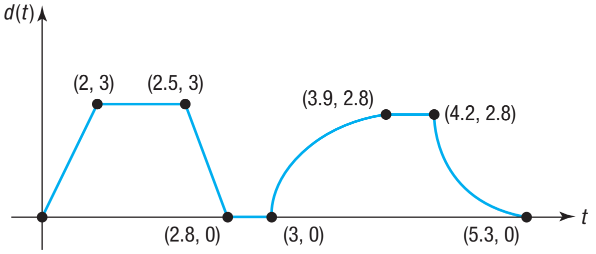


(I) (II) (III) (IV) (V)

47. Consider the following scenario: Barbara decides to take a walk. She leaves home, walks 2 blocks in 5 minutes at a constant speed, and realizes that she forgot to lock the door. So Barbara runs home in 1 minute. While at her doorstep, it takes her 1 minute to find her keys and lock the door. Barbara walks 5 blocks in 15 minutes and then decides to jog home. It takes her 7 minutes to get home. Draw a graph of Barbara's distance from home (in blocks) as a function of time.

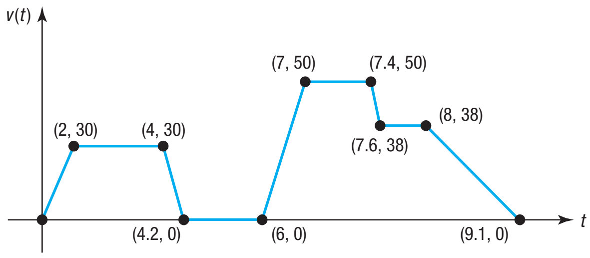
48. Consider the following scenario: Jayne enjoys riding her bicycle through the woods. At the forest preserve, she gets on her bicycle and rides up a 2000-foot incline in 10 minutes. She then travels down the incline in 3 minutes. The next 5000 feet is level terrain, and she covers the distance in 20 minutes. She rests for 15 minutes. Jayne then travels 10,000 feet in 30 minutes. Draw a graph of Jayne's distance traveled (in feet) as a function of time.

49. The graph below represents the distance d (in miles) that Kevin was from home as a function of time t (in hours). Answer the questions by referring to the graph. In parts (a)–(g), how many hours elapsed and how far was Kevin from home during the times listed?



(a) From $t = 0$ to $t = 2$ **(b)** From $t = 2$ to $t = 2.5$
(c) From $t = 2.5$ to $t = 2.8$ **(d)** From $t = 2.8$ to $t = 3$
(e) From $t = 3$ to $t = 3.9$ **(f)** From $t = 3.9$ to $t = 4.2$
(g) From $t = 4.2$ to $t = 5.3$
(h) What is the farthest distance that Kevin was from home?
(i) How many times did Kevin return home?

50. The graph below represents the speed v (in miles per hour) of Michael's car as a function of time t (in minutes).



(a) Over what interval of time was Michael traveling fastest?
(b) Over what interval(s) of time was Michael's speed zero?
(c) What was Michael's speed between 0 and 2 minutes?
(d) What was Michael's speed between 4.2 and 6 minutes?
(e) What was Michael's speed between 7 and 7.4 minutes?
(f) When was Michael's speed constant?

51. Graph a function whose domain is

$$\{x \mid -3 \le x \le 8, \quad x \ne 5\}$$

and whose range is

$$\{y \mid -1 \le y \le 2, \quad y \ne 0\}$$

What point(s) in the rectangle $-3 \le x \le 8$, $-1 \le y \le 2$ cannot be on the graph? Compare your graph with those of other students. What differences do you see?

52. Is there a function whose graph is symmetric with respect to the x-axis? Explain.

53. Explain why the vertical-line test works.

Retain Your Knowledge

Problems 54–63 are based on material learned earlier in the course. The purpose of these problems is to keep the material fresh in your mind so that you are better prepared for the final exam.

54. If $f(x) = -x^2 + x - 3$, find $f(x-2)$.

55. Find the distance between the points $(3, -6)$ and $(1, 0)$.

56. Write an equation of the line with slope $\frac{2}{3}$ that contains the point $(-6, 4)$.

57. Find the domain of $g(x) = \sqrt[3]{x+4} - 5$.

58. Find the number that must be added to $m^2 - 12m$ to complete the square.

59. Rationalize the numerator: $\dfrac{\sqrt{x} - \sqrt{6}}{x - 6}$

60. Two cars leave an intersection at the same time, one traveling north at 25 mph and the other traveling west at 35 mph. How long will it take for the cars to be 40 miles apart?

61. Find the numbers x that satisfy both of the inequalities

$$3x + 4 \le 7 \text{ and } 5 - 2x < 13$$

62. Simplify $(5x^2 - 7x + 2) - (8x - 10)$.

63. Write the inequality $-3 \le x \le 10$ in interval notation.

'Are You Prepared?' Answers

1. $(-4, 0), (4, 0), (0, -2), (0, 2)$ **2.** False

2.3 Properties of Functions

PREPARING FOR THIS SECTION *Before getting started, review the following:*

- Intervals (Section A.9, pp. A72–A74)
- Intercepts (Section 1.2, pp. 48–49)
- Slope of a Line (Section 1.3, pp. 56–59)
- Point-Slope Form of a Line (Section 1.3, pp. 60–61)
- Symmetry (Section 1.2, pp. 49–51)

Now Work the **'Are You Prepared?'** problems on page 117.

OBJECTIVES
1. Identify Even and Odd Functions from a Graph (p. 109)
2. Identify Even and Odd Functions from an Equation (p. 110)
3. Use a Graph to Determine Where a Function Is Increasing, Decreasing, or Constant (p. 111)
4. Use a Graph to Locate Local Maxima and Local Minima (p. 112)
5. Use a Graph to Locate the Absolute Maximum and the Absolute Minimum (p. 113)
6. Use a Graphing Utility to Approximate Local Maxima and Local Minima and to Determine Where a Function Is Increasing or Decreasing (p. 115)
7. Find the Average Rate of Change of a Function (p. 115)

To obtain the graph of a function $y = f(x)$, it is often helpful to know properties of the function and the impact of these properties on the graph of the function.

1 Identify Even and Odd Functions from a Graph

The words *even* and *odd*, when discussing a function *f*, describe the symmetry of the graph of the function.

A function *f* is even if and only if, whenever the point (x, y) is on the graph of *f*, the point $(-x, y)$ is also on the graph. Using function notation, we define an even function as follows:

DEFINITION Even Function

A function *f* is **even** if, for every number x in its domain, the number $-x$ is also in the domain and

$$\boxed{f(-x) = f(x)}$$

A function *f* is odd if and only if, whenever the point (x, y) is on the graph of *f*, the point $(-x, -y)$ is also on the graph. Using function notation, we define an odd function as follows:

DEFINITION Odd Function

A function *f* is **odd** if, for every number x in its domain, the number $-x$ is also in the domain and

$$\boxed{f(-x) = -f(x)}$$

Refer to page 50, where the tests for symmetry are listed. The results below follow.

THEOREM Graphs of Even and Odd Functions

- A function is even if and only if its graph is symmetric with respect to the y-axis.
- A function is odd if and only if its graph is symmetric with respect to the origin.

EXAMPLE 1 Identifying Even and Odd Functions from a Graph

Determine whether each graph given in Figure 21 is the graph of an even function, an odd function, or a function that is neither even nor odd.

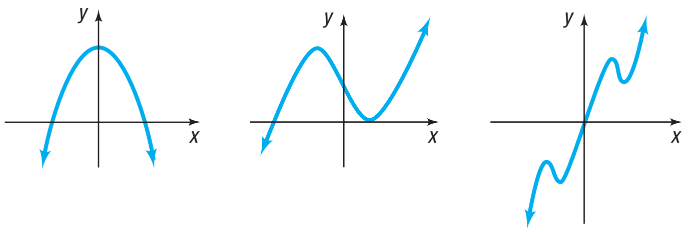


Figure 21 (a) (b) (c)

Solution (a) The graph in Figure 21(a) is that of an even function, because the graph is symmetric with respect to the y-axis.

(b) The function whose graph is given in Figure 21(b) is neither even nor odd, because the graph is neither symmetric with respect to the y-axis nor symmetric with respect to the origin.

(c) The function whose graph is given in Figure 21(c) is odd, because its graph is symmetric with respect to the origin.

Now Work PROBLEMS 25(a), (b), AND (d)

2 Identify Even and Odd Functions from an Equation

EXAMPLE 2 Identifying Even and Odd Functions Algebraically

Determine whether each of the following functions is even, odd, or neither. Then determine whether the graph is symmetric with respect to the y-axis, with respect to the origin, or neither.

(a) $f(x) = x^2 - 5$ (b) $g(x) = x^3 - 1$

(c) $h(x) = 5x^3 - x$ (d) $F(x) = |x|$

Solution (a) To determine whether f is even, odd, or neither, replace x by $-x$ in $f(x) = x^2 - 5$.

$$f(-x) = (-x)^2 - 5 = x^2 - 5 = f(x)$$

Since $f(-x) = f(x)$, the function is even, and the graph of f is symmetric with respect to the y-axis.

(b) Replace x by $-x$ in $g(x) = x^3 - 1$.

$$g(-x) = (-x)^3 - 1 = -x^3 - 1$$

Since $g(-x) \neq g(x)$ and $g(-x) \neq -g(x) = -(x^3 - 1) = -x^3 + 1$, the function is neither even nor odd. The graph of g is not symmetric with respect to the y-axis, nor is it symmetric with respect to the origin.

(c) Replace x by $-x$ in $h(x) = 5x^3 - x$.

$$h(-x) = 5(-x)^3 - (-x) = -5x^3 + x = -(5x^3 - x) = -h(x)$$

Since $h(-x) = -h(x)$, h is an odd function, and the graph of h is symmetric with respect to the origin.

(d) Replace x by $-x$ in $F(x) = |x|$.

$$F(-x) = |-x| = |-1| \cdot |x| = |x| = F(x)$$

Since $F(-x) = F(x)$, F is an even function, and the graph of F is symmetric with respect to the y-axis.

Now Work PROBLEM 37

3 Use a Graph to Determine Where a Function Is Increasing, Decreasing, or Constant

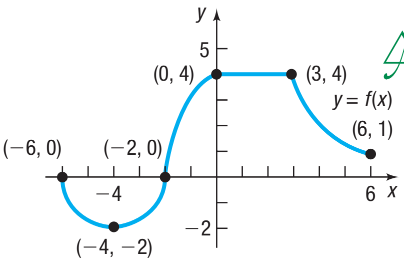


Figure 22

Consider the graph given in Figure 22. If you look from left to right along the graph of the function, you will notice that parts of the graph are going up, parts are going down, and parts are horizontal. In such cases, the function is described as *increasing*, *decreasing*, or *constant*, respectively.

In Words

- If a function is increasing, then as the values of x get bigger, the values of the function also get bigger.
- If a function is decreasing, then as the values of x get bigger, the values of the function get smaller.
- If a function is constant, then as the values of x get bigger, the values of the function remain unchanged.

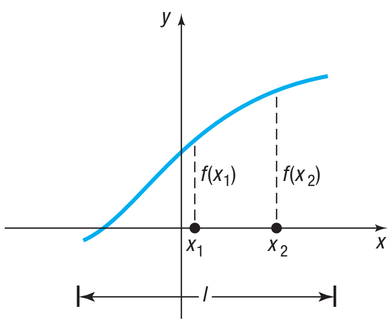
DEFINITIONS Increasing Function, Decreasing Function, Constant Function

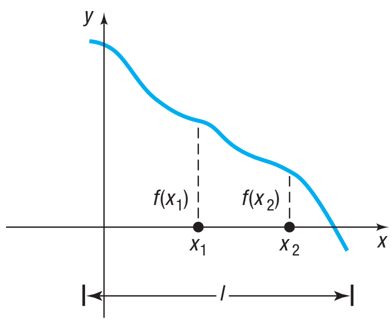
- A function f is **increasing** on an interval I if, for any choice of x_1 and x_2 in I, with $x_1 < x_2$, then $f(x_1) < f(x_2)$.
- A function f is **decreasing** on an interval I if, for any choice of x_1 and x_2 in I, with $x_1 < x_2$, then $f(x_1) > f(x_2)$.
- A function f is **constant** on an interval I if, for all choices of x in I, the values $f(x)$ are equal.

Need to Review?

Interval Notation is discussed in Section A.9, pp. A72–A74.

Figure 23 illustrates the definitions. The graph of an increasing function goes up from left to right, the graph of a decreasing function goes down from left to right, and the graph of a constant function remains at a fixed height. The interval I on which a function is increasing, decreasing, or constant may be open, closed, or half-open/half-closed depending on whether the endpoints of the interval satisfy the required inequality or not.





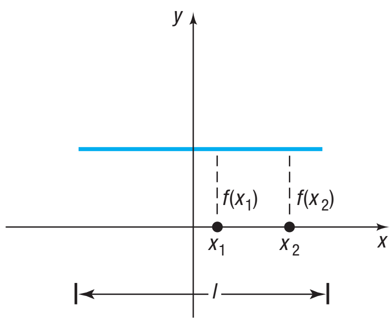


Figure 23

(a) For $x_1 < x_2$ in I, $f(x_1) < f(x_2)$; f is increasing on I.

(b) For $x_1 < x_2$ in I, $f(x_1) > f(x_2)$; f is decreasing on I.

(c) For all x in I, the values of f are equal; f is constant on I.

EXAMPLE 3 Determining Where a Function Is Increasing, Decreasing, or Constant from Its Graph

Determine the values of x for which the function in Figure 24 is increasing. Where is it decreasing? Where is it constant?

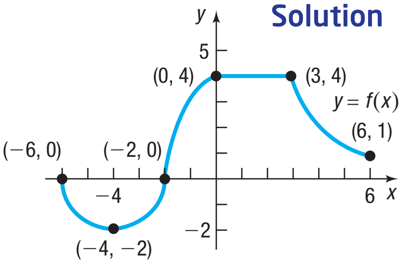


Figure 24

Solution When determining where a function is increasing, where it is decreasing, and where it is constant, we use intervals involving the independent variable x.

The function whose graph is given in Figure 24 is decreasing on the interval $[-6, -4]$ because for any choice of x_1 and x_2 in the interval for which $x_1 < x_2$, we have $f(x_1) > f(x_2)$. For example, -6 is included in the interval where f is decreasing because if x is any number for which $-6 < x \leq -4$, then $f(-6) > f(x)$.

Similarly, f is increasing on the interval $[-4, 0]$ because for any choice of x_1 and x_2 in the interval for which $x_1 < x_2$, we have $f(x_1) < f(x_2)$.

Finally, the function f is constant on the interval [0, 3] and is decreasing on the interval [3, 6].

Now Work PROBLEMS 13, 15, 17, AND 25(c)

4 Use a Graph to Locate Local Maxima and Local Minima

Suppose f is a function defined on an open interval I containing c. If the value of f at c is greater than or equal to the other values of f on I, then f has a *local maximum* at c.† See Figure 25(a).

If the value of f at c is less than or equal to the other values of f on I, then f has a *local minimum* at c. See Figure 25(b).

NOTE "Maxima" is the plural of "maximum"; "minima" is the plural of "minimum."

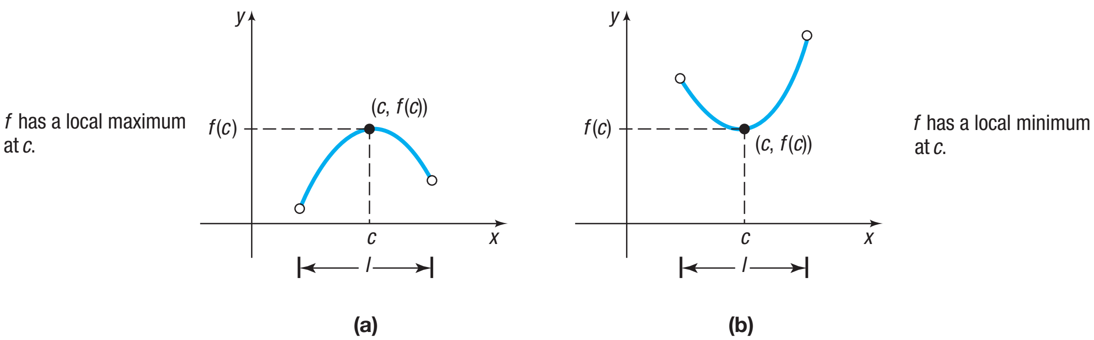


Figure 25 Local maximum and local minimum

DEFINITIONS Local Maximum/Minimum

Let f be a function defined on some interval I and let c be a number in I.

- A function f has a **local maximum** at c if there is an open interval in I containing c so that $f(c) \geq f(x)$ for all x in this open interval. The number $f(c)$ is called a **local maximum value of** $\boldsymbol{f}$.
- A function f has a **local minimum** at c if there is an open interval in I containing c so that $f(c) \leq f(x)$ for all x in this open interval. The number $f(c)$ is called a **local minimum value of** $\boldsymbol{f}$.

If f has a local maximum at c, then the value of f at c is greater than or equal to the values of f near c. If f has a local minimum at c, then the value of f at c is less than or equal to the values of f near c. The word *local* is used to suggest that it is only near c, not necessarily over the entire domain, that the value $f(c)$ has these properties.

†Some texts use the term *relative* instead of *local*.

EXAMPLE 4

Finding Local Maxima and Local Minima from the Graph of a Function and Determining Where the Function Is Increasing, Decreasing, or Constant

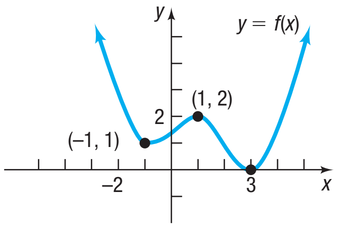


Figure 26

Figure 26 shows the graph of a function f.

(a) At what numbers x, if any, does f have a local maximum? List the local maximum value(s).

(b) At what numbers x, if any, does f have a local minimum? List the local minimum value(s).

(c) Find the intervals on which f is increasing. Find the intervals on which f is decreasing.

Solution

The domain of f is the set of real numbers.

(a) f has a local maximum at 1, since for all x close to 1, we have $f(x) \leq f(1)$. The local maximum value is $f(1) = 2$.

(b) f has local minima at -1 and at 3. The local minimum values are $f(-1) = 1$ and $f(3) = 0$.

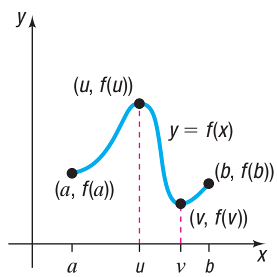
(c) The function whose graph is given in Figure 26 is increasing on the intervals $[-1, 1]$ and $[3, \infty)$, or for $-1 \leq x \leq 1$ and $x \geq 3$. The function is decreasing on the intervals $(-\infty, -1]$ and [1, 3], or for $x \leq -1$ and $1 \leq x \leq 3$.

WARNING The y-value is the local maximum value or local minimum value, and it occurs at some number x. For example, in Figure 26, we say f has a local maximum at 1 and the local maximum value is 2. ■

Now Work PROBLEMS 19 AND 21

5 Use a Graph to Locate the Absolute Maximum and the Absolute Minimum

Look at the graph of the function f given in Figure 27. The domain of f is the closed interval $[a, b]$. Also, the largest value of f is $f(u)$ and the smallest value of f is $f(v)$. These are called, respectively, the *absolute maximum* and the *absolute minimum* of f on $[a, b]$.



domain: $[a, b]$
for all x in $[a, b]$, $f(x) \leq f(u)$
for all x in $[a, b]$, $f(x) \geq f(v)$
absolute maximum: $f(u)$
absolute minimum: $f(v)$

Figure 27

DEFINITIONS Absolute Maximum and Absolute Minimum

Let f be a function defined on some interval I.

- If there is a number u in I for which $f(u) \geq f(x)$ for all x in I, then f has an **absolute maximum at u**, and the number $f(u)$ is the **absolute maximum of f on I**.
- If there is a number v in I for which $f(v) \leq f(x)$ for all x in I, then f has an **absolute minimum at v**, and the number $f(v)$ is the **absolute minimum of f on I**.

The absolute maximum and absolute minimum of a function f are sometimes called the **absolute extrema** or the **extreme values** of f on I.

The absolute maximum or absolute minimum of a function f may not exist. Let's look at some examples.

EXAMPLE 5 Finding the Absolute Maximum and the Absolute Minimum from the Graph of a Function

For the graph of each function $y = f(x)$ in Figure 28, find the absolute maximum and the absolute minimum, if they exist. Also, find any local maxima or local minima.

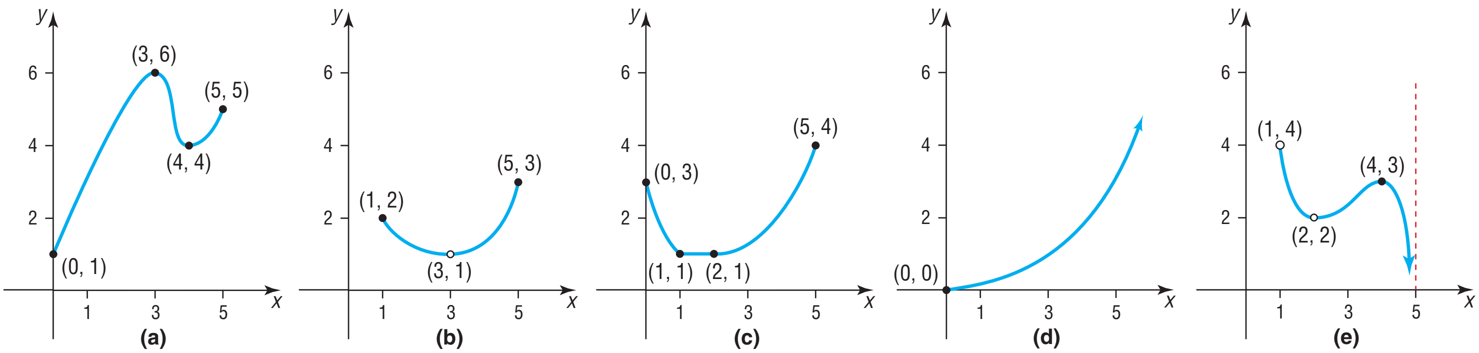


Figure 28

Solution

(a) The function f whose graph is given in Figure 28(a) has the closed interval $[0, 5]$ as its domain. The largest value of f is $f(3) = 6$, the absolute maximum. The smallest value of f is $f(0) = 1$, the absolute minimum. The function has a local maximum value of 6 at $x = 3$ and a local minimum value of 4 at $x = 4$.

(b) The function f whose graph is given in Figure 28(b) has domain $\{x \mid 1 \le x \le 5, x \ne 3\}$. Note that we exclude 3 from the domain because of the "hole" at $(3, 1)$. The largest value of f on its domain is $f(5) = 3$, the absolute maximum. There is no absolute minimum. Do you see why? As you trace the graph, getting closer to the point $(3, 1)$, there is no single smallest value. [As soon as you claim a smallest value, we can trace closer to $(3, 1)$ and get a smaller value!] The function has no local maxima or minima.

(c) The function f whose graph is given in Figure 28(c) has the interval $[0, 5]$ as its domain. The absolute maximum of f is $f(5) = 4$. The absolute minimum is 1. Notice that the absolute minimum occurs at any number in the interval $[1, 2]$. The function has a local minimum value of 1 at every x in the interval $[1, 2]$, but it has no local maximum.

(d) The function f given in Figure 28(d) has the interval $[0, \infty)$ as its domain. The function has no absolute maximum; the absolute minimum is $f(0) = 0$. The function has no local maxima or local minima.

(e) The function f in Figure 28(e) has domain $\{x \mid 1 < x < 5, x \ne 2\}$. The function has no absolute maximum and no absolute minimum. Do you see why? The function has a local maximum value of 3 at $x = 4$, but no local minimum value.

In calculus, there is a theorem with conditions that guarantee a function will have an absolute maximum and an absolute minimum.

THEOREM Extreme Value Theorem

If a function f is continuous* on a closed interval $[a, b]$, then f has an absolute maximum and an absolute minimum on $[a, b]$.

The absolute maximum (minimum) can be found by selecting the largest (smallest) value of f from the following list:

- The values of f at any local maxima or local minima of f in $[a, b]$.
- The value of f at each endpoint of $[a, b]$—that is, $f(a)$ and $f(b)$.

*Although a precise definition requires calculus, we'll agree for now that a function is continuous if its graph has no gaps or holes and can be traced without lifting the pencil from the paper.

For example, the graph of the function f given in Figure 28(a) is continuous on the closed interval $[0, 5]$. The Extreme Value Theorem guarantees that f has extreme values on $[0, 5]$. To find them, we list

- The value of f at the local extrema: $f(3) = 6, f(4) = 4$
- The value of f at the endpoints: $f(0) = 1, f(5) = 5$

The largest of these, 6, is the absolute maximum; the smallest of these, 1, is the absolute minimum.

Now Work PROBLEM 49

6 Use a Graphing Utility to Approximate Local Maxima and Local Minima and to Determine Where a Function Is Increasing or Decreasing

To locate the exact value at which a function f has a local maximum or a local minimum usually requires calculus. However, a graphing utility may be used to approximate these values using the MAXIMUM and MINIMUM features.

EXAMPLE 6 Using a Graphing Utility to Approximate Local Maxima and Minima and to Determine Where a Function Is Increasing or Decreasing

(a) Use a graphing utility to graph $f(x) = 6x^3 - 12x + 5$ for $-2 \le x \le 2$. Approximate where f has a local maximum and where f has a local minimum.

(b) Determine where f is increasing and where it is decreasing.

Solution (a) Graphing utilities have a feature that finds the maximum or minimum point of a graph within a given interval. Graph the function f for $-2 \le x \le 2$. Using MAXIMUM on a TI-84 Plus C, we find that the local maximum value is 11.53 and that it occurs at $x = -0.82$, rounded to two decimal places. See Figure 29(a). Using MINIMUM, we find that the local minimum value is -1.53 and that it occurs at $x = 0.82$, rounded to two decimal places. See Figure 29(b). Figure 29(c) shows the two extrema in Desmos.

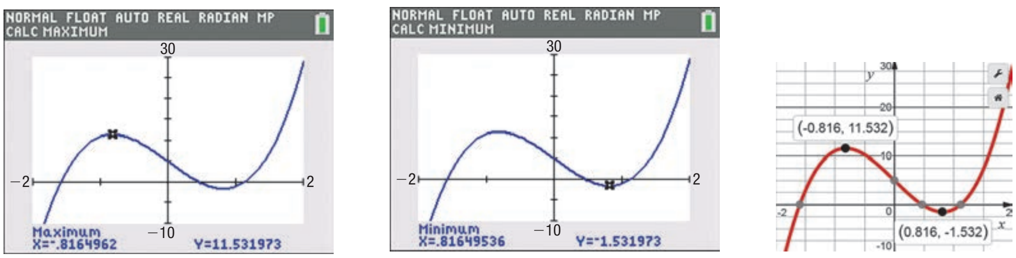


Figure 29 **(a)** Local maximum **(b)** Local minimum **(c)** Local extrema

(b) Looking at Figure 29, we see that f is increasing on the intervals $[-2, -0.82]$ and $[0.82, 2]$, or for $-2 \le x \le -0.82$ and $0.82 \le x \le 2$. And f is decreasing on the interval $[-0.82, 0.82]$, or for $-0.82 \le x \le 0.82$.

Now Work PROBLEM 57

7 Find the Average Rate of Change of a Function

In Section 1.3, we said that the slope of a line can be interpreted as the average rate of change. To find the average rate of change of a function between any two points on its graph, calculate the slope of the line containing the two points.

DEFINITION Average Rate of Change

If a and b, $a \neq b$, are in the domain of a function $y = f(x)$, the **average rate of change of f** from a to b is defined as

$$\text{Average rate of change} = \frac{\Delta y}{\Delta x} = \frac{f(b) - f(a)}{b - a} \qquad a \neq b \qquad \textbf{(1)}$$

The symbol Δy in equation (1) is the "change in y," and Δx is the "change in x." The average rate of change of f is the change in y divided by the change in x.

EXAMPLE 7

Finding the Average Rate of Change

Find the average rate of change of $f(x) = 3x^2$:

(a) From 1 to 3 (b) From 1 to 5 (c) From 1 to 7

Solution (a) The average rate of change of $f(x) = 3x^2$ from 1 to 3 is

$$\frac{\Delta y}{\Delta x} = \frac{f(3) - f(1)}{3 - 1} = \frac{27 - 3}{3 - 1} = \frac{24}{2} = 12$$

(b) The average rate of change of $f(x) = 3x^2$ from 1 to 5 is

$$\frac{\Delta y}{\Delta x} = \frac{f(5) - f(1)}{5 - 1} = \frac{75 - 3}{5 - 1} = \frac{72}{4} = 18$$

(c) The average rate of change of $f(x) = 3x^2$ from 1 to 7 is

$$\frac{\Delta y}{\Delta x} = \frac{f(7) - f(1)}{7 - 1} = \frac{147 - 3}{7 - 1} = \frac{144}{6} = 24$$

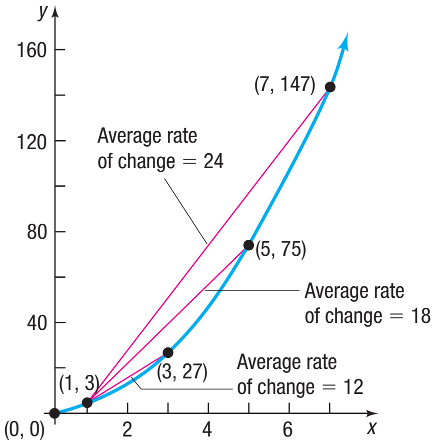


Figure 30 $f(x) = 3x^2$

See Figure 30 for a graph of $f(x) = 3x^2$. The function f is increasing on the interval $[0, \infty)$. The fact that the average rate of change is positive for any $x_1, x_2, x_1 \neq x_2$, in the interval $[1, 7]$ indicates that f is increasing on $1 \leq x \leq 7$.

Now Work PROBLEM 65

The Secant Line

The average rate of change of a function has an important geometric interpretation. Look at the graph of $y = f(x)$ in Figure 31. Two points are labeled on the graph: $(a, f(a))$ and $(b, f(b))$. The line containing these two points is called a **secant line**; its slope is

$$m_{\text{sec}} = \frac{f(b) - f(a)}{b - a}$$

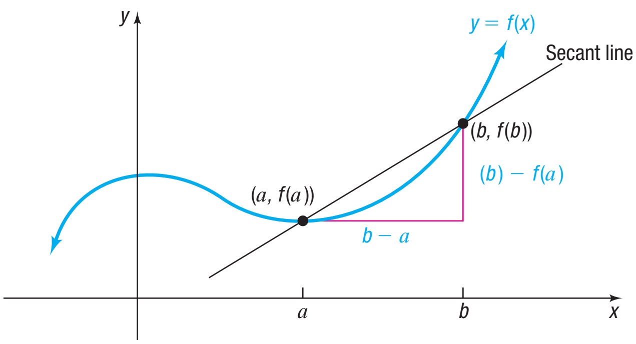


Figure 31 Secant line

THEOREM Slope of a Secant Line

The average rate of change of a function from a to b equals the slope of the secant line containing the two points $(a, f(a))$ and $(b, f(b))$ on its graph.

EXAMPLE 8

Finding an Equation of a Secant Line

Suppose that $g(x) = 3x^2 - 2x + 3$.

(a) Find the average rate of change of g from -2 to 1.

(b) Find an equation of the secant line containing $(-2, g(-2))$ and $(1, g(1))$.

(c) Using a graphing utility, draw the graph of g and the secant line obtained in part (b) on the same screen.

Solution

(a) The average rate of change of $g(x) = 3x^2 - 2x + 3$ from -2 to 1 is

$$\text{Average rate of change} = \frac{g(1) - g(-2)}{1 - (-2)}$$

$$= \frac{4 - 19}{3} \qquad g(1) = 3(1)^2 - 2 \cdot 1 + 3 = 4; \quad g(-2) = 3(-2)^2 - 2(-2) + 3 = 19$$

$$= -\frac{15}{3} = -5$$

(b) The slope of the secant line containing $(-2, g(-2)) = (-2, 19)$ and $(1, g(1)) = (1, 4)$ is $m_{\text{sec}} = -5$. Use the point-slope form to find an equation of the secant line.

$$y - y_1 = m_{\text{sec}}(x - x_1) \qquad \text{Point-slope form of a secant line}$$

$$y - 19 = -5(x - (-2)) \qquad x_1 = -2, y_1 = g(-2) = 19, m_{sec} = -5$$

$$y - 19 = -5x - 10 \qquad \text{Distribute.}$$

$$y = -5x + 9 \qquad \text{Slope-intercept form of the secant line}$$

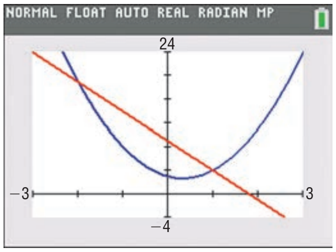


Figure 32 Graph of g and the secant line

(c) Figure 32 shows the graph of g along with the secant line $y = -5x + 9$ on a TI-84 Plus C.

Now Work PROBLEM 71

2.3 Assess Your Understanding

'Are You Prepared?' *Answers are given at the end of these exercises. If you get a wrong answer, read the pages listed in red.*

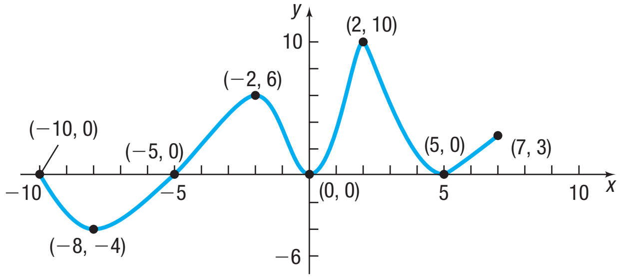
1. The interval (2, 5) can be written as the inequality ________. (pp. A72–A74)
2. The slope of the line containing the points $(-2, 3)$ and (3, 8) is ________. (pp. 56–59)
3. Test the equation $y = 5x^2 - 1$ for symmetry with respect to the x-axis, the y-axis, and the origin. (pp. 49–51)
4. Write the point-slope form of the line with slope 5 containing the point $(3, -2)$. (pp. 60–61)
5. The intercepts of the graph of $y = x^2 - 9$ are ________. (pp. 48–49)

Concepts and Vocabulary

6. A function f is ________ on an interval I if, for any choice of x_1 and x_2 in I, with $x_1 < x_2$, then $f(x_1) < f(x_2)$.
7. A(n) ________ function f is one for which $f(-x) = f(x)$ for every x and $-x$ in the domain of f; a(n) ________ function f is one for which $f(-x) = -f(x)$ for every x and $-x$ in the domain of f.
8. ***True or False*** A function f is decreasing on an interval I if, for any choice of x_1 and x_2 in I, with $x_1 < x_2$, then $f(x_1) > f(x_2)$.
9. ***True or False*** A function f has a local minimum at c if there is an open interval I containing c so that $f(c) \le f(x)$ for all x in this open interval.
10. ***True or False*** Even functions have graphs that are symmetric with respect to the origin.
11. ***Multiple Choice*** An odd function is symmetric with respect to ________.
    (a) the x-axis **(b)** the y-axis
    (c) the origin **(d)** the line $y = x$
12. ***Multiple Choice*** A function that is continuous on the interval ________ is guaranteed to have both an absolute maximum and an absolute minimum.
    (a) (a, b) **(b)** $(a, b]$ **(c)** $[a, b)$ **(d)** $[a, b]$

Skill Building

In Problems 13–24, use the graph on the right of the function f.



13. Is f increasing on the interval $[-8, -2]$?

14. Is f decreasing on the interval $[-8, -4]$?

15. Is f increasing on the interval $[-2, 6]$?

16. Is f decreasing on the interval $[2, 5]$?

17. List the interval(s) on which f is increasing.

18. List the interval(s) on which f is decreasing.

19. Is there a local maximum at 2? If yes, what is it?

20. Is there a local maximum at 5? If yes, what is it?

21. List the number(s) at which f has a local maximum. What are the local maximum values?

22. List the number(s) at which f has a local minimum. What are the local minimum values?

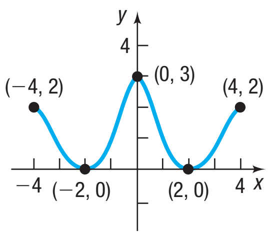
23. Find the absolute maximum of f on $[-10, 7]$.

24. Find the absolute minimum of f on $[-10, 7]$.

In Problems 25–32, the graph of a function is given. Use the graph to find:

(a) The intercepts, if any
(b) The domain and range
(c) The intervals on which the function is increasing, decreasing, or constant
(d) Whether the function is even, odd, or neither

25.



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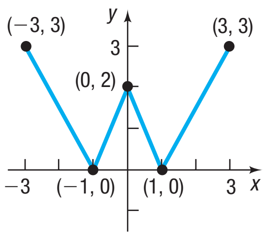

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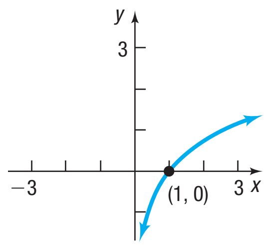

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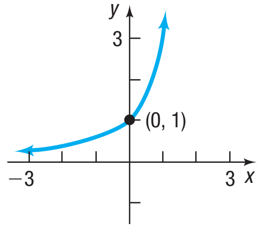

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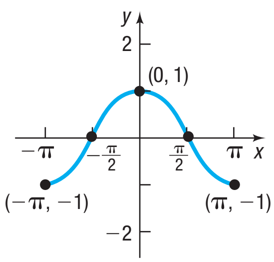

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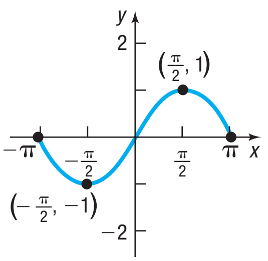

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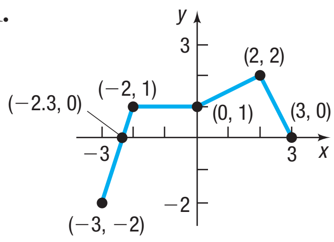

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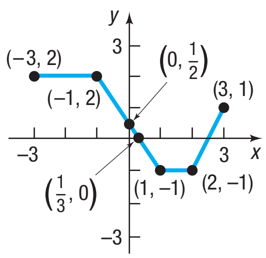
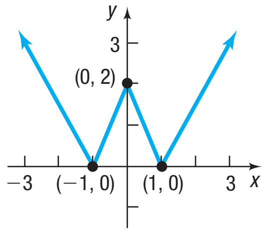


In Problems 33–36, the graph of a function f is given. Use the graph to find:

(a) The numbers, if any, at which f has a local maximum. What are the local maximum values?
(b) The numbers, if any, at which f has a local minimum. What are the local minimum values?

33.



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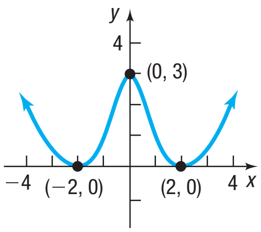

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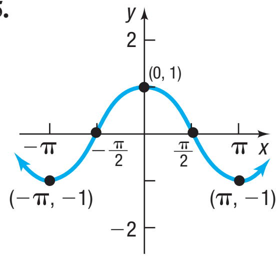

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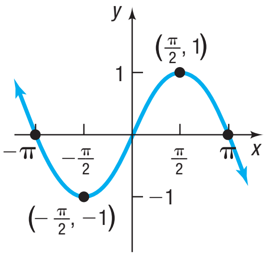


In Problems 37–48, determine algebraically whether each function is even, odd, or neither.

37. $f(x) = 4x^3$

38. $f(x) = 2x^4 - x^2$

39. $h(x) = 3x^3 + 5$

40. $g(x) = 10 - x^2$

41. $G(x) = \sqrt{x}$

42. $F(x) = \sqrt[3]{4x}$

43. $f(x) = \sqrt[3]{2x^2 + 1}$

44. $f(x) = x + |x|$

45. $h(x) = \dfrac{x}{x^2 - 1}$

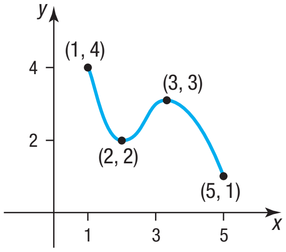
46. $g(x) = \dfrac{1}{x^2 + 8}$

47. $F(x) = \dfrac{2x}{|x|}$

48. $h(x) = \dfrac{-x^3}{3x^2 - 9}$

In Problems 49–56, for each graph of a function $y = f(x)$, find the absolute maximum and the absolute minimum, if they exist. Identify any local maximum values or local minimum values.

49.



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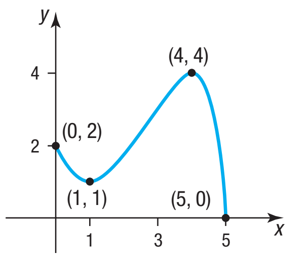

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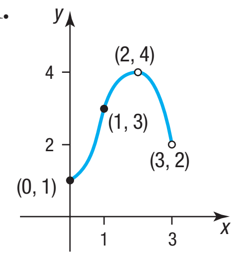

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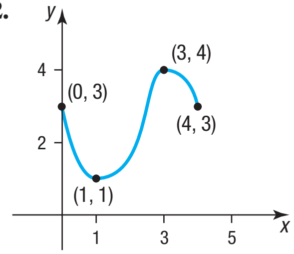

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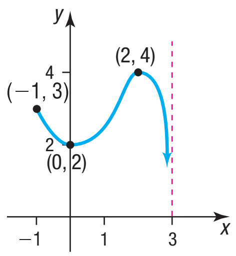

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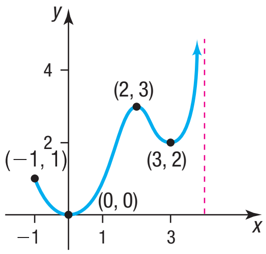

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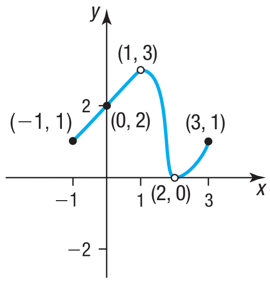

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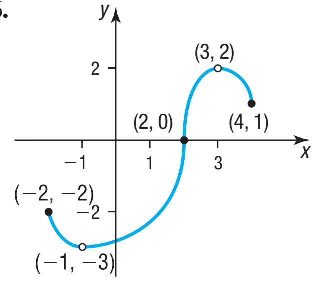

In Problems 57–64, use a graphing utility to graph each function over the indicated interval and approximate any local maximum values and local minimum values. Determine where the function is increasing and where it is decreasing. Round answers to two decimal places.

57. $f(x) = x^3 - 3x + 2 \quad [-2, 2]$

58. $f(x) = x^3 - 3x^2 + 5 \quad [-1, 3]$

59. $f(x) = x^4 - x^2 \quad [-2, 2]$

60. $f(x) = x^5 - x^3 \quad [-2, 2]$

61. $f(x) = -0.4x^3 + 0.6x^2 + 3x - 2 \quad [-4, 5]$

62. $f(x) = -0.2x^3 - 0.6x^2 + 4x - 6 \quad [-6, 4]$

63. $f(x) = -0.4x^4 - 0.5x^3 + 0.8x^2 - 2 \quad [-3, 2]$

64. $f(x) = 0.25x^4 + 0.3x^3 - 0.9x^2 + 3 \quad [-3, 2]$

65. Find the average rate of change of $f(x) = -2x^2 + 4$:
(a) From 0 to 2 **(b)** From 1 to 3 **(c)** From 1 to 4

66. Find the average rate of change of $f(x) = -x^3 + 1$:
(a) From 0 to 2 **(b)** From 1 to 3 **(c)** From −1 to 1

67. Find the average rate of change of $h(x) = x^2 - 2x + 3$:
(a) From −1 to 1 **(b)** From 0 to 2 **(c)** From 2 to 5

68. Find the average rate of change of $g(x) = x^3 - 4x + 7$:
(a) From −3 to −2 **(b)** From −1 to 1 **(c)** From 1 to 3

69. $f(x) = -4x + 1$
(a) Find the average rate of change from 2 to 5.
(b) Find an equation of the secant line containing $(2, f(2))$ and $(5, f(5))$.

70. $f(x) = 5x - 2$
(a) Find the average rate of change from 1 to 3.
(b) Find an equation of the secant line containing $(1, f(1))$ and $(3, f(3))$.

71. $g(x) = x^2 - 2$
(a) Find the average rate of change from −2 to 1.
(b) Find an equation of the secant line containing $(-2, g(-2))$ and $(1, g(1))$.

72. $g(x) = x^2 + 1$
(a) Find the average rate of change from −1 to 2.
(b) Find an equation of the secant line containing $(-1, g(-1))$ and $(2, g(2))$.

73. $h(x) = -2x^2 + x$
(a) Find the average rate of change from 0 to 3.
(b) Find an equation of the secant line containing $(0, h(0))$ and $(3, h(3))$.

74. $h(x) = x^2 - 2x$
(a) Find the average rate of change from 2 to 4.
(b) Find an equation of the secant line containing $(2, h(2))$ and $(4, h(4))$.

75. ***Mixed Practice*** $f(x) = -x^3 + 12x$
(a) Determine whether f is even, odd, or neither.
(b) There is a local maximum value of 16 at 2. Determine the local minimum value.

76. ***Mixed Practice*** $g(x) = x^3 - 27x$
(a) Determine whether g is even, odd, or neither.
(b) There is a local minimum value of −54 at 3. Determine the local maximum value.

Applications and Extensions

77. $F(x) = -x^4 + 8x^2 + 9$
(a) Determine whether F is even, odd, or neither.
(b) There is a local maximum value of 25 at $x = 2$. Find a second local maximum value.
(c) Suppose the area of the region enclosed by the graph of F and the x-axis between $x = 0$ and $x = 3$ is 50.4 square units. Using the result from (a), determine the area of the region enclosed by the graph of F and the x-axis between $x = -3$ and $x = 0$.